BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

Effect of Computerised Knowledge-Based Clinical Decision Support Systems on Patient-Reported and Clinical Outcomes of Chronic Disease Patients Managed in Primary Care Settings
A Systematic Review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-054659
Article Type:	Original research
Date Submitted by the Author:	23-Jun-2021
Complete List of Authors:	El Asmar, Marie line; Imperial College London, Self Care Academic Research Unit (SCARU) Dharmayat, Kanika ; Imperial College London, Imperial Centre for Cardiovascular Disease Prevention (ICCP), Department of Primary Care and Public Health Vallejo-Vaz, Antonio; Imperial College London, Imperial Centre for Cardiovascular Disease Prevention (ICCP), Department of Primary Care and Public Health, School of Public Health Irwin, Ryan; University of Birmingham, Institute of Applied Health Research Mastellos, Nikolaos; Imperial College London, Department of Primary Care and Public Health, School of Public Health
Keywords:	GENERAL MEDICINE (see Internal Medicine), PRIMARY CARE, PREVENTIVE MEDICINE, Health informatics < BIOTECHNOLOGY & BIOINFORMATICS

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Effect of Computerised Knowledge-Based Clinical Decision Support Systems on Patient-Reported and Clinical Outcomes of Chronic Disease Patients Managed in Primary Care Settings

A Systematic Review

Marie Line El Asmar,[1] MD, MPH, Kanika I Dharmayat,[2] MPH, BSc (hons), Antonio J. Vallejo-Vaz,[2] MD, PHD, Ryan Irwin,[3,*] PhD MPA MSc BSc (hons), Nikolaos Mastellos PhD, MSc, BSc (hons)[4,*]

1- Self Care Academic Research Unit (SCARU), Imperial College London. London, United Kingdom.
2- Imperial Centre for Cardiovascular Disease Prevention (ICCP), Department of Primary Care and Public Health, School of Public Health, Imperial College London. London, United Kingdom
3- Institute of Applied Health Research, University of Birmingham.
4- Department of Primary Care and Public Health, School of Public Health, Imperial College London. London, United Kingdom

* Joint principal/last authors. These authors contributed equally to the work.

Corresponding author:
Dr Marie Line El Asmar, MD, MPH
me719@ic.ac.uk

For peer review only

Key words
Clinical decision support system
Electronic health records
Primary care
Chronic disease
Health outcomes

Word Count:

- Abstract: 300 words
- Main text: 4,505 words

Number of Figures: 2

Number of Tables: 4

Number of references: 59

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Abstract

Objectives: Chronic diseases are the leading cause of disability globally. Most chronic disease management occurs in primary care, with studies showing that outcomes vary across primary care providers. Computerised Clinical Decision Support Systems (CDSS) have been shown to positively affect clinician behaviour by improving adherence to clinical guidelines. This study provides a summary of the available evidence on the effect of CDSS embedded in Electronic Health Records on patient-reported and clinical outcomes of adult chronic disease patients managed in primary care.

Design and eligibility criteria: Systematic review including randomised and quasi-randomised trials on the effect of CDSS (vs. usual care) on patient-reported or clinical outcomes of adult patients with selected common chronic diseases (asthma, chronic obstructive pulmonary disease, heart failure, myocardial ischemia, hypertension, type 2 diabetes mellitus, hyperlipidaemia, arthritis, and osteoporosis) managed in primary care.

Data sources: Medline, Embase, Cochrane Central Register of Controlled Trials, Health Management Information Consortium, and trial register clinicaltrials.gov, searched from inception to 24/06/2021.

Data extraction and synthesis: Screening, data extraction, and quality assessment were performed by two reviewers independently. The Cochrane Risk of Bias tool was used for quality appraisal.

Results: From 5,430 articles retrieved, eight studies met the inclusion criteria. Studies were heterogeneous in population characteristics, intervention components, and outcome measurements and focused on diabetes, asthma, hyperlipidaemia, and hypertension. Most outcomes were clinical with one study reporting on quality of life and disease-specific patient-reported outcomes. Quality of the evidence was impacted mostly by methodological biases of the studies.

Conclusions: There is inconclusive evidence in support of CDSS. A firm inference on the intervention effect was not possible due to methodological biases and study heterogeneity. Further research is needed to provide evidence on the intervention effect and the interplay between health care setting features, CDSS characteristics and the implementation process.

Review registration: Prospero ID: CRD42020218184

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Strengths and limitations

- In contrast to other reviews, this review focusses on interventions involving a specific type of CDSS that are knowledge-based, coupled with EHR systems, that provide electronic alerts, and are used by physicians in primary care.
- the electronic search strategy was piloted, and hand-searching of reference lists of previous reviews permitted a sufficient level of confidence that all relevant articles, against the eligibility criteria, were included for synthesis.
- Some limitations arose related to studies being of 1-year duration on average, making it difficult to deduce long-term effects on patient outcomes.
- Nine common chronic diseases were included in the syntax. However, there was an absence of eligible studies in five of them. The included studies were also of low-to-moderate quality of evidence.
- Additionally, the review does not report on process outcomes.

For peer review only

Introduction

Non-communicable chronic diseases are the leading cause of disability and mortality worldwide, affecting one in three adults globally.[1] Rising healthcare expenditure because of chronic disease prevalence has led to an ever-growing demand for effective primary care services for the long-term management of chronic conditions.[2-4] Considering that health systems have mainly adapted to providing periodic care for acute problems, an even greater need to achieve efficient, comprehensive and coordinated long-term care persists.[5] Models such as the Chronic Care Model (CCM) have been proposed to improve the quality of care in chronic care systems.[6] This model recommends 'decision support' as one of six applicable changes in primary care settings that promote better patient outcomes, if successfully implemented.[6-8] Within this and other models of primary care and chronic condition management, clinicians are encouraged to provide patient care according to evidence-based clinical practice guidelines (CPGs) to support improved patient outcomes.[9,10] Given the frequent updates to CPGs resulting from rapid advancements in medical knowledge, a major challenge is the maintenance of clinicians' knowledge of evidence-based CPGs.[10] Consequently, clinicians' adherence to CPGs may be variable and frequently insufficient, which may result in sub-optimal disease management outcomes.[10,11]

Clinical decision support is an umbrella term that includes various care structures, processes and tools.[12] Clinical decision support systems (CDSS) range from non-computerised to computerised systems and basic to advanced systems.[12,13] Currently, most attention is driven toward computerised advanced CDSS, which perform complex decision-making functions from electronic health record (EHR) patient data.[12] Such systems are termed knowledge-based, as they use software algorithms to generate actionable patient-specific outputs in the form of recommendations to the practitioner using conditional (IF-THEN) rule statements derived from CPGs.[14 15] They also differ in their technical and clinical interface, whether passive (providing

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

user-initiated prompts to generate clinical advice, e.g., by clicking buttons), or active (automatic system-initiated prompts).[12,16,17] CDSS also differ in system-user communication, which could be consulting (i.e., provide appropriate 'next-steps' clinical recommendations) or of critiquing nature (verify decisions taken by practitioners).[12,16] Given the differences in interfaces, the scope of this review includes computerised, knowledge-based CDSS embedded in EHRs that are of either passive or active, consulting or critiquing, nature.

Although CDSS are ubiquitous, few have been successfully adopted.[18] Previous reviews appraised the efficacy of different types of CDSS in primary care, presenting a greater consensus on the useful role of CDSS on practitioner-related outcomes such as guideline adherence.[19-23] However, the impact of CDSS on patient outcomes remains largely uncertain.[19-23] *Bryan et al* demonstrated a positive association between the use of CDSS in primary care and patient outcomes.[23] However, a need to further evaluate the effect of CDSS on patient outcomes was identified, particularly in primary care and specifically for chronic disease management, to inform efficacious implementation strategies and attract the industry's consideration of CDSS use in primary care settings.[23] There remains a need to assess the capabilities of CDSS in distinctive clinical professional areas and disease entities, particularly on patient outcomes.[18] A recent review on EHR-embedded CDSS focused on process delivery (practitioner outcomes) and was not limited to primary care settings and chronic conditions.[24] Despite obvious heterogeneity, clinically meaningful improvements in care processes were shown.[24] However, it was unclear under what clinical circumstances these improvements could be replicated in every clinical context to affect clinical outcomes presumptively.[24]

In this systematic review, patient outcomes refer to quantifiable clinical biomarkers such as glycated haemoglobin (HbA1c) and low-density lipoprotein-cholesterol (LDL-C) levels, clinical outcomes such as mortality and morbidity, and patient-reported outcomes, such as health-related quality of life (HRQoL). To our knowledge, and in the view of rapid upgrades in the world of CDSS, no review exists solely on the effect of CDSS embedded within EHRs on patient outcomes for chronic disease patients managed in primary care. This review aims to fill this knowledge gap by collating all the available evidence on this classification of CDSS.

Methods

Study design and eligibility criteria

A systematic review of the existing literature was conducted following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines.[25,26] The research question, inclusion, and exclusion criteria were defined using the PICOS framework.[27] Nine common chronic diseases were selected as part of the inclusion criteria, following a review by *Reynolds et al,*[28] namely asthma, chronic obstructive pulmonary disease (COPD), heart failure, myocardial ischemia, hypertension, type 2 diabetes mellitus, hyperlipidaemia, arthritis, and osteoporosis. Studies reporting on patient-related and clinical outcomes managed by primary care clinicians using CDSS-EHR compared to usual care were included. Studies involving participants <18 years of age, multidisciplinary (non-clinician) system users, and other varieties of CDSS (not

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

linked with EHR) were excluded. A detailed view of the criteria is provided in Supplementary Table 1.

Figure 1 depicts an analytic framework of the intervention adapted from a framework published by *Lobach et al*.[22] The framework provides a theoretical context for possible features that could underpin the success, or lack of success, of a complex intervention such as implementing CDSS in a primary care setting. The framework guided the extraction of relevant data from studies to inform the narrative synthesis.

Search Strategy

Electronic databases Medline, Embase, Scopus, Cochrane Central Register of Controlled Trials (CENTRAL) and Health Management Information Consortium (HMIC), and one clinical trial registry (clinicaltrials.gov), were searched from inception to 24th June 2020. Cochrane reviews were screened for relevant search terms and subject headings (MeSH) terms (Supplementary Table 2). The search was limited to articles in English. The strategy was then tested in Medline to ensure its efficacy in capturing relevant studies. Supplementary Table 3 shows the full Medline search syntax.

Terms for patient-reported outcome measures (PROM) were excluded. Existing PROM filters are not comprehensive, and the use of those filters in the search strategy could increase the risk of missing relevant studies.[29] However, we manually screened papers to identify any "patient-reported outcomes" measured in studies. PROM may be described in various ways, thus attempting to list specific patient outcomes for each entity could introduce bias and lead to missing relevant studies. Previous systematic reviews around CDSS in primary care were also searched for potentially eligible studies.[19-21,23,24]

Study Screening and Selection

Search results were exported to RefWorks, then screened using Covidence. After removal of duplicates, screening was conducted by two reviewers independently (MA, KID). Disagreements were resolved by consensus within the review group. After an initial screening of titles and abstracts, the full text of potentially relevant articles were assessed.

Data Extraction

A data extraction form was adapted using the 'Cochrane data collection form for intervention reviews: randomised controlled trials (RCTs) and non-RCTs' and aided by *Lobach et al*'s data abstraction guidance.[30,31] The form was piloted and two reviewers (MA, KID) completed data extraction independently.

Quality Assessment of Individual Studies

Risk of bias was assessed using the Cochrane risk of bias (RoB) tool,[32,33] by two reviewers independently (MA, KD). The methodological risk of bias was assessed per the following elements in RCTs: selection bias, performance bias, detection bias, attrition bias, reporting bias, and other biases.[33,34]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Quasi-RCTs were reported as high risk for random sequence generation. Cluster-RCTs were assessed for an additional element 'selective recruitment of cluster participants.' All judgments were made with guidance from the Cochrane Handbook and the guidelines of the Cochrane Consumers and Communication Review Group.[33-35]

Studies were considered as high risk if they scored 'high' or 'unclear' for either random sequence generation or allocation concealment, based on the rising evidence that these two elements are particularly significant sources of bias.[34] A summary 'risk of bias' figure was generated for all risk assessments of elements with a narrative commentary for each study. Additionally, a detailed adapted table was generated to visualise sub-elements that fit under each element in the Cochrane RoB tool.[34]

Quality Assessment of Outcomes

The GRADE (Grading of Recommendations, Assessment, Development and Evaluations) tool was used to assess the quality of outcomes from individual studies. [36] The tool involves five elements: risk of bias, inconsistency, indirectness, imprecision, and publication bias. The tool is primarily used for meta-analyses; however, it can be adapted for use in narrative reviews. Study outcomes were appraised, ranked as high, moderate, low or very low and justifications were provided, in line with Cochrane guidance.[37]

Data Synthesis

Treatment effect was reported as odds ratios (OR) with 95% confidence intervals (CIs) and p-values for dichotomous outcomes, where available. Continuous data were presented as mean differences with 95% CIs and p-values.

Heterogeneity in the results was anticipated due to the wide spectrum of possible patient-related outcomes reported by the studies. Methodological heterogeneity was assessed using Cochrane's RoB tool. Heterogeneity in included studies varied in terms of intervention components, population characteristics, and outcome measurements; thus, conducting a meta-analysis was not appropriate.

Patient and Public Involvement

No patients were involved.

Ethics Approval

The present work relied on already published aggregated data and ethics committee approval was unnecessary.

For Peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Results

The search identified 5,430 articles. Following removal of duplicates, 4,498 articles remained for screening by title and abstract. Of these, 266 articles were selected for full-text review and 10 articles met the inclusion criteria. Among them, two articles were published protocols of included articles; thus, 8 different studies were finally included. The detailed search results are presented in the PRISMA flowchart (Figure 2).

Study Characteristics

Characteristics of included studies are presented in Table 1. Included studies comprised of patients in four disease areas: T2DM, asthma, hyperlipidaemia, and hypertension.[38-45] No studies on COPD, heart failure, myocardial ischemia, arthritis, and osteoporosis were identified. Six studies were cluster-RCTs, one study was parallel-group RCT, and one was quasi-RCT of stepped-wedge design. Most studies were conducted in the United States, and three studies took place in the United Kingdom, Spain, and Belgium. Studies, both among the disease conditions and within each disease entity, differed appreciably in choice of study outcomes, participant covariates, setting attributes, and CDSS characteristics. Hence, reported outcomes were not fit for statistical pooling (meta-analysis). Follow-up ranged from 6 months to 1.5 years. No study incorporated financial reimbursement before the implementation of interventions. Further details are shown in Table 1.

CDSS Characteristics

CDSS interventions were characterised based on their origin (commercial/local), delivery mode (user or system-initiated), user response (optional/mandatory), events (system functions), coupled EHR, and auxiliary features (local clinician involvement in system development process, system accessibility to other team members, provision of performance feedback, or coupled with user education). Additional information such as user-training, user adherence, and reported usability were recorded. Details are provided in Table 2.

Risk of Bias and Quality of the Evidence

Supplementary Tables 3 and 4 provide risk assessments details. Overall, the included studies were impacted by risk of bias, mainly due to incomplete outcome data and uncertainty on blinding of outcome assessment and personnel. In most studies, it was unclear if outcome assessors were blinded. Additionally, blinding of physicians (CDSS users) was not possible across all studies, as they were the intended system users. Most studies reported substantial withdrawal rates due to poor follow-up. These limitations resulted in analysing subgroups (rather than the initially intended full cohorts) in some trials, hence most studies were considered of high risk for incomplete outcome data.

Overall, quality of the evidence was rated as either moderate or low, with the most impact resulting from methodological risks of bias within studies. Supplementary Table 5 presents the quality ovidence evaluated using GRADE, with reasons stated for quality downgrading.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Narrative Synthesis

Diabetes Mellitus

Three studies assessed the effect of CDSS on clinical and patient-reported outcomes in T2DM patients.[38-40] Among them, *Gill et al* retrieved small, yet statistically significant, effect with the CDSS intervention compared to the control group, with a small between-group difference in HbA1c and LDL-C levels of 0.12% and 3.57 mg/dL, respectively (Table 4). Though statistically significant, the reported reductions in HbA1c and LDL-C were clinically very small; however, the intervention group had 52% higher odds of achieving personalised HbA1c levels, 56% higher odds of achieving HbA1c <7.0%, and 34% higher odds of achieving an LDL-C <100 mg/dL compared to usual care patients over 1 year of follow-up. Nevertheless, compared to the control group, the intervention group had better-controlled baseline glycaemia and LDL-C. The three studies involved commercial systems. However, the study by *Gill et al* involved a system-initiated "push" CDSS, compared to the other two studies that involved user-initiated CDSS. *Gill et al* reported the highest usability among the three studies. Additionally, unlike the other two studies, in *Gill et al* the CDSS was integrated into EHRs already operating at participating practices. Reported limitations were of EHR-CDSS interoperability difficulties, which resulted in missing patient lab values. Other missing lab values also resulted from patients not returning for lab tests. In view of these limitations, a subset of randomised patients without missing HbA1c and LDL-C values were analysed (Table 4).

Schnipper et al (RCT) reported the odds of patients meeting management goals (Table 4). Intention-to-treat analysis showed non-significant changes in patients' odds of meeting management goals; a per-protocol analysis, however, showed slightly significant odds of attaining management goals in the intervention group compared to the control group (Table 4). Among the three studies, this intervention is the only one that involved user training (Table 2)**.** Auxiliary features also included feedback on performance and local user (physician) involvement in the CDSS development process; yet system usability was reported to be used for only 5.6% of patients (Table 2).

Heselmans et al (cluster RCT) reported non-significant between-group differences in HbA1c, LDL-C, diastolic (DBP) and systolic (SBP) blood pressure (BP) (Table 4). Levels of HbA1c were relatively well-controlled**,** with reasonable mean values for HbA1c in both groups at baseline, especially in the control group (Table 3). Subsequent analysis of a subgroup of 601 patients with HbA1c >7.0% retrieved a statistically significant between-group difference in HbA1c favouring the intervention, though not clinically relevant (-0.40% [95% CI -0.70 to -0.09]). The same analysis retrieved non-significant differences in LDL-C and BP. User adherence and usability were not measured.

In all three studies, there was no mention of the provision of reimbursement for the implementation of the CDSS. All CDSS were commercial, and user response to the system alerts was optional.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Asthma and Angina

One study assessed the impact of CDSS on self-reported quality of life and condition-specific outcomes in patients with asthma and angina.[41] The angina intervention group acted as the control group for the asthma intervention group and vice versa (justified by authors to eliminate a possible Hawthorne effect in practitioner behaviour on the primary outcome). Questionnaires were administered in three rounds (a year before the intervention, beginning of the intervention, and a year after the intervention), which showed non-significant differences in both intervention groups post-intervention (Table 4). Completion of the three rounds of questionnaires occurred in 46% and 35% of patients with angina and asthma, respectively. The implemented CDSS was locally produced and embedded in two widely used health systems (Table 2). The CDSS was modified four months into the study from system to user-initiated, as requested by participating physicians. User training was limited, and system usability was very low, measured using a log with a median of zero user-system interaction during most of the study (Table 2).

Hyperlipidaemia

Two studies involved participants with hyperlipidaemia.[42,43] Both studies reported non-superiority for an improvement of LDL-C values with the CDSS intervention compared to usual care. In *Cobos et al*, despite a usability of 71%, the intervention had no superiority over usual care in terms of between-group differences in lipid values (Table 4). The study had a lower number of participants compared to *Gill et al*'s study and additionally it reported >25% patient lost to follow-up for clinic visits or LDL-C testing (Table 3).

By comparison, *Gill et al* involved a considerably larger sample size (Table 3). The study involved implementing a commercial CDSS in participating practices of one institution. All patients enlisted under participating practices with recorded LDL-C values were enrolled. A high correlation in outcomes was found at baseline and end measurements, causing patients not meeting LDL-C levels after randomisation to be analysed only. The control group also had a lower risk of disease and was significantly younger. Patients were stratified into three guideline-based risk groups, and the percentage of patients reaching the pre-defined goals at baseline and end were recorded (Table 3). All categories in the intervention and control groups showed a significant increase in the proportion of patients attaining lipid goals, except in the moderate-risk group. Calculated differences were not presented; however, p-values of differences within the intervention and control groups suggested no significant improvement in any risk category.

Hypertension

Two studies assessed the effect of CDSS on BP in hypertensive patients.[44,45] Both studies employed CDSS that were system-initiated with optional user responses (Table 2). *Hicks et al* reported no differences between intervention groups in BP control (Table 4). Differences in baseline and end SBP and DBP between groups were not significant. Nonetheless, the intervention group had a significantly higher percentage of patients with controlled BP (Table 3). No user training was provided, and system usability was not measured (Table 2). Less than 80% of recruited patients were followed-up due to withdrawals caused by the study's inclusion criteria, which required patients to have a recorded ethnicity in the EHR. Subsequently, patients without

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

BP recordings were considered uncontrolled. However, a subset analysis excluding these patients still retrieved a non-significant difference in BP values.

Lopez et al study was a quasi-RCT and reported significant odds of BP improvement in the intervention group compared to the control group (Table 4). Notably, extensive training was provided for participating clinicians compared to all other studies, and semi-structured interviews revealed that all practices appeared to use the system components (Table 3).

Discussion

Main findings

As primary care providers are increasingly required to monitor and manage chronic disease, there is a need to assess the capabilities of CDSS on patient outcomes. This systematic review included the eight available studies on the effect of computerised, knowledge-based, EHR-embedded CDSS on clinical and patient-reported outcomes. Two studies reported a statistically significant impact of CDSS on patient outcomes,[39,44] though of unclear clinical significance due to the very small changes achieved. Another two studies reported statistically significant findings but in subgroup analyses.[38,40] One study reported statistical significance in partial outcomes and three studies did not report any statistically significant impact of CDSS on outcomes at all.[42-44] Overall, evidence was not conclusive to imply a differential effect, either positive or negative, of clinician use of CDSS on patient or clinical outcomes compared to usual care, mainly due to the quality of evidence impacted by the prominent risk of bias in the included studies. The systematic review also noted a limited availability of studies evaluating the CDSS of interest directed to clinicians in primary care for the management of the common chronic diseases. This is especially true for COPD, arthritis, and osteoporosis, which were eligible conditions, yet no such studies were available.

Implications of the review

The systematic review infers the importance of considering how the interplay of factors such as the nature of a health care setting (academic, community) and CDSS interact with clinicians (or users per se). System-initiated CDSS seemed to have a better reflection on patient outcomes compared to user-initiated systems; however, this relationship was confounded by many factors such as poor process measures, namely low system usability and EHR-CDSS interoperability difficulties. A common limitation reported across most studies was poor patient follow-up, which highlights the importance of patient behaviour as a confounding factor to the performance of CDSS interventions in clinical settings. Some interventions included helpful auxiliary features such as involvement of local physicians in CDSS development and feedback on performance, yet these features did not seem to reflect positively on patient outcomes. Understanding user (clinician) experiences qualitatively in the context of using such CDSS to manage chronic diseases in primary care may be of benefit to inform on their practical use. This review assessed CDSS functionality when used by clinicians; however, CDSS accessibility to other members of a multidisciplinary team may have an added positive effect which could translate into better patient outcomes. Multidisciplinary teams in primary care have been shown to aid successful implementation of

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

CDSS by increasing adherence to CPGs and improving patient outcomes in the absence of CDSS.[46,47]

The healthcare setting and intervention implementation itself may dictate the success or failure of an intervention. Large academic care settings may be better equipped with IT and financial resources than community centres to ensure successful implementations of CDSS.[48] Included studies revealed limitations due to poor implementation of the CDSS, namely interoperability issues, lack of IT support, poor usability, and lack of user training, which could have hindered the intervention from reflecting significant outcomes compared to usual care. Models have been offered to help explicate how structural elements of healthcare settings and various levels of healthcare organisations levelling up to the industry may influence clinician behaviour and uptake of health IT to influence patient outcomes.[17,49] Finally, it is imperative to consider the effect of non-clinical factors such as patient behaviour outside clinic visits, their perceptions, and adherence to clinical advice.[50]..

Comparison with previous reviews

Previous reviews of CDSS of multiple types have indicated impact on patient outcomes in some instances.[44,48,85,86] A recent review showed that CDSS systems achieve small to moderate effects on processes of care, yet the extent of translation to patient outcomes and under what clinical contexts and conditions this was achieved was not determined.[24] Previous reviews were not limited to primary care settings and included various other CDSS, namely paper-based, web-based, standalone electronic systems, and patient-directed systems that do not necessarily alert a clinician at the point-of-care.[19,21,51] However, reviews assessing the impact of various types of CDSS on chronic disease patient outcomes in primary care identified inconclusive results, with a need to define more clearly the exact role of CDSS compared to usual care in subsequent studies.[20,23] Reviews evaluating patient outcomes noted conceptual between-study heterogeneity, as this review did, especially when several diseases and clinical outcomes were considered.[52] In terms of patient outcomes, CDSS seem to affect clinical outcomes for cardiovascular risk factors such as hyperlipidaemia and diabetes.[11,53] Nevertheless, a much more recent review on the effect of CDSS on cardiovascular risks revealed no definitive clinical benefits after their implementation.[53] On the other hand, the evidence is less robust for other non-cardiovascular risk-related chronic conditions, such as cancer, and other primary care sensitive cases (PCSC) such as vaccination, which were not part of the eligibility criteria in this review.[54]

A review of CDSS embedded within EHRs in all-purpose healthcare settings on overall mortality and morbidity suggested no effect of CDSS on mortality. However, it noted that potential benefits or risks would highly depend on setting and disease characteristics, which is an evident implication in this review.[52] A review by *Bryan et al* validated the use of CDSS in primary care to improve outcomes; however, they reported substantial variability in intervention effectiveness related to the type and implementation of the different types of CDSS.[23] Additionally, other reviews incorporated studies that involved the use of CDSS by multidisciplinary care members, reflecting positive clinical outcomes with shared decision making.[19,51,55]

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Unanswered questions and future research

More research is needed to establish the effectiveness of EHR-embedded CDSS on clinical outcomes of patients with chronic diseases treated in primary care settings. Further studies involving patients with chronic diseases are needed for robust evidence. Moreover, it may not be reasonable to evaluate the intervention's efficacy based on older studies, given the exponential growth and rapid advancements in the field of CDSS.[56] Future research could include practitioner-related, cost-effectiveness, and patient-related outcomes to present an ampler perspective on the effect of this type of CDSS to inform policymaking and clinicians. Other primary care sensitive conditions (PCSCs) can be studied for a broader scope, including cancer, mental health illnesses and preventative interventions such as vaccination.[57] Research around prolonged evaluations of CDSS is needed. Longer-term studies may account for potential implementation barriers that could take place. Additionally, it would be beneficial to synthesise findings on the effect of this particular type of CDSS on patient outcomes, when operated within a multidisciplinary team of non-clinician providers. It is worth noting that patient outcomes are not always addressed efficiently in trials, among the reasons is the possibility of having subjective, composite, or surrogate outcomes, which do not necessarily translate into real-life patient outcome improvements.[58] Hence, including observational studies in updated reviews may be useful. Additionally, no studies of registries embedded with CDSS fit the eligibility criteria, suggesting a potentially greater inclination to embed CDSS with EHRs currently. There also seems to be a dearth of studies evaluating non-cardiovascular risk-related chronic diseases post-implementation of CDSS linked with EHRs, as no studies evaluating patients with COPD, osteoporosis, and arthritis were eligible.

Strengths and limitations

This systematic review focusses on interventions involving a specific type of CDSS –those that are knowledge-based, coupled with EHR systems, that provide electronic alerts, and are used by physicians in primary care. To the authors' knowledge, no such review exists considering the attention these systems are increasingly gaining.[29] Application of the eligibility criteria identified eight studies for inclusion, which may be a small number given the recent growth of interest in CDSS in healthcare settings.[59] Nevertheless, piloting the electronic search strategy and supplementing it with hand-searching of reference lists of previous reviews in the field permitted a sufficient level of confidence that all relevant articles, against the eligibility criteria, were included for synthesis. The review also contributes to the knowledge provided by a recent review that focuses on process outcomes.[24]

Some limitations arose from methodological constraints. Studies were of an average of 1-year duration, which makes it difficult to deduce long-term effects on patient outcomes. Nine common chronic diseases were included in the syntax.[28] However, there was an absence of eligible studies in five of them. The included studies were also of low-to-moderate quality of evidence. Additionally, our review does not report on process outcomes. The decision to exclude process outcomes was made to channel the research question to focus on where the most significant knowledge gap lies, as the effect of CDSS on patient outcomes remains uncertain.[19-21,23,24] Nevertheless, capturing those outcomes would have provided a more comprehensive assessment on the effect of CDSS for the management of chronic disease patients in primary care.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Conclusion

This systematic review aimed to assess the impact of CDSS integrated with EHRs on patient-reported and clinical outcomes of chronic disease patients in primary care settings. A definite conclusion on the intervention effect on outcomes could not be drawn due to between-study heterogeneity and methodological risks, reflecting an immature research area. Intervention characteristics and implementation approaches may have an impact on outcomes from CDSS use and we make reference to the literature in this area that may guide and help evaluate successful implementations of CDSS in practice. Further studies are needed to formulate more robust inferences and validate the intervention's effect on patient outcomes. With the continuous rise in the prevalence of chronic diseases, it is essential to understand how enhancing primary care management using IT support could support improved outcomes. Practical applications of CDSS evaluations in the healthcare workflow and setting should be sustained to suitably determine the actual effectiveness of CDSS and support improved implementation efforts into practice. Strengthening the clinical benefit of CDSS and determining their cost-effectiveness could inform policymaking of enhanced primary healthcare of chronic disease management.

For peer review only

Summary of Implications

- There is inconclusive evidence in support of the clinical use of CDSS for chronic disease management in primary care.
- Further and longer-term studies involving EHR-integrated CDSS use in primary care for chronic disease patients are needed.
- Including observational studies may be of value when studying patient outcomes.
- There seems to be a lack of studies involving non-cardiovascular risk-related chronic diseases and studies involving CDSS linked with patient registries
- It is imperative to consider the interplay of health care setting features and CDSS characteristics with the CDSS implementation process.
- Limitations arising from implementing CDSS in the primary care setting were common among studies, suggesting a potential benefit of systematically assessing CDSS implementation using published evaluation methods.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Acknowledgements

The authors would like to thank the library services team at Imperial College London. Special thanks to Ms Rebecca Jones for her support in reviewing the search strategy.

Ms KI Dharmayat acknowledges support from a PhD Studentship from the National Institute for Health Research (NIHR) under the Applied Health Research (ARC) programme for North West London, United Kingdom (the views expressed in this publication are those of the authors and not necessarily those of the NHS, the NIHR or the Department of Health).

Authors' statement

RI conceived the original study concept with input from NM. NM and AJV-V led the overall study design and supervised the subsequent steps of the study. MA and KD designed the search strategy with supervision from the other authors. MA and KD conducted the literature search, data extraction and quality assessment of studies. MA wrote the first draft of the manuscript. All authors participated in the interpretation of data, have critically reviewed the manuscript and provided edits and comments, and approved its final submission.

Conflict of interest

Ryan Irwin and Nikolas Mastellos are employees of Cerner UK Ltd –a health technology organisation. Dr AJ Vallejo-Vaz reports past or current participation in research grants to Imperial College London from Pfizer, Amgen, Sanofi, MSD, Daiichi Sankyo and Regeneron, honoraria for lectures from Amgen, Mylan and Akcea, and consulting fees from Bayer, all outside the submitted work. MS KI Dharmayat reports past or current participation in research grants to Imperial College London from Amgen, Sanofi, MSD, Daiichi Sankyo and Regeneron and consulting fees from Bayer, all outside the submitted work.

Funding

No funds were received for conducting the present study.

Data Sharing Statement

Not applicable. Data on which this study is based on is publicly available in electronic databases of studies.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1: Characteristics of included studies

							Participants		
Study	**Disease**	**Study Design**	**Country**	**Duration**	**Sponsorship**	**Setting**	**Patient identification**	**Intervention**	**Control**	**Outcomes**
Schnipper et al. (2010)	Type II Diabetes Mellitus	RCT	United States	6 months	The Agency for Healthcare Research and Quality	Partners HealthCare System, 10 practices at Brigham and Women's Hospital and Massachusetts General Hospital (Academic)	NS[1]	3431 patients	3578 patients	Difference in proportion of patients meeting management goals compared to baseline.
Gill et al. (2017)	Type II Diabetes Mellitus	Cluster -RCT	United States	1 year	Sanofi, U.S	Primary care practices in Delaware and Maryland (Community)	International classification of diseases (ICD-9CM) EHR code	5 offices 29 PCPs 4484 patients **Analysed (by subgroups)**: HbA1c : 2041 LDL-C: 2793	5 offices 23 PCPs 1902 patients **Analysed (by subgroups):** HbA1c: 723 patients LDL-C: 931 patients	• Difference in HbA1c [2] from baseline • Difference in LDL-C [3] from baseline • % patients achieving HbA1c goals
Heselmans et al. (2020)	Type II Diabetes Mellitus	Cluster -RCT	Belgium	1 year	None	Belgian primary care practices using OneHealth EHR (Community)	International classification of primary care (ICPC-2) EHR code or confirming lab tests	25 offices 60 PCPs **Analysed:** 14 offices 21 PCPs 1351 patients	26 offices 60 PCPs **Analysed:** 15 offices 31 PCPs 2464 patients	Mean value differences of HbA1c, LDL-C, and blood pressure of the previous year at 6 and 12 months compared to baseline
Eccles et al. (2002)	Asthma and Angina	Cluster -RCT	United Kingdom	1 year	UK NHS research & development	General practices in north east England. (Community)	Asthma or angina EHR code in prior 12 months	**Asthma**: 31 offices **Angina**: 29 offices	**Angina:** 29 offices, **Asthma**: 31 offices	Quality of life and condition specific outcomes using questionnaires
Gill et al. (2009)	Hyperlipidaemia	Cluster -RCT	United States	1 year	None	Medical quality improvement consortium (MQIC) of primary care practices across the country (Academic and Community)	Identified as having LDL levels recorded in EMR	12 offices 49 PCPs 26,696 patients	13 offices 56 PCPs 37,454 patients	The percentage of patients with most recent LDL-C at goal

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Cobos et al. (2006)	hyperlipidaemia	Cluster RCT	Spain	1 year	Novartis	General practices in Catalonia	Identified as having total cholesterol (TC) between 200mg/dL and 400 mg/dL	20 offices 1046 patients	22 offices 1145 patients	Between-group differences in means of lipid values
Hicks et al. (2008)	Hypertension	Cluster RCT	United States	1.5 years	The Agency of Healthcare Research and Quality	Primary care practices affiliated with the Brigham and Women's Hospital	Identified as having 2 or more hypertension-related outpatient visits using EHR codes	7 offices 786 patients	7 offices 1048 patients	• Percentage of patients with controlled blood pressure • Odds of blood pressure control at outcome visit
Lopez et al. (2019)-2016	Hypertension	Non- RCT	United States	1 year (2016)	Centre for Disease Control	Primary care practices in New York	Identified as having at least 1 ICD code for hypertensio n in EHR	7 offices	7 offices	Odds of blood pressure control post-intervention

[1] NS abbreviates not stated
[2] HbA1c refers to Glycated haemoglobin
[3] LDL-C refers to Low-density lipoprotein

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 2: CDSS characteristics of interventions in included studies

Study	Disease	Origin	Delivery Mode	User Response	Events	Guidelines	Integrated EHR (name)	User Training	User Adherence	Usability	Auxiliary features
Schnipper et al. (2010)	Diabetes mellitus	Commercial	User- initiated	Optional	**1)** Requests missing data. **2)** Alerts outputs of assessments of clinical care **3)** Recommends orders for medication changes, laboratory studies, appointments, and referrals	NS	Longitudinal Medical Record	• Brief instructions at start • Monthly tips	Tracked use by clinician and sent customised emails	Used for 5.6% of eligible patients	Local user involvement in development process
Gill et al. (2017)	Diabetes mellitus	Commercial	System-initiated	Optional	**1)** Generates reports for patient-specific recommendations at the POC	• American diabetes Association • European Association for the study of Diabetes & ICD 9 coding	Multiple EHRs (previously employed)	Not provided	Qualitatively assessed through informal interviews	4/5 practices reported use	Recommendations also received by non-clinicians in care team
Heselmans et al. (2020)	Diabetes mellitus	Commercial	User- initiated	Unclear	**1)** Alerts for reminders, therapeutic suggestions, and diagnosis-specific guideline links	Belgian and International Guidelines	HealthOne	Not provided	Not assessed	Not assessed	None
Eccles et al, (2002)	Asthma	Local	System- initiated (4 months) then altered to user-initiated	Optional	1) Alerts management and prescription options and patient referrals	NS	AAH Meditel or EMIS systems	1-day training	Measured by a usage log	Median number of system interactions was zero	Education on system guidelines
Gill et al. (2009)	Hyperlipidaemia	Commercial	System- initiated	Optional	**1)** Alerts for overdue lipid tests **2)** Alerts if patient has not met personalised lipid goals 3)Generates disease risk categories	Adult Treatment Panel-III (ATP-III) guidelines	EMR (Centricity)	1-year familiarity with system	Not measured	Not measured	Generates a list of patients who have not met lipid goals, outside office visits. Listed patients were sent a letter at the start and after 6 months of the study.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Cobos et al. (2006)	Hyper lipidaemia	Local	System- initiated	Optional	1) Recommends therapeutic options 2) Recommends follow- ups 3) Recommends lab testing	The European Society of Cardiology and other societies for Hypercholesterolemia management (ESCHM)	Not specified	None	Monitored by CDSS	71%	None
Hicks et al. (2008)	Hypertension	Local	System- initiated	Optional	Alerts for therapeutic options	American Heart Association/American College of Cardiology (AHA/ACC) 2001 guidelines	Not specified	None	Assessed prescription of a recommended drug class within 1 week of clinic visit	NS	None
Lopez et al (2019)	Hypertension	Local	System- initiated	Optional	1)Alerts for missing BP readings 2) Alerts for therapeutic options	Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure (JNC- 7) guidelines	eClinicalWorks or MD Land	2-day training at start, 7 training sessions during second year, ad-hoc sessions as needed	Semi-structured interviews	Reported all practices have used all CDSS components	Generates reports of patients by status to manage visits

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 3: Characteristics of participants

					Participants		Comparability		
Study	**Disease**	**Country**	**Patient identification**	**Unit of randomisation**	**Intervention**	**Control**	**Age**	**Sex**	**Disease Severity**
Schnipper et al. (2010)	Diabetes	US	NS [1]	Patients	3431 patients	3578 patients	Comparable	**IG** [2]: more females (statistically significant)	Comparable
Gill et al. (2017)	Diabetes	US	International classification of diseases (ICD-9CM) EHR code	Practices	5 offices 29 PCPs 4484 patients **Analysed (by subgroups)**: HbA1c: 2041 LDL-C: 2793	5 offices 23 PCPs 1902 patients **Analysed (by subgroups):** HbA1c: 723 patients LDL-C: 931 patients	**Overall**: 18-75 **IG**: older (statistically significant)	Comparable	**IG:** Better glycaemic control, lower LDL-C
Heselmans et al. (2020)	Diabetes	Belgium	International classification of primary care (ICPC-2)(EHR code or lab tests)	Practices	25 offices 60 PCPs **Analysed:** 14 offices 21 PCPs 1351 patients	26 offices 60 PCPs **Analysed:** 15 offices 31 PCPs 2464 patients	**Overall**: ≥ 18 years **IG**: older (significant statistically)	**CG** 3: more females (statistically significant)	**Comparable for:** mean HbA1c, mean BP, % patients meeting target for HbA1c , BP and LDL-C
Eccles et al (2002)	Asthma	UK	Identified as having asthma or angina EHR code in prior 12 months	Practices	**Asthma**: 31 offices **Angina**: 29 offices	**Angina:** 29 offices, **Asthma**: 31 offices	≥ 18 years no details presented	NS	NS
Gill et al. (2009)	hyperlipidaemia	US	Identified as having LDL levels recorded in EMR	Practices	12 offices 49 PCPs 26,696 patients	13 offices 56 PCPs 37,454 patients	Overall: 20-79 years **CG** younger	**CG:** more males (statistically significant)	**CG** lower disease risk

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Cobos et al. (2006)	hyperlipidaemia	Spain	Identified as having total cholesterol (TC) between 200mg/dL and 400 mg/dL	Practices	20 offices 1046 patients	22 offices 1145 patients	Overall: 18-94 years	**CG:** more females (statistically significant)	comparable
Hicks et al. (2008)	Hypertension	US	Identified as having at least 2 hypertension-related visits using EHR codes	Practices	7 offices 1048 patients	7 offices 786 patients	Comparable	Comparable >20 years	**IG:** significantly higher % of patients with controlled BP at first visit
Lopez et al. (2019)	Hypertension	US	Identified as having at least 1 ICD code for hypertension in EHR	Practices	7 offices	7 offices	18-85	NS	NS

[1] NS refers to not stated
[2] IG refers to intervention group
[3] CG refers to control group

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 4: Description of Outcomes

Study	Disease	Outcome	Outcome definition	Absolute Effects: Intervention	Absolute Effects: Usual Care	Relative Effects	Conclusions	Quality
Schnipper et al. (2010)	Diabetes	Odds of patients meeting management goals at the end compared to baseline	Targets undefined	Not reported		**ITT**: OR = 1.02; 95% CI, 0.99-1.04, P not reported **PP:** OR = 1.16; 95% CI, 1.08-1.25; P 0<.001	**ITT**: non- significant odds of intervention patients meeting management goals compared to control patients at 6 months. **PP**: significant odds of intervention patients meeting management goals compared to control patients after follow-up period.	⊕⊕◯◯ LOW
Gill et al. (2017)	Diabetes	• Difference in $A1_c$ from baseline • Difference in LDL-C from baseline % patients achieving $A1_c$ goals	**$HbA1_c$ goals**: <7% and <7.5% in patients with microvascular complications. **LDL-C goals:** < 100mg/dL	**Difference in HbA1c from baseline (%)**: −0.08% ±1.15% **Difference in LDL-C from baseline (mg/dL)**: −4.35± 24.54 mg/dL	**Difference in HbA1c from baseline (%)**: −0.14% ± 1.51% **Difference in LDL-C from baseline (mg/dL):** −1.70 ± 26.16 mg/dL	**Adjusted HbA1c between-group difference (%)**: 0.12%, 95% CI 0.02-0.22; P = 0.02 **Adjusted LDL-C between-group difference (mg/dL):** 3.57 mg/dL 95% CI 1.80-5.34; P < 0.0001 **Odds of achieving personalized HbA1c goals:** adjusted odds ratio [aOR] 1.52, 95% CI 1.24-1.86; P < 0.0001 **Odds of achieving HbA1c < 7%:** 1.56, 95% CI 1.27-1.91. P < 0.0001 **Odds of achieving HbA1c ≤ 9%:** 1.07, 95% CI 0.75-1.52; P = 0.71 **Odds of achieving LDL-C <**	Clinically small, yet statistically significant between-group reductions in HbA1c and LDL-C	⊕⊕⊕◯ MODERATE

For peer review only

						100mg/dL: 1.34; 95% CI 1.11-1.61. P = 0.002		
Heselmans et al. (2020)	Diabetes	• Mean change of HbA1c • Mean change in LDL-C levels • Mean change in BP	NA	Not reported		**Between-group difference in HbA1c means** -0.09%, 95% CI − 0.18;0.01, p-value = 0.06) **Between-group change difference in systolic BP:** 0.13% (95% CI − 0.91; 1.16) p=0.81 **Between-group change difference in diastolic BP**: 0.12% (95%CI − 1.25; 1.49) p=0.86 **Between-group change difference in** LDL-C: 1.76% (95%CI − 0.46; 3.98) p=0.12	No significant changes in between-group means for HbA1c, BP, or LDL-C	⊕⊕⊕◯ MODERATE
Eccles et al (2002)	Asthma and angina	**Patient reported outcomes using**: **generic measures**: SF36, EQ-5D **Condition specific measures**: The Seattle Angina Questionnaire, The Newcastle Asthma Symptoms Questionnaire		Not reported			No effect on any patient outcome	⊕⊕⊕◯ MODERATE
Gill et al. (2009)	Hyperlipidaemia	The percentage of patients whose most recent low-density lipoprotein. Cholesterol (LDL-C) was at goal (%)	**Defined by the ATP III guidelines** LDL goals: < 100 for high-risk < 130 for moderate risk <160 for low risk patients	**High risk:** Baseline: 50% End: 53.3% P-value: <0.001 **Moderate risk**: Baseline: 64.1% End: 64.7% P value: >0.1 **Low risk:** Baseline: 88.5% End: 87.9% P value <0.05	**High risk:** Baseline: 52.3% End: 56.1% P value: <0.001 **Moderate risk**: Baseline:67.1 % End: 68.5% P <0.001 **Low risk:** Baseline: 90.2% End: 90.9% P value <0.001	**High risk:** OR = 1.17 (95% CI not reported), p-value <0.05 **Moderate risk:** OR = 0.29 (95% CI not reported), p-value <0.05 **Low risk:** OR =1.74 (95% CI not reported), p-value <0.05	A significant increase in proportion of patients attaining goals for all categories except the moderate-risk intervention group. No significant difference in improvement in any category compared to the control group	⊕⊕◯◯ LOW

For peer review

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Cobos et al. (2006)	Hyperlipidaemia	Between- group differences in means of lipid values (mg/dL)	NA	**Total cholesterol:** 233.8 **LDL-C:** 149.2 **HDL-C:** 58.0 **Triglycerides:** 136.6	**Total cholesterol:** 231.0 **LDL-C:** 146.5 **HDL-C:** 56.3 **Triglycerides:** 135.2	**Difference, 95% CI, P- value** **Total cholesterol:** 2.8 (–1.7, –7.3**),** 0.218 **LDL-C:** 2.7 (–1.7, –7.1), 0.227 **HDL-C:** 1.6 (–0.6, –3.6), 0.142 **Triglycerides:** 1.4 (–8.3, –11.2), 0.766	No impact detected on final lipid values	⊕⊕⊕◯ MODERATE
Hicks et al. (2008)	Hypertension	**Absolute effects**: • Percentage of patients with controlled BP (%) • Mean baseline and end SBP and DBP (mmHg) **Relative effects**: • Odds of blood pressure control at outcome visit • Baseline and End difference in SBP and DBP (mmHg)	**Controlled BP:** <130/80mmHg for patients with diabetes or renal failure <140/90 mmHg for other patients	• 48% • Baseline SBP: 136 mmHg • End SBP: 138 mmHg • Baseline DBP: 78 mmHg • End DBP: 77 mmHg	• 45% • BaselineSBP:138mmHg • End SBP: 137mmHg • Baseline DBP: 80 mmHg • End DBP: 78 mmHg	**OR**: 0.96 Baseline SBP: 2 mmHg (p-value) 0.59 Baseline DBP: 2mmg Hg (p-value): 0.07 End SBP: 0 mm Hg (p-value): 0.67 End DBP: 1 mmHg (p-value): 0.05	No differences between intervention groups in mean systolic or diastolic blood pressure readings at (end) outcome visits	⊕⊕◯◯ LOW
Lopez et al. (2019)	Hypertension	**Relative effects:** odds of blood pressure control post-intervention	**Controlled BP:** <140/90 mmHg for all patients	Not reported		**OR**: 1.36 95% CI (1.08-1.71) (p-value: 0.009)	Significant odds of improvement of BP in the intervention group compared to usual care	⊕⊕◯◯ LOW

For peer review only

References

1) Viswanathan M, Golin CE, Jones CD, Ashok M, Blalock SJ, Wines RCM, et al. Interventions to improve adherence to self-administered medications for chronic diseases in the United States: a systematic review. *Annals of Internal Medicine.* 2012; 157 (11): 785-795. doi: 10.7326/0003-4819-157-11-201212040-00538

2) *Health and Economic Costs of Chronic Diseases | CDC.* Available from: https://www.cdc.gov/chronicdisease/about/costs/index.htm

3) Van Oostrom SH, Picavet HSJ, de Bruin SR, et al. Multimorbidity of chronic diseases and health care utilization in general practice. *BMC family practice.* 2014; 15 61. doi: 10.1186/1471-2296-15-61

4) Lehnert T, Heider D, Leicht H et al. Review: health care utilization and costs of elderly persons with multiple chronic conditions. *Medical care research and review: MCRR.* 2011; 68 (4): 387-420. doi: 10.1177/1077558711399580

5) *WHO | Innovative Care for Chronic Conditions: Building Blocks for Action.* Available from: http://www.who.int/chp/knowledge/publications/icccreport/en/

6) Francesconi P, Ballo P, Profili F, et al. Chronic Care Model for the Management of Patients with Heart Failure in Primary Care. *Health Services Insights.* 2019; 12 1178632919866200. Doi: 10.1177/1178632919866200

7) Tellefson M, Dipnarine K, Stopka C. The chronic care model and diabetes management in US primary care settings: a systematic review. *Preventing chronic disease.* 2013; 10 E26. Doi:10.5888/pcd10.120180

8) Oprea L, Cojocaru D, Sandu A, et al. The Chronic Care Model (CCM) and the Social Gradient in Health. *Revista de Cercetare si Interventie Sociala, ISSN-e 1584-5397, Vol 41, 2013, págs 176-189*. 2013

9) Murad MH. Clinical Practice Guidelines: A Primer on Development and Dissemination. *Mayo Clinic Proceedings.* 2017; 92 (3): 423-433. doi: 10.1016/j.mayocp.2017.01.001

10) Fürthauer J, Flamm M, Sönnichsen A. Patient and physician related factors of adherence to evidence based guidelines in diabetes mellitus type 2, cardiovascular disease and prevention: a cross sectional study. *BMC family practice.* 2013; 14 47. doi: 10.1186/1471-2296-14-47

11) Nichol MB, Knight TK, Priest JL, et al. Nonadherence to clinical practice guidelines and medications for multiple chronic conditions in a California Medicaid population. *Journal of*

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

the American Pharmacists Association: JAPhA. 2010; 50 (4): 496-507. doi: 10.1331/JAPhA.2010.09123

12) Wasylewicz ATM, Scheepers-Hoeks, Clinical Decision Support Systems. *Fundamentals of Clinical Data Science,* 2018. doi:10.1007/978-3-319-99713-1_11

13) Jao C. Efficient Decision Support Systems: Practice and Challenges in Biomedical Related Domain. BoD–Books on Demand; 2011 Sep 6; doi:10.5772/937

14) Haynes RB, Wilczynski NL. Effects of computerized clinical decision support systems on practitioner performance and patient outcomes: methods of a decision-maker-researcher partnership systematic review. *Implementation science: IS.* 2010; 5 12. doi: 10.1186/1748-5908-5-12

15) Reed TS, David P, Daniel CB, et al. An overview of clinical decision support systems: benefits, risks, and strategies for success. *NPJ Digital Medicine.* 2020; 3 17-17. doi: 10.1038/s41746-020-0221-y

16) Shortliffe EH, Shortliffe EH, Cimino JJ, et al. Biomedical informatics: computer applications in health care and biomedicine. *Springer;* 2014.

17) Kaushal JSA, Alison E, Sarah N, Diane H, et al. Effects of workload, work complexity, and repeated alerts on alert fatigue in a clinical decision support system. *BMC Medical Informatics and Decision Making*. 2017-04-10 2017;17(1):1-9. doi:10.1186/s12911-017-0430-8

18) Shahsavarani AM, Azad Marz Abadi E, Hakimi Kalkhoran M,et al. Clinical Decision Support Systems (CDSS): State of the art Review of Literature. *International Journal of Medical Reviews.* 2015; 2 (4): 299-308.

19) Bright TJ, Wong A, Dhurjati R, et al. Effect of clinical decision-support systems: a systematic review. *Annals of Internal Medicine.* 2012; 157 (1): 29-43. doi: 10.7326/0003-4819-157-1-201207030-00450

20) Souza NM, Sebaldt RJ, Mackay JA, et al. Computerized clinical decision support systems for primary preventive care: a decision-maker-researcher partnership systematic review of effects on process of care and patient outcomes. : *Centre for Reviews and Dissemination (UK)*; 2011. doi:10.1186/1748-5908-6-87

21) Garg AX, Adhikari NKJ, McDonald H, et al. Effects of computerized clinical decision support systems on practitioner performance and patient outcomes: a systematic review. *JAMA.* 2005; 293 (10): 1223-1238. doi: 10.1001/jama.293.10.1223

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

22) Lobach D, Sanders GD, Bright TJ, et al. *Methods.:* Agency for Healthcare Research and Quality (US); 2012.

23) Bryan C, Boren S. The use and effectiveness of electronic clinical decision support tools in the ambulatory/primary care setting: a systematic review of the literature. *Journal of Innovation in Health Informatics.* 2008; 16 (2): 79-91. doi: 10.14236/jhi.v16i2.679

24) Kwan JL, Lo L, Ferguson J, et al. Computerised clinical decision support systems and absolute improvements in care: meta-analysis of controlled clinical trials. bmj. 2020 Sep 17;370. doi:10.1136/bmj.m3216

25) Moher D, Liberati A, Tetzlaff J, et al. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. *PLOS Medicine.* 2009; 6 (7): e1000097. doi: 10.1371/journal.pmed.1000097

26) Moher AL, Douglas GA, Jennifer T, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. 2009-07-21 2009;doi:10.1136/bmj.b2700

27) Howard C. *Subject and Course Guides: Evidence Based Medicine: PICO.* Available from: https://researchguides.uic.edu/c.php?g=252338&p=3954402

28) Zwar RR, Sarah D, Iqbal H, et al. A systematic review of chronic disease management interventions in primary care. OriginalPaper. *BMC Family Practice*. 2018-01-09 2018;19(1):1-13. doi:doi:10.1186/s12875-017-0692-3

29) Jansma EP, Vries RD. Patient reported outcome measures (PROMs*)*. bmi-online search blocks. Available from: https://blocks.bmi-online.nl/catalog/248.

30) Data extraction forms. https://dplp.cochrane.org/data-extraction-forms

31) Lobach D, Sanders GD, Bright TJ, et al. *Data Abstraction Guidance.* : Agency for Healthcare Research and Quality (US); 2012.

32) Zhao KF, Kelsey Y, Matthew CT,et al. Risk of bias tools in systematic reviews of health interventions: an analysis of PROSPERO-registered protocols. *Systematic Reviews*. 2019-11-15 2019;8(1):1-9. doi:doi:10.1186/s13643-019-1172-8

33) Higgins JP, Altman DG, Gøtzsche PC,et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *Bmj.* 2011; d5928 (343): doi:10.1136/bmj.d5928

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

34) Higgins JP, Savović J, Page MJ, Elbers RG, Sterne JA. Assessing risk of bias in a randomized trial. Cochrane Handbook for Systematic Reviews of Interventions. 2019 Sep 23:205-28. doi:10.1136/bmj.d5928

35) Ryan R, Hill S, Prictor M,et al. *Cochrane Consumers & Communication Review Group Study Quality Guide, Guide for Review Authors on Assessing Study Quality.* 2013.

36) *What is GRADE? | BMJ Best Practice.* Available from: https://bestpractice.bmj.com/info/toolkit/learn-ebm/what-is-grade/

37) Meader N, Kristel K, Llewellyn A, et al. A checklist designed to aid consistency and reproducibility of GRADE assessments: development and pilot validation. *Systematic reviews*. 07/24/2014 2014;3doi:10.1186/2046-4053-3-82

38) Schnipper JL, Linder JA, Palchuk MB, et al. Effects of documentation-based decision support on chronic disease management. *The American Journal of Managed Care.* 2010; 16 (12 Suppl HIT) (pp SP72-81): ate of Pubaton: e 2010.

39) Gill J, Kucharski K, Turk B, Pan C, Wei W. Using Electronic Clinical Decision Support in Patient-Centered Medical Homes to Improve Management of Diabetes in Primary Care: The DECIDE Study. *Journal of Ambulatory Care Management.* 2019; 42 (2): 105-115. doi:10.1097/JAC.0000000000000267

40) Heselmans A, Delvaux N, Laenen A, Van de Velde S, Ramaekers D, Kunnamo I, et al. Computerized clinical decision support system for diabetes in primary care does not improve quality of care: a cluster-randomized controlled trial. *Implementation science : IS.* 2020; 15 (1): 5. doi:10.1186/s13012-019-0955-6

41) Eccles M, McColl E, Steen N, Liaw ST. Computerised evidence-based guidelines may not improve asthma or angina management in primary care. *Evidence-Based Healthcare.* 2003; 7 (2): 81-82.

42) Gill JM, Chen YX, Glutting JJ, et al. Impact of decision support in electronic medical records on lipid management in primary care. *Population health management.* 2009; 12 (5): 221-226. doi:10.1089/pop.2009.0003. doi:10.1089/pop.2009.0003

43) Cobos A, Vilaseca J, Asenjo C, et al. Cost effectiveness of a clinical decision support system based on the recommendations of the European Society of Cardiology and other societies for the management of hypercholesterolemia: Report of a cluster-randomized trial. *Disease Management and Health Outcomes.* 2005; 13 (6): 421-432. doi:10.2165/00115677-200513060-00007

For peer review only

44) Hicks LS, Sequist TD, Ayanian JZ, et al. Impact of computerized decision support on blood pressure management and control: A randomized controlled trial. *Journal of General Internal Medicine.* 2008; 23 (4): 429-441. doi:10.1007/s11606-007-0403-1

45) Lopez PM, Divney A, Goldfeld K, et al. Feasibility and Outcomes of an Electronic Health Record Intervention to Improve Hypertension Management in Immigrant-serving Primary Care Practices. *Medical care.* 2019; 57 (Supplement 62) (pp S164-S171): ate of Pubaton: 01 Jun 2019. doi:10.1097/MLR.0000000000000994

46) Goud R, de Keizer NF, ter Riet G, et al. Effect of guideline based computerised decision support on decision making of multidisciplinary teams: cluster randomised trial in cardiac rehabilitation. *The BMJ.* 2009; 338. doi: 10.1136/bmj. b1440

47) Manns BJ, Tonelli M, Zhang J, et al. Enrolment in primary care networks: impact on outcomes and processes of care for patients with diabetes. *CMAJ: Canadian Medical Association journal = journal de l'Association medicale canadienne.* 2012; 184 (2): 144.: doi: 10.1503/cmaj.110755

48) McClellan SR, Casalino LP, Shortell SM, et al. When does adoption of health information technology by physician practices lead to use by physicians within the practice? *Journal of the American Medical Informatics Association: JAMIA.* 2013; 20 (e1): e26-e32. doi: 10.1136/amiajnl-2012-001271

49) Latoszek-Berendsen A, Tange H, van den Herik, H. J.,et al. From clinical practice guidelines to computer-interpretable guidelines. A literature overview. *Methods of Information in Medicine.* 2010; 49 (6): 550-570. doi: 10.3414/ME10-01-0056

50) Hajjaj FM, Salek MS, Basra MK,et al. Non-clinical influences on clinical decision-making: a major challenge to evidence-based practice. Journal of the Royal Society of Medicine. 2010 May 1;103(5):178-87. doi:10.1258/jrsm.2010.100104

51) Varghese J, Kleine M, Gessner SI,et al. Effects of computerized decision support system implementations on patient outcomes in inpatient care: a systematic review. *Journal of the American Medical Informatics Association: JAMIA.* 2018; 25 (5): 593-602. doi: 10.1093/jamia/ocx100

52) Moja L, Kwag KH, Lytras T, Bertizzolo L, Brandt L, Pecoraro V, et al. Effectiveness of Computerized Decision Support Systems Linked to Electronic Health Records: A Systematic Review and Meta-Analysis. *American Journal of Public Health.* 2014; 104 (12): e12-e22. Available from: doi: 10.2105/AJPH.2014.302164

53) Groenhof TKJ, Asselbergs FW, Groenwold RHH,et al. The effect of computerized decision support systems on cardiovascular risk factors: a systematic review and meta-analysis. *BMC*

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

medical informatics and decision making. 2019; 19 (1): 108. doi: 10.1186/s12911-019-0824-x

54) Roshanov PS, Misra S, Gerstein HC, et al. Computerized clinical decision support systems for chronic disease management: A decision-maker-researcher partnership systematic review. *Implementation science : IS.* 2011; 6 (1): 92. doi: 10.1186/1748-5908-6-92

55) Kooij L, Groen WG, van Harten WH. The Effectiveness of Information Technology-Supported Shared Care for Patients With Chronic Disease: A Systematic Review. *Journal of Medical Internet Research.* 2017; 19 (6).doi: 10.2196/jmir.7405

56) Castaneda C, Nalley K, Mannion C, et al. Clinical decision support systems for improving diagnostic accuracy and achieving precision medicine. *Journal of Clinical Bioinformatics.* 2015; 5 .doi: 10.1186/s13336-015-0019-3

57) Gibbons RJ, Chatterjee K, Daley J,et al. ACC/AHA/ACP-ASIM guidelines for the management of patients with chronic stable angina: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines (Committee on Management of Patients With Chronic Stable Angina). *Journal of the American College of Cardiology*. 1999 Jun;33(7):2092-197. doi:10.1016/s0735-1097(99)00150-3

58) Heneghan C, Goldacre B, Mahtani KR. Why clinical trial outcomes fail to translate into benefits for patients. *Trials.* 2017 Dec;18(1):1-7. doi:10.1186/s13063-017-1870-2

59) Afzal M, Hussain M, Ali T, et al Knowledge-based query construction using the CDSS knowledge base for efficient evidence retrieval. *Sensors*. 2015 Sep;15(9):21294-314. doi:10.3390/s150921294

For peer review only

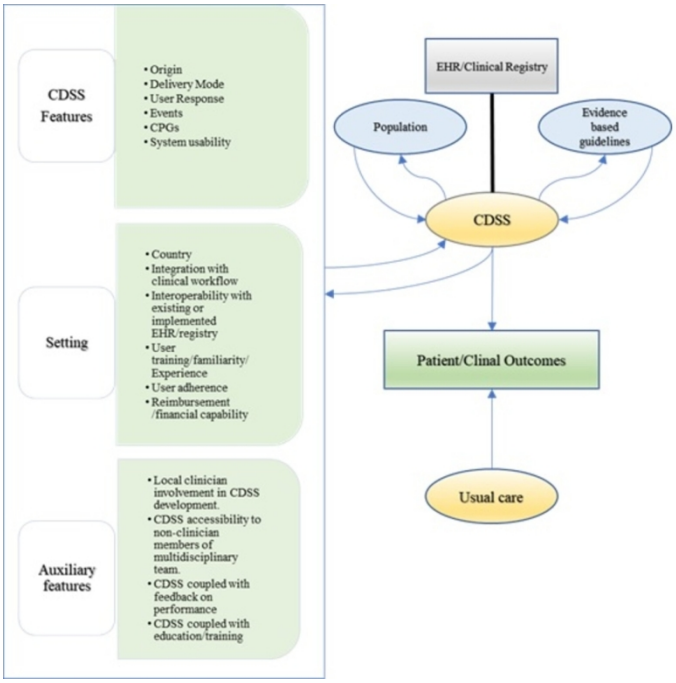


Figure 1: Analytic Framework

133x90mm (300 x 300 DPI)

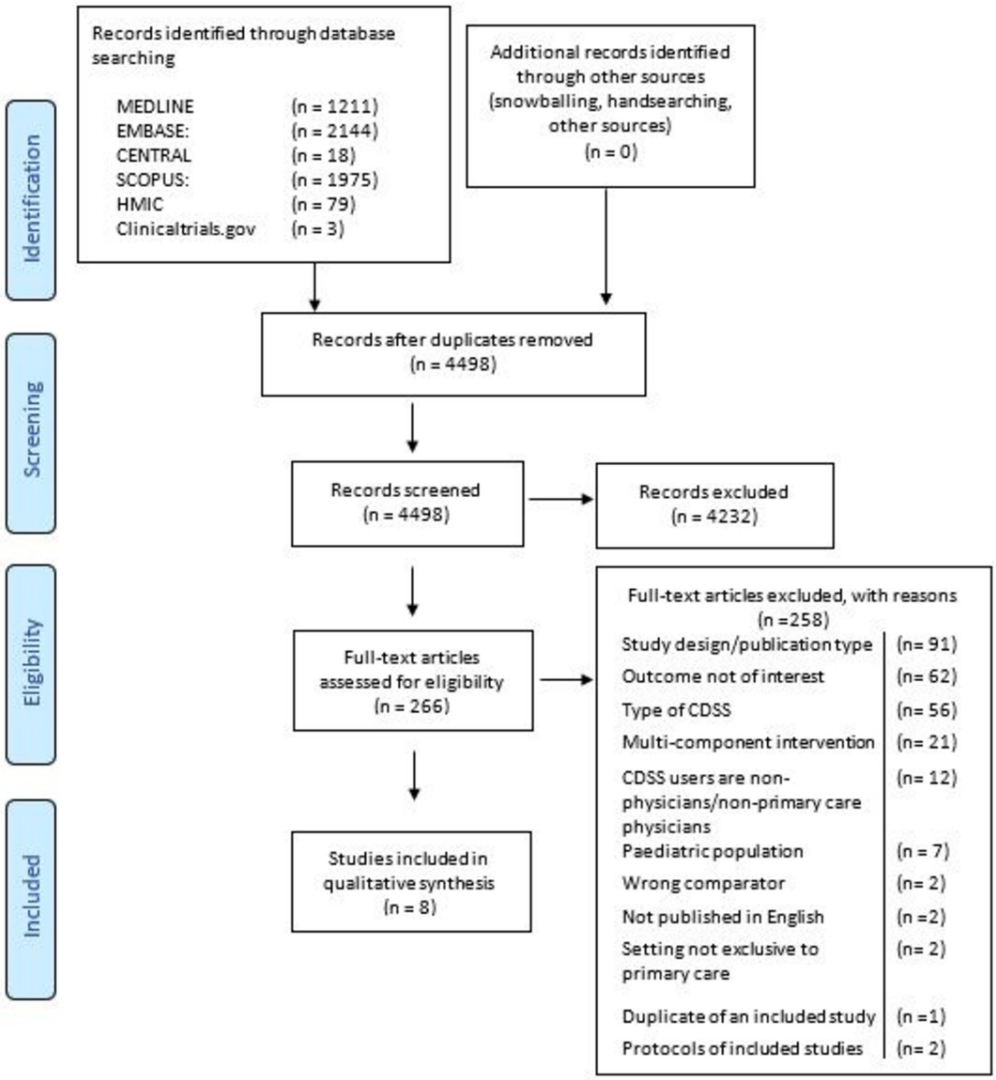


Figure 2: PRISMA flowchart detailing the systematic search results

90x97mm (300 x 300 DPI)

Supplementary Material

Supplementary Table 1: Eligibility criteria defined by PICOS

P	**Participants:** - Patients above 18 years with one or more concurrent chronic condition, regardless of sex, education, marital status, employment status, income, or other socioeconomic factors who were managed in a primary care setting using a EHR or clinical registry) supported by a computerised, knowledge-based CDSS. - Chronic conditions were defined and limited to the following conditions: o Asthma, chronic obstructive pulmonary disease (COPD), heart failure, myocardial ischemia, hypertension, type II diabetes mellitus, hyperlipidaemia, arthritis, and osteoporosis.
I	**Interventions:** Computerised knowledge based CDSS embedded EHRs or clinical registries in primary care settings as an intervention with the following inclusion and exclusion criteria: **Studies will be included if:** - They focus on evaluating a CDSS for medical management of the selected chronic diseases - The CDSS is of knowledge-based type (informed by CPGs). - Included CDSS is synchronous, meaning alerts are electronically presented to the clinician in real-time during the physician-patient encounter. - The CDSS users are primary care physicians (PCPs). - They report at least: one quantifiable patient clinical outcome/patient-related measure in adult populations in the specified chronic disease entities **Studies will be excluded if**: - Not published in English. - Not conducted in primary care settings. - Include care provided by subspecialized physicians (not PCPs). - Evaluates a CDSS for other outcome measures such as system usability, or other quality outcomes, not including patient or clinical outcomes. - Interventions that are used for other direct care purposes other than chronic disease management or non-direct care purposes, such as research purposes. - Evaluate the effect of a CDSS in other diseases not specified in the inclusion criteria. - Include a CDSS that does not produce real-time recommendations based on CPGs (non-knowledge based CDSS). - Web-based CDSS, paper-based CDSS, standalone CDSS (not integrated with EHRs), CDSS integrated with EHR but not producing electronic alerts, and other CDSS that use e-mail, phone, or pagers to notify the user. - Studies without a comparator group - Comparator group is another type of CDSS.
C	**Comparators:** Usual care not utilising CDSS
O	**Outcomes:** - Patient-related health and clinical outcomes in chronic disease management (such as HBA1C, $FEV_{1,}$ or other biomarkers of chronic diseases), morbidity, mortality, HRQoL, preventable co-morbidities, or disease activity) - Practitioner-related, patient-related, and cost-effectiveness outcomes were not included.
S	**Study types**: Randomised controlled trials (RCTs), cluster RCTs, quasi-RCTs, interrupted time series (ITS) and controlled before-and-after (CBA) studies

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Table 2: List of search terms

CDSS terms	Dashboard*, registry, clinical decision support system*, medical order entry system*, computeri?ed physician order entry system*, CPOE, order set*, computer assisted decision making, (computer$ adj3 decision$), knowledge management system*, case-finding algorithm*, advisory decision support, (computer* adj2 therap*)

AND

General Practice/Primary care	general practi*, family practice, family clinic*, family practi*, family physician*, family doctor*, primary care physician*, primary care clinic*, primary care practi*,

OR each of the below individual entities combined with AND

Chronic Disease Entities	
Asthma	asthma*, antiasthma*, wheez*, (bronch* adj3 spasm*), bronchoconstrict*, (bronch$ adj3 constrict$), ((bronchial$ or respiratory or airway$ or lung$) adj3 (hypersensitiv$ or hyperreactiv$ or allerg$ or insufficiency)),
Heart Failure/ Myocardial Ischemia	(heart* adj2 decomp*), ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*)), ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*))
Hypertension	((elevat$ or increas$ or lower or high or rais$ or rising) adj2 (arterial pressure or blood pressure or diastolic pressure or systolic pressure)), hypertens$ or prehypertens$),
Diabetes Mellitus	(IDDM or T2DM or T2D), (insulin* depend* or insulin?depend*)., (non insulin* depend* or noninsulin* depend* or non insulin?depend* or noninsulin?depend*)., diabet*.
Hyperlipidemia	LDL, (dyslipid?emia* or lipoprotein* or dyslipidprotein?emia* or hyperlipoprotein?emia), (hyperlipid?emia* or cholesterol* or lip?emia* or hypercholesterol?emia* or hypercholester?emia* or hyperlip?emia* or triglycerid* or hypertriglycerid?emia*).,
COPD	(COPD or COAD or COBD or AECB), ((obstruct$ adj3 lung$) or (obstruct$ adj 3 airway$)), (obstruct$ adj3 pulmonary), (chronic* adj3 bronchiti*), emphysema*.,
Arthritis	Osteoarthritis, undifferentiated oligoarthritis, ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$))., (joint pain or chronic joint symptoms or gonarthrosis or osteoarthrosis or astoarthrosis or degeneretive arthritis or joint diseases or arthralgia),
Osteoporosis	Densitometry, osteopenia, bone mass, bone densit*, osteoporos?s, (osteoporosis or osteopenia or (bone and (density or mass or loss)) or hepatic osteodystrophy),

For peer review only

Supplementary Table 3 - Full Medline Search Strategy

1 Dashboard*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1228)
2 registry.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (109312)
3 registries.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (104552)
4 exp Registries/ (95452)
5 1 or 2 or 3 or 4 (173943)
6 exp Decision Support Systems, Clinical/ (7925)
7 clinical decision support system*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (2132)
8 medical order entry system*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (2297)
9 computeri?ed physician order entry system*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (224)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

10 CPOE.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1113)

11 order set*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1009)

12 exp Decision Making, Computer-Assisted/ (144386)

13 computer assisted decision making.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (76)

14 (computer$ adj3 decision$).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (4723)

15 knowledge management system*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (109)

16 case-finding algorithm*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (84)

17 advisory decision support.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1)

For peer review only

18 exp Therapy, Computer-Assisted/ (63908)

19 (computer* adj2 therap*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (8758)

20 exp Reminder Systems/ (3443)

21 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 (159108)

22 5 or 21 (331941)

23 exp Asthma/ (127501)

24 asthma*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (179286)

25 antiasthma*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1458)

26 exp Respiratory Sounds/ (12994)

27 wheez*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (13724)

28 exp Bronchial Spasm/ (4303)

29 (bronch* adj3 spasm*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (4523)

30 exp Bronchoconstriction/ (4131)

For peer review only

31 bronchoconstrict*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (11491)

32 (bronch$ adj3 constrict$).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (715)

33 exp Bronchial Hyperreactivity/ (7348)

34 exp Respiratory Hypersensitivity/ (154474)

35 ((bronchial$ or respiratory or airway$ or lung$) adj3 (hypersensitiv$ or hyperreactiv$ or allerg$ or insufficiency)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (66254)

36 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 (261794)

37 exp Heart Failure/ (120835)

38 (heart* adj2 decomp*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (4261)

39 ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (664015)

40 ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept

For peer review only

word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (664015)

41 exp Myocardial Ischemia/ (428216)

42 exp Coronary Disease/ (216812)

43 37 or 38 or 39 or 40 or 41 or 42 (842183)

44 exp Hypertension/ (253393)

45 exp Essential Hypertension/ (2255)

46 (hypertens$ or prehypertens$).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (510526)

47 ((elevat$ or increas$ or lower or high or rais$ or rising) adj2 (arterial pressure or blood pressure or diastolic pressure or systolic pressure)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (63244)

48 44 or 45 or 46 or 47 (537839)

49 exp Diabetes Mellitus/ (423542)

50 diabet*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (693674)

51 (non insulin* depend* or noninsulin* depend* or non insulin?depend* or noninsulin?depend*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (12372)

52 (insulin* depend* or insulin?depend*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word,

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (29531)

53 (IDDM or T2DM or T2D).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (37329)

54 exp Diabetes Mellitus, Type 2/ (132230)

55 49 or 50 or 51 or 52 or 53 or 54 (697363)

56 exp Hyperlipidemias/ (65918)

57 exp Dyslipidemias/ (79414)

58 exp Cholesterol/ (159670)

59 (hyperlipid?emia* or cholesterol* or lip?emia* or hypercholesterol?emia* or hypercholester?emia* or hyperlip?emia* or triglycerid* or hypertriglycerid?emia*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (389543)

60 (dyslipid?emia* or lipoprotein* or dyslipidprotein?emia* or hyperlipoprotein?emia*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (209412)

61 LDL.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (101128)

62 56 or 57 or 58 or 59 or 60 or 61 (479855)

63 exp Pulmonary Disease, Chronic Obstructive/ (55377)

64 exp Lung Diseases, Obstructive/ (210481)

For peer review only

65 emphysema*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (35893)

66 (chronic* adj3 bronchiti*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (11493)

67 (obstruct$ adj3 pulmonary).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (67392)

68 ((obstruct$ adj3 lung$) or (obstruct$ adj 3 airway$)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (24368)

69 (COPD or COAD or COBD or AECB).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (46979)

70 63 or 64 or 65 or 66 or 67 or 68 or 69 (256014)

71 exp Arthritis/ (258204)

72 exp Osteoarthritis/ (62805)

73 osteoarthritis.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (88630)

74 exp Joint Diseases/ (378029)

75 exp Arthralgia/ (13069)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

76 (joint pain or chronic joint symptoms or gonarthrosis or osteoarthrosis or astoarthrosis or degeneretive arthritis or joint diseases or arthralgia).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (49768)

77 ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (176792)

78 undifferentiated oligoarthritis.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (36)

79 71 or 72 or 73 or 74 or 75 or 76 or 77 or 78 (466351)

80 exp Osteoporosis/ (55499)

81 (osteoporosis or osteopenia or (bone and (density or mass or loss)) or hepatic osteodystrophy).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (229156)

82 osteoporos?s.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (86861)

83 exp Bone Density/ (52984)

84 bone densit*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary

For peer review only

concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (70284)

85 exp Bone Demineralization, Pathologic/ (652)

86 exp Bone Diseases, Metabolic/ (76180)

87 bone mass.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (18851)

88 osteopenia.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (9555)

89 exp Densitometry/ (34775)

90 densitometry.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (20649)

91 80 or 81 or 82 or 83 or 84 or 85 or 86 or 87 or 88 or 89 or 90 (266936)

92 exp General Practitioners/ (7841)

93 exp General Practice/ (74873)

94 general practi*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (91362)

95 exp Physicians, Family/ (16348)

96 exp Family Practice/ (65161)

97 family practice.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (68324)

98 family clinic*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (470)

99 family physician*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (14677)

100 family practi*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (69798)

101 family doctor*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (4649)

102 primary care physician*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (19694)

103 primary care clinic*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (7532)

104 exp Physicians, Primary Care/ (3421)

105 primary care practi*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism

For peer review only

supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (7576)

106 92 or 93 or 94 or 95 or 96 or 97 or 98 or 99 or 100 or 101 or 102 or 103 or 104 or 105 (180127)

107 22 and 106 (4326)

108 91 and 107 (50)

109 79 and 107 (76)

110 70 and 107 (184)

111 62 and 107 (206)

112 55 and 107 (568)

113 48 and 107 (239)

114 43 and 107 (343)

115 36 and 107 (148)

116 108 or 109 or 110 or 111 or 112 or 113 or 114 or 115 (1211)

117 from 116 keep 1-1211 (1211)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Table 3: Risk of bias in included studies as per the Cochrane tool

Trial	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data	Selective outcome reporting	Other bias	Selective recruitment of cluster members
Shnipper et al.(2010)	✓	✗	!	!	!	✓	✓	NA
Gill et al. (2017)	✗	✓	✓	✓	✗	✗	✓	✓
Hesselman et al.(2020)	✓	✓	✓	✓	✗	✓	✓	✓
Eccles et al. (2002)	✓	✓	✓	✓	✗	✗	✓	✓
Gill et al. (2009)	✗	✓	!	!	✗	✓	✓	✓
Cobos et al. (2006)	✓	✓	!	!	✓	✓	✓	✓
Hicks et al. (2008)	✗	✓	!	!	✗	✓	✓	✓
Lopez et al. (2019)	✗	✗	✗	✗	✓	✓	✓	✓

Key
✓ Low risk
✗ High risk
! Unclear

For peer review only

Supplementary Table 4: Adapted risk of bias table detailing methodological risks and characteristics. PCPs refers to primary care practitioners (clinicians)

	Randomization				Baseline Comparability				Blinding			Withdrawal		
Trial	Truly Random	Allocation Concealment	Number Stated		Presented	Achieved	Inlcusion/Exclusion Criteria Specified		Assessors	Patients	PCPs	>80% in final analysis	Reason stated	Intention to treat
			Patients	PCPs			Patients	PCPs						
Shnipper et al.(2010)	✓	✗	✓	✗	✓	✓	✓	✓	NS	NS	NA	✗	NS	✓
Gill et al. (2017)	✗	✗	✓	✓	✓	✗	✓	✓	✓	✗	NA	✗	✓	✗
Hesselman et al.(2020)	✓	✓	✓	✓	✗	✓	✓	✓	✓	✓	NA	✗	✓	✓
Eccles et al. (2002)	✓	✓	✓	✓	!	!	✓	✓	✓	✓	NA	✓	✓	✓
Gill et al. (2009)	!	✗	✓	✓	✓	✗	✓	✓	✗	✗	NA	!	✓	✗
Cobos et al. (2006)	✓	✓	✓	✓	✓	✓	✓	✓	NS	NS	NA	✗	✓	✓
Hicks et al. (2008)	!	✗	!	✓	✗	✓	✓	✓	NS	NS	NA	✗	✓	✓
Lopez et al. (2019)	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗	NA	!	NA	!

Key	
✓	Adequately addressed
!	Partially addressed
✗	Not adequately addressed
NS	Not stated
NA	Not applicable

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Risk of Bias Assessment: Type 2 Diabetes Mellitus

Referring to **Supplementary Tables 3 and 4**, Schnipper et al was rated at 'low risk' for random sequence generation as a random number generation on Excel was used [1].Allocation concealment was rated as 'high' after considering the randomisation an open allocation method. Blinding of personnel was not cited. Thus, it was rated as unclear. There was no mention of withdrawals; hence, the risk was unclear for incomplete outcome reporting. All outcomes set be studied were reported, and no other risks related to study funding were detected.

Gill et al's study was rated at 'high risk' for incomplete and selective outcome reporting due to analysing subgroups of patients for A1C and LDL-C for the narrated study limitations in the following section [2].Randomisation was at 'high risk' since clusters of practices were allocated based on the number of clinicians. Although the randomisation method could be considered open to predictability, eligibility criteria for patients within clusters of practices were pre-defined, which could not have been subverted by allocation knowledge. All other elements were considered at 'low risk.'

Heselmans et al's study was at 'low risk' for all elements except incomplete outcome data due to the withdrawal of a significant number of practices, with less than 80% of practices remaining after randomisation for analysis[3]. Reasons for system concerns were attributed to withdrawals.

3.2.2 Risk of Bias Assessment: Asthma and Angina

Referring to **Supplementary Tables 3 and 4**, the risk of bias in Eccles et al. 's study was 'low' for all elements due to the use of a computerised randomisation process and blinding of assessors[4]. However, less than 80% of eligible patients completed the provided questionnaires, which could have introduced attrition bias. Baseline characteristics of patients and practitioners were not sufficiently addressed, but there was a mention that participating practices were of similar size and training. Selective recruitment of clusters was adequately prevented by recruiting practices and enrolling all patients who met the inclusion criteria.

Risk of Bias Assessment: Hyperlipidaemia

Referring to Supplementary T**ables 3 and 4**, Cobos et al. 's study was rated at 'low-risk' for random sequence generation and allocation concealment due to the use of a random number table to randomise, which does not allow predictable sequences[5]. There was no mention of blinding of assessors, statisticians, or patients. Although <80% of randomised patients had follow-up visits and LDL-C follow-up tests, the study was rated as 'low risk' for attrition and reporting bias because the ITT analysis yielded insignificant outcomes as did the PP analyses. Study funding did not seem to introduce methodological bias since the outcome was patient-reported. Selective recruitment of clusters was adequately prevented by recruiting practices and enrolling all patients of randomised practices who met the inclusion criteria.

Gill et al 's trial was rated at 'high risk' for random sequence generation but 'low risk' for allocation concealment due to randomising clusters of practices based on practice similarities, which is not preferable randomisation[6]. The Eligibility criteria for patients within clusters of practices were pre-defined, which could not have been subverted by allocation knowledge. Blinding of assessors and

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

statisticians was not mentioned. The number of patients analysed was not reported. However, all outcomes set to be studied were undertaken. The first author is a consultant researcher at the company that holds the concerned EHR's licensing rights, but this conflict of interest did not seem to reflect in methodological risks. Similarly, selective recruitment of clusters was prevented.

Risk of Bias Assessment: Hypertension

Referring to **Supplementary Tables 3 and 4,** Hick's et al 's study was rated 'high risk' for random sequence generation, as clusters of practices were randomised by size and type of practice, which may not be a desirable method of randomisation [7]. Although randomisation could have been open to predictability, patient eligibility criteria practices were pre-defined, which could not have been subverted by allocation knowledge. Thus, the study was rated at 'low risk' for allocation concealment. Blinding of assessors or patients was not mentioned. Thus, the study was rated 'unclear' for blinding participants and outcome assessments. Due to the high number of randomised patients that were withdrawn due to stated reasons, it was rated at 'high risk' for attrition bias. All outcomes were explored, and there was a 'low risk' for selective recruitment of cluster members.

Lopez et al. 's study was rated at 'high risk' for random sequence generation and allocation concealment due to its non-randomised design[8]. There was no mention of blinding personnel, hence it was rated 'unclear' for blinding. There was no mention of withdrawals, and all outcomes were reported. Thus, it was rated as 'low risk' for attrition and selective outcome reporting. No other risks related to funding were significant.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Table 5: Quality of Evidence from individual studies evaluated using GRADE

Study	Outcome	Risk of bias	Level of inconsist-ency	Level of indirect-ness	Level of imprecise-on	Publication bias	Quality	Reason for downgrading
Schnipper et al. (2010)	Difference in proportion of patients meeting management goals at the end compared to the beginning of the study	severe	NA	some	low	NA	⊕⊕◯◯ LOW	**RoB:** High risk for allocation concealment and unclear whether personnel blinding occurred **Indirectness:** population identification and management goals were not pre-defined
Gill et al. (2017)	• Difference in HbA1c from baseline • Difference in LDL-C from baseline • % patients achieving HbA1c goals	some	NA	low	low	NA	⊕⊕⊕◯ MODERATE	**RoB:** High risk for random sequence generation and selective outcome reporting
Heselmans et al. (2020)	• Difference between the mean value of HbA1c, LDL-C, and blood pressure of the previous year at the moment of the study start and the mean value of HbA1c, LDL-C, and blood pressure of the previous year at 6 and 12 months.	low	NA	some	low	NA	⊕⊕⊕◯ MODERATE	**Indirectness:** patients had reasonable HbA1c mean values, which does not necessarily reflect diabetic populations with a higher severity of illness
Eccles et al. (2002)	Quality of life and condition specific outcomes using validated questionnaires	some	NA	low	low	NA	⊕⊕⊕◯ MODERATE	**RoB:** High risk for incomplete and selective outcome reporting
Gill et al. (2009)	The percentage of patients with most recent LDL-C at goal	severe	NA	low	low	NA	⊕⊕◯◯ LOW	**RoB:** High risk for random sequence generation and incomplete outcome data. Unclear for personnel blinding
Cobos et al. (2006)	• Between-group differences in means of lipid values	some	NA	some	low	NA	⊕⊕⊕◯ MODERATE	**ROB**: Unclear for blinding of personnel **Indirectness:** excludes general populations with more severe elevations of LDL
Hicks et al. (2008)	• Percentage of patients with controlled blood pressure • Odds of blood pressure control at outcome visit	severe	NA	low	low	NA	⊕⊕◯◯ LOW	**RoB:** High risk for random sequence allocation, incomplete outcome reporting and unclear for personnel blinding
Lopez et al. (2019)	• Odds of blood pressure control post-intervention	severe	NA	low	low	NA	⊕⊕◯◯ LOW	**RoB:** NRCT, non-randomised and high risk for blinding of personnel

For peer review only

References

1) Schnipper JL, Linder JA, Palchuk MB, et al. Effects of documentation-based decision support on chronic disease management. *The American Journal of Managed Care.* 2010; 16 (12 Suppl HIT) (pp SP72-81): ate of Pubaton: e 2010.

2) Gill J, Kucharski K, Turk B, Pan C, Wei W. Using Electronic Clinical Decision Support in Patient-Centered Medical Homes to Improve Management of Diabetes in Primary Care: The DECIDE Study. *Journal of Ambulatory Care Management.* 2019; 42 (2): 105-115. doi:10.1097/JAC.0000000000000267

3) Heselmans A, Delvaux N, Laenen A, Van de Velde S, Ramaekers D, Kunnamo I, et al. Computerized clinical decision support system for diabetes in primary care does not improve quality of care: a cluster-randomized controlled trial. *Implementation science : IS.* 2020; 15 (1): 5. doi:10.1186/s13012-019-0955-6

4) Eccles M, McColl E, Steen N, Liaw ST. Computerised evidence-based guidelines may not improve asthma or angina management in primary care. *Evidence-Based Healthcare.* 2003; 7 (2): 81-82.

5) Cobos A, Vilaseca J, Asenjo C, et al. Cost effectiveness of a clinical decision support system based on the recommendations of the European Society of Cardiology and other societies for the management of hypercholesterolemia: Report of a cluster-randomized trial. *Disease Management and Health Outcomes.* 2005; 13 (6): 421-432. doi:10.2165/00115677-200513060-00007

6) Gill JM, Chen YX, Glutting JJ, et al. Impact of decision support in electronic medical records on lipid management in primary care. *Population health management.* 2009; 12 (5): 221-226. doi:10.1089/pop.2009.0003. doi:10.1089/pop.2009.0003

7) Hicks LS, Sequist TD, Ayanian JZ, et al. Impact of computerized decision support on blood pressure management and control: A randomized controlled trial. *Journal of General Internal Medicine.* 2008; 23 (4): 429-441. doi:10.1007/s11606-007-0403-1

8) Lopez PM, Divney A, Goldfeld K, et al. Feasibility and Outcomes of an Electronic Health Record Intervention to Improve Hypertension Management in Immigrant-serving Primary Care Practices. *Medical care.* 2019; 57 (Supplement 62) (pp S164-S171): ate of Pubaton: 01 Jun 2019. doi:10.1097/MLR.0000000000000994

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 3/4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 3/4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 4
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 5, supplementary material table 3
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 5
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 5
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 5/6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 5/6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 6
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 6
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g., tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 6
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 6
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 6
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 6
Reporting bias	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 6

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 6
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 6/7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 6/7
Study characteristics	17	Cite each included study and present its characteristics.	Page 6/7
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 7 & supplementary section
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Page 7/8/9
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page 7 & supplementary section
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 7/8/9
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 7 & supplementary section
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page 7
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 10/11/12
	23b	Discuss any limitations of the evidence included in the review.	Page10/11/12
	23c	Discuss any limitations of the review processes used.	Page10/11/12
	23d	Discuss implications of the results for practice, policy, and future research.	Page10/11/12
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 2
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 2
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Page 2
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 14
Competing	26	Declare any competing interests of review authors.	Page 14

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
interests			
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 14

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

For peer review only

BMJ Open

Effect of computerised, knowledge-based, clinical decision support systems on patient-reported and clinical outcomes of patients with chronic disease managed in primary care settings

A Systematic Review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-054659.R1
Article Type:	Original research
Date Submitted by the Author:	21-Oct-2021
Complete List of Authors:	El Asmar, Marie line; Imperial College London, Self Care Academic Research Unit (SCARU) Dharmayat, Kanika ; Imperial College London, Imperial Centre for Cardiovascular Disease Prevention (ICCP), Department of Primary Care and Public Health Vallejo-Vaz, Antonio J.; Imperial College London, Imperial Centre for Cardiovascular Disease Prevention (ICCP), Department of Primary Care and Public Health, School of Public Health Irwin, Ryan; University of Birmingham, Institute of Applied Health Research Mastellos, Nikolaos; Imperial College London, Department of Primary Care and Public Health, School of Public Health
Primary Subject Heading:	General practice / Family practice
Secondary Subject Heading:	General practice / Family practice, Health informatics, Evidence based practice
Keywords:	GENERAL MEDICINE (see Internal Medicine), PRIMARY CARE, PREVENTIVE MEDICINE, Health informatics < BIOTECHNOLOGY & BIOINFORMATICS

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Effect of computerised, knowledge-based, clinical decision support systems on patient-reported and clinical outcomes of patients with chronic disease managed in primary care settings

A Systematic Review

Marie Line El Asmar,[1] MD, MPH, Kanika I Dharmayat,[2] MPH, BSc (hons), Antonio J. Vallejo-Vaz,[2,3,4] MD, PHD, Ryan Irwin,[5,*] PhD MPA MSc BSc (hons), Nikolaos Mastellos[6,*] PhD, MSc, BSc (hons)

1- Self Care Academic Research Unit (SCARU), Imperial College London. London, United Kingdom.
2- Imperial Centre for Cardiovascular Disease Prevention (ICCP), Department of Primary Care and Public Health, School of Public Health, Imperial College London. London, United Kingdom.
3- Department of Medicine, Faculty of Medicine, University of Seville. Seville, Spain.
4- Clinical Epidemiology and Vascular Risk, Instituto de Biomedicina de Sevilla, IBiS/Hospital Universitario Virgen del Rocío/Universidad de Sevilla/CSIC. Seville, Spain.
5- Institute of Applied Health Research, University of Birmingham, United Kingdom.
6- Department of Primary Care and Public Health, School of Public Health, Imperial College London. London, United Kingdom.

* Joint principal/last authors. These authors contributed equally to the work.

Corresponding author:
Dr Marie Line El Asmar, MD, MPH
me719@ic.ac.uk

Key words
Clinical decision support system
Electronic health records
Primary care
Chronic disease
Health outcomes

Word Count:

- Abstract: 300 words
- Main text: 4,646 words

Number of Figures: 2
Number of Tables: 3
Number of references: 59

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Abstract

Objectives: Chronic diseases are the leading cause of disability globally. Most chronic disease management occurs in primary care with outcomes varying across primary care providers. Computerised Clinical Decision Support Systems (CDSS) have been shown to positively affect clinician behaviour by improving adherence to clinical guidelines. This study provides a summary of the available evidence on the effect of CDSS embedded in electronic health records on patient-reported and clinical outcomes of adult patients with chronic disease managed in primary care.

Design and eligibility criteria: Systematic review including randomised controlled trials (RCTs), cluster RCTs, quasi-RCTs, interrupted time series, and controlled before-and-after studies , assessing the effect of CDSS (vs. usual care) on patient-reported or clinical outcomes of adult patients with selected common chronic diseases (asthma, chronic obstructive pulmonary disease, heart failure, myocardial ischemia, hypertension, diabetes mellitus, hyperlipidaemia, arthritis, and osteoporosis) managed in primary care.

Data sources: Medline, Embase, Cochrane Central Register of Controlled Trials, Scopus, Health Management Information Consortium, and trial register clinicaltrials.gov, searched from inception to June 24, 2020.

Data extraction and synthesis: Screening, data extraction, and quality assessment were performed by two reviewers independently. The Cochrane Risk of Bias tool was used for quality appraisal.

Results: From 5,430 articles, eight studies met the inclusion criteria. Studies were heterogeneous in population characteristics, intervention components, and outcome measurements and focused on diabetes, asthma, hyperlipidaemia, and hypertension. Most outcomes were clinical with one study reporting on patient-reported outcomes. Quality of the evidence was impacted by methodological biases of studies.

Conclusions: There is inconclusive evidence in support of CDSS. A firm inference on the intervention effect was not possible due to methodological biases and study heterogeneity. Further research is needed to provide evidence on the intervention effect and the interplay between healthcare setting features, CDSS characteristics, and implementation processes.

Review registration: Prospero ID: CRD42020218184

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Strengths and limitations

- In contrast to other reviews, this review focusses on interventions involving CDSSs that are knowledge-based, embedded within EHR systems, provide electronic alerts, and are used by physicians in primary care.
- The electronic search strategy was piloted, and hand-searching of reference lists of previous reviews permitted a sufficient level of confidence that all relevant articles were included.
- Some limitations arose related to studies being of 1-year duration on average, making it difficult to deduce long-term effects on patient outcomes.
- The scope of the review was restricted to nine chronic diseases; however, the selection included the most common chronic diseases, yet there were no eligible studies in five of them.
- The included studies were of low-to-moderate quality of evidence, and heterogeneity among the included studies limited the ability to conduct a meta-analysis.

Introduction

Non-communicable chronic diseases are the leading cause of disability and mortality worldwide, affecting one in three adults globally.[1] Rising healthcare expenditure because of chronic disease prevalence has led to an ever-growing demand for effective primary care services for the long-term management of chronic conditions.[2-4] Considering that health systems have mainly adapted to providing periodic care for acute problems, an even greater need to achieve efficient, comprehensive and coordinated long-term care persists.[5] Models such as the Chronic Care Model (CCM) have been proposed to improve the quality of care in chronic care systems.[6] This model recommends 'decision support' as one of six applicable changes in primary care settings that promote better patient outcomes, if successfully implemented.[6-8] Within this and other models of primary care and chronic condition management, clinicians are encouraged to provide patient care according to evidence-based clinical practice guidelines (CPGs) to support improved patient outcomes.[9, 10] Given the frequent updates to CPGs resulting from rapid advancements in medical knowledge, a major challenge is the maintenance of clinicians' knowledge of evidence-based CPGs.[10] Consequently, clinicians' adherence to CPGs may be variable and frequently insufficient, which may result in sub-optimal disease management outcomes.[10, 11]

Clinical decision support is an umbrella term that includes various care structures, processes and tools.[12] Clinical decision support systems (CDSS) range from non-computerised to computerised systems and basic to advanced systems.[12, 13] Currently, most attention is driven toward computerised advanced CDSS, which perform complex decision-making functions from electronic health record (EHR) patient data.[12] Such systems are termed knowledge-based, as they use software algorithms to generate actionable patient-specific outputs in the form of recommendations to the practitioner using conditional (IF-THEN) rule statements derived from CPGs.[14, 15] They also differ in their technical and clinical interface, whether passive (providing user-initiated prompts to generate clinical advice, e.g., by clicking buttons), or active (automatic system-initiated prompts).[12, 16, 17] CDSS also differ in system-user communication, which could be consulting (i.e., provide appropriate 'next-steps' clinical recommendations) or of critiquing nature (verify decisions taken by practitioners).[12, 16] Given the differences in interfaces, the scope of this review includes computerised, knowledge-based CDSS embedded in EHRs that are of either passive or active, consulting or critiquing, nature.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Although CDSS are ubiquitous, few have been successfully adopted.[18] Previous reviews appraised the efficacy of different types of CDSS in primary care, presenting a greater consensus on the useful role of CDSS on practitioner-related outcomes such as guideline adherence.[19-23] However, the impact of CDSS on patient outcomes remains largely uncertain.[19-23] *Bryan et al* demonstrated a positive association between the use of CDSS in primary care and patient outcomes.[23] However, a need to further evaluate the effect of CDSS on patient outcomes was identified, particularly in primary care and specifically for chronic disease management. The novel focus of this review is in the combined focus on the effect of computerised, knowledge-based, clinical decision support systems *and* their effect on patient-reported *and* clinical outcomes of patients *and* with chronic disease *and* managed in primary care settings. This particular focus was chosen to inform more specific and efficacious implementation strategies in primary care management of chronic disease and attract the industry's consideration of CDSS use in primary care settings, which a more generalised focus on multiple care settings, multiple diseases beyond just chronic disease, or lack of focus on patient-reported outcomes may not achieve. There remains a need to assess the capabilities of CDSS in distinctive clinical professional areas and disease entities, particularly on patient outcomes.[18] A recent review on EHR-embedded CDSS focused on process delivery (practitioner outcomes) and was not limited to primary care settings and chronic conditions.[24] Despite obvious heterogeneity, clinically meaningful improvements in care processes were shown.[24] However, it was unclear under what clinical circumstances these improvements could be replicated in every clinical context to affect clinical outcomes presumptively.[24]

In this systematic review, patient outcomes refer to quantifiable clinical biomarkers such as glycated haemoglobin (HbA1c) and low-density lipoprotein-cholesterol (LDL-C) levels, clinical outcomes such as mortality and morbidity, and patient-reported outcomes, such as health-related quality of life (HRQoL). To our knowledge, and in the view of rapid upgrades in the world of CDSS, no review exists solely on the effect of CDSS embedded within EHRs on patient outcomes for patients with chronic disease managed in primary care. This review aims to fill this knowledge gap by collating all the available evidence on this classification of CDSS.

Methods

Study design and eligibility criteria

A systematic review of the existing literature was conducted following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines.[25, 26] The research question, inclusion, and exclusion criteria were defined using the PICOS framework.[27] Nine common chronic diseases were selected as part of the inclusion criteria, following a review by *Reynolds et al,*[28] namely asthma, chronic obstructive pulmonary disease (COPD), heart failure, myocardial ischemia, hypertension, type 2 diabetes mellitus, hyperlipidaemia, arthritis, and osteoporosis. Studies reporting on patient-related and clinical outcomes managed by primary care clinicians using CDSS-EHR compared to usual care were included. Studies involving participants <18 years of age, multidisciplinary (non-clinician) system users, and other varieties of CDSS (not linked with EHR) were excluded. A detailed view of the criteria is provided in Supplementary Table 1.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Figure 1 depicts an analytic framework of the intervention adapted from a framework published by *Lobach et al*.[22] The framework provides a theoretical context for possible features that could underpin the success, or lack of success, of a complex intervention such as implementing CDSS in a primary care setting. The framework guided the extraction of relevant data from studies to inform the narrative synthesis.

Search Strategy

Electronic databases Medline, Embase, Scopus, Cochrane Central Register of Controlled Trials (CENTRAL) and Health Management Information Consortium (HMIC), and one clinical trial registry (clinicaltrials.gov), were searched from inception to June 24th 2020. Cochrane reviews were screened for relevant search terms and subject headings (MeSH) terms under the following categories 'registries supported by CDSS' AND 'General Practice/Primary care' OR each of the nine chronic diseases combined with AND 'Asthma', 'heart failure/myocardial ischemia', 'hypertension', 'diabetes mellitus', 'hyperlipidaemia', 'COPD', 'arthritis', 'osteoporosis' (Supplementary Table 2 and 3). The search was limited to articles in English.

Terms for patient-reported outcome measures (PROM) were excluded. Existing PROM filters are not comprehensive, and the use of those filters in the search strategy could increase the risk of missing relevant studies.[29] However, we manually screened papers to identify any "patient-reported outcomes" measured in studies. PROM may be described in various ways, thus attempting to list specific patient outcomes for each entity could introduce bias and lead to missing relevant studies. Previous systematic reviews around CDSS in primary care were also searched for potentially eligible studies.[19-21, 23, 24]

Study Screening and Selection

Search results were exported to RefWorks, then screened using Covidence. After removal of duplicates, screening was conducted by two reviewers independently (MA, KID). Disagreements were resolved by consensus within the review group. After an initial screening of titles and abstracts, the full text of potentially relevant articles were assessed.

Data Extraction

A data extraction form was adapted using the 'Cochrane data collection form for intervention reviews: randomised controlled trials (RCTs) and non-RCTs' and aided by *Lobach et al*'s data abstraction guidance.[30, 31] The form was piloted and two reviewers (MA, KID) completed data extraction independently.

Quality Assessment of Individual Studies

Risk of bias was assessed using the Cochrane risk of bias (RoB) tool,[32, 33] by two reviewers independently (MA, KD). The methodological risk of bias was assessed per the following elements in RCTs: selection bias, performance bias, detection bias, attrition bias, reporting bias, and other biases.[33, 34]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Quasi-RCTs were reported as high risk for random sequence generation. Cluster-RCTs were assessed for an additional element 'selective recruitment of cluster participants.' All judgments were made with guidance from the Cochrane Handbook and the guidelines of the Cochrane Consumers and Communication Review Group.[33-35]

Studies were considered as high risk if they scored 'high' or 'unclear' for either random sequence generation or allocation concealment, based on the rising evidence that these two elements are particularly significant sources of bias.[34] A summary 'risk of bias' figure was generated for all risk assessments of elements with a narrative commentary for each study. Additionally, a detailed adapted table was generated to visualise sub-elements that fit under each element in the Cochrane RoB tool.[34]

Quality Assessment of Outcomes

The GRADE (Grading of Recommendations, Assessment, Development and Evaluations) tool was used to assess the quality of outcomes from individual studies.[36] The tool involves five elements: risk of bias, inconsistency, indirectness, imprecision, and publication bias. The tool is primarily used for meta-analyses; however, it can be adapted for use in narrative reviews. Study outcomes were appraised, ranked as high, moderate, low or very low and justifications were provided, in line with Cochrane guidance.[37]

Data Synthesis

Treatment effect was reported as odds ratios (OR) with 95% confidence intervals (CIs) and p-values for dichotomous outcomes, where available. Continuous data were presented as mean differences with 95% CIs and p-values.

Heterogeneity in the results was anticipated due to the wide spectrum of possible patient-related outcomes reported by the studies. Methodological heterogeneity was assessed using Cochrane's RoB tool. Heterogeneity in included studies varied in terms of intervention components, population characteristics, and outcome measurements; thus, conducting a meta-analysis was deemed not appropriate by the four-person author group.

Patient and Public Involvement

No patients were involved.

Ethics Approval

The present work relied on already published aggregated data and ethics committee approval was unnecessary.

Results

The search identified 5,430 articles. Following removal of duplicates, 4,498 articles remained for screening by title and abstract. Of these, 266 articles were selected for full-text review and 10

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

articles met the inclusion criteria. Among them, two articles were published protocols of included articles; thus, 8 different studies were finally included. The detailed search results are presented in the PRISMA flowchart (Figure 2).

Study Characteristics

Characteristics of included studies are presented in Table 1. Included studies comprised of patients in four disease areas: T2DM, asthma, hyperlipidaemia, and hypertension.[38-45] No studies on COPD, heart failure, myocardial ischemia, arthritis, and osteoporosis were identified. Six studies were cluster-RCTs, one study was parallel-group RCT, and one was quasi-RCT of stepped-wedge design. Most studies were conducted in the United States, and three studies took place in the United Kingdom, Spain, and Belgium. Studies, both among the disease conditions and within each disease entity, differed appreciably in participant covariates, setting attributes, and CDSS characteristics. These factors, compounded with the heterogeneity of the studies (as described above in the methods section) suggested that the reported outcomes were not fit for statistical pooling, hence, these factors justified not conducted a meta-analysis. Follow-up ranged from 6 months to 1.5 years. No study incorporated financial reimbursement before the implementation of interventions. Further details are shown in Table 1.

CDSS Characteristics

CDSS interventions were characterised based on their origin (commercial/local), delivery mode (user or system-initiated), user response (optional/mandatory), events (system functions), coupled EHR, and auxiliary features (local clinician involvement in system development process, system accessibility to other team members, provision of performance feedback, or coupled with user education). Additional information such as user-training, user adherence, and reported usability were recorded. Details are provided in Table 2.

Risk of Bias and Quality of the Evidence

Supplementary Tables 4 and 5 provide risk assessments details. Overall, the included studies were impacted by risk of bias, mainly due to incomplete outcome data and uncertainty on blinding of outcome assessment and personnel. In most studies, it was unclear if outcome assessors were blinded. Additionally, blinding of physicians (CDSS users) was not possible across all studies, as they were the intended system users. Most studies reported substantial withdrawal rates due to poor follow-up. These limitations resulted in analysing subgroups (rather than the initially intended full cohorts) in some trials, hence most studies were considered of high risk for incomplete outcome data.

Overall, quality of the evidence was rated as either moderate or low, with the most impact resulting from methodological risks of bias within studies. Supplementary Table 6 presents the quality of evidence evaluated using GRADE, with reasons stated for quality downgrading.

For Peer Review Only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Narrative Synthesis

Diabetes Mellitus

Three studies assessed the effect of CDSS on clinical and patient-reported outcomes in T2DM patients.[38-40]

Gill *et al* retrieved slight, yet statistically significant, reduced HbA1c and LDL-C with the CDSS intervention compared to the control group, with a small between-group difference in HbA1c and LDL-C levels of 0.12% and 3.57 mg/dL, respectively (Table 3).[39]The reported reductions in HbA1c and LDL-C were clinically small; however, statistically, the intervention group had 52% higher odds of achieving personalised HbA1c levels, 56% higher odds of achieving HbA1c <7.0%, and 34% higher odds of achieving an LDL-C <100 mg/dL compared to usual care patients over 1 year of follow-up. Nevertheless, compared to the control group, the intervention group had better-controlled baseline levels of HbA1c and LDL-C. The study by *Gill et al* involved a system-initiated "push" CDSS which was integrated into EHRs already operating at participating practices. Reported limitations were of EHR-CDSS interoperability difficulties, which resulted in missing patient lab values.[39] Other missing lab values also resulted from patients not returning for lab tests. In view of these limitations, a subset of randomised patients without missing HbA1c and LDL-C values were analysed (Table 3).

Schnipper *et al* (RCT) reported the odds of patients meeting management goals (Table 3).[38] Intention-to-treat analysis showed non-significant changes in patients' odds of meeting management goals; a per-protocol analysis, however, showed slightly significant odds of attaining management goals in the intervention group compared to the control group (Table 3). Among the three studies, this intervention is the only one that involved user training (Table 2)**.** Auxiliary features also included feedback on performance and local user (physician) involvement in the CDSS development process; yet system usability was reported to be used for only 5.6% of patients (Table 2).

Heselmans *et al* (cluster RCT) reported non-significant between-group differences in HbA1c, LDL-C, diastolic (DBP) and systolic (SBP) blood pressure (BP) (Table 3).[40] Levels of HbA1c were relatively well-controlled**,** with reasonable mean values for HbA1c in both groups at baseline, especially in the control group (Table 1).[40] Subsequent analysis of a subgroup of 601 patients with HbA1c >7.0% retrieved a statistically significant between-group difference in HbA1c favouring the intervention, though not clinically relevant (-0.40% [95% CI -0.70 to -0.09]). The same analysis retrieved non-significant differences in LDL-C and BP. User adherence and usability were not measured.

In all three studies, there was no mention of the provision of reimbursement for the implementation of the CDSS. All CDSS were commercial, and user response to the system alerts was optional.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Asthma and Angina

One study assessed the impact of CDSS on self-reported quality of life and condition-specific outcomes in patients with asthma and angina.[41] The angina intervention group acted as the control group for the asthma intervention group and vice versa (justified by authors to eliminate a possible Hawthorne effect in practitioner behaviour on the primary outcome). Questionnaires were administered in three rounds (a year before the intervention, beginning of the intervention, and a year after the intervention), which showed non-significant differences in both intervention groups post-intervention (Table 3). Completion of the three rounds of questionnaires occurred in 46% and 35% of patients with angina and asthma, respectively. The implemented CDSS was locally produced and embedded in two widely used health systems (Table 2). The CDSS was modified four months into the study from system to user-initiated, as requested by participating physicians. User training was limited, and system usability was very low, measured using a log with a median of zero user-system interaction during most of the study (Table 2).

Hyperlipidaemia

Two studies involved participants with hyperlipidaemia.[42, 43]

The study by *Gill et al* involved implementing a commercial CDSS in participating practices of one institution.[42] All patients enlisted under participating practices with recorded LDL-C values were enrolled. A high correlation in outcomes was found at baseline and end measurements, causing patients not meeting LDL-C levels after randomisation to be analysed only. The control group also had a lower risk of disease and was significantly younger. Patients were stratified into three guideline-based risk groups, and the percentage of patients reaching the pre-defined goals at baseline and end were recorded (Table 1). All categories in the intervention and control groups showed a significant increase in the proportion of patients attaining lipid goals, except in the moderate-risk group. Calculated differences were not presented; however, p-values of differences within the intervention and control groups suggested no significant improvement in any risk category.

In *Cobos et al*, despite a usability of 71%, the intervention had no superiority over usual care in terms of between-group differences in lipid values (Table 3).[43] The study reported >25% patient lost to follow-up for clinic visits or LDL-C testing (Table 1).

Hypertension

Two studies assessed the effect of CDSS on BP in hypertensive patients.[44, 45] Both studies employed CDSS that were system-initiated with optional user responses (Table 2).

Hicks et al reported no differences between intervention groups in BP control (Table 3).[44] Differences in baseline and end SBP and DBP between groups were not significant. Nonetheless, the intervention group had a significantly higher percentage of patients with controlled BP (Table 1). No user training was provided, and system usability was not measured (Table 2). Less than 80% of recruited patients were followed-up due to withdrawals caused by the study's inclusion

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

criteria, which required patients to have a recorded ethnicity in the EHR. Subsequently, patients without BP recordings were considered uncontrolled. However, a subset analysis excluding these patients still retrieved a non-significant difference in BP values.

Lopez et al study was a quasi-RCT and reported significant odds of BP improvement in the intervention group compared to the control group (Table 3).[45] Notably, extensive training was provided for participating clinicians compared to all other studies, and semi-structured interviews revealed that all practices appeared to use the system components (Table 1).

Discussion

Main findings

As primary care providers are increasingly required to monitor and manage chronic diseases, there is a need to assess the capabilities of CDSS on patient outcomes. This systematic review included the eight available studies on the effect of computerised, knowledge-based, EHR-embedded CDSS on clinical and patient-reported outcomes. Several study designs were included in the eligibility criteria; however, studies fit for inclusion were trials, namely RCTs, cluster-RCTs, and non-RCTs (supplementary table 1, table 1). Two studies reported a statistically significant impact of CDSS on patient outcomes,[39, 44] one reporting clinically small yet significant reductions in $HbA1_C$ and LDL-C and another reporting significant increases in the proportion of hypertensive patients who achieve goals, which may be clinically relevant.[45] Another two studies reported statistically significant findings but in subgroup analyses.[38, 40] One study reported statistical significance in partial outcomes and three studies did not report any statistically significant impact of CDSS on outcomes at all.[42-44] Overall, evidence was not conclusive to imply a differential effect, either positive or negative, of clinician use of CDSS on patient or clinical outcomes compared to usual care, mainly due to the quality of evidence impacted by the prominent risk of bias in the included studies. The systematic review also noted a limited availability of studies evaluating the CDSS of interest directed to clinicians in primary care for the management of the common chronic diseases. This is especially true for COPD, arthritis, and osteoporosis, which were eligible conditions, yet no such studies were available.

Implications of the review

The systematic review infers the importance of considering how the interplay of factors such as the nature of a health care setting (academic, community) and CDSS interact with clinicians (or users per se). System-initiated CDSS seemed to have a better reflection on patient outcomes compared to user-initiated systems; however, this relationship was confounded by many factors such as poor process measures, namely low system usability and EHR-CDSS interoperability difficulties. A common limitation reported across most studies was poor patient follow-up, which highlights the importance of patient behaviour as a confounding factor to the performance of CDSS interventions in clinical settings. Some interventions included helpful auxiliary features such as involvement of local physicians in CDSS development and feedback on performance, yet these features did not seem to reflect positively on patient outcomes. Understanding user (clinician) experiences qualitatively in the context of using such CDSS to manage chronic diseases in primary care may be of benefit to inform on their practical use. This review assessed CDSS functionality

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

when used by clinicians; however, CDSS accessibility to other members of a multidisciplinary team may have an added positive effect which could translate into better patient outcomes. Multidisciplinary teams in primary care have been shown to aid successful implementation of CDSS by increasing adherence to CPGs and improving patient outcomes in the absence of CDSS.[46, 47]

The healthcare setting and intervention implementation itself may dictate the success or failure of an intervention. Large academic care settings may be better equipped with IT and financial resources than community centres to ensure successful implementations of CDSS.[48] Included studies revealed limitations due to poor implementation of the CDSS, namely interoperability issues, lack of IT support, poor usability, and lack of user training, which could have hindered the intervention from reflecting significant outcomes compared to usual care. Models have been offered to help explicate how structural elements of healthcare settings and various levels of healthcare organisations levelling up to the industry may influence clinician behaviour and uptake of health IT to influence patient outcomes.[17, 49] Finally, it is imperative to consider the effect of non-clinical factors such as patient behaviour outside clinic visits, their perceptions, and adherence to clinical advice.[50]

Comparison with previous reviews

Previous reviews of CDSS of multiple types have indicated impact on patient outcomes in some instances. A recent review showed that CDSS systems achieve small to moderate effects on processes of care, yet the extent of translation to patient outcomes and under what clinical contexts and conditions this was achieved was not determined.[24] Previous reviews were not limited to primary care settings and included various other CDSS, namely paper-based, web-based, standalone electronic systems, and patient-directed systems that do not necessarily alert a clinician at the point-of-care.[19, 21, 51] However, reviews assessing the impact of various types of CDSS on chronic disease patient outcomes in primary care identified inconclusive results, with a need to define more clearly the exact role of CDSS compared to usual care in subsequent studies.[20, 23] Reviews evaluating patient outcomes noted conceptual between-study heterogeneity, as this review did, especially when several diseases and clinical outcomes were considered.[52] In terms of patient outcomes, CDSS seem to affect clinical outcomes for cardiovascular risk factors such as hyperlipidaemia and diabetes.[11, 53] Nevertheless, a much more recent review on the effect of CDSS on cardiovascular risks revealed no definitive clinical benefits after their implementation.[53] On the other hand, the evidence is less robust for other non-cardiovascular risk-related chronic conditions, such as cancer, and other primary care sensitive cases (PCSC) such as vaccination, which were not part of the eligibility criteria in this review.[54]

A review of CDSS embedded within EHRs in all-purpose healthcare settings on overall mortality and morbidity suggested no effect of CDSS on mortality. However, it noted that potential benefits or risks would highly depend on setting and disease characteristics, which is an evident implication in this review.[52] A review by *Bryan et al* validated the use of CDSS in primary care to improve outcomes; however, they reported substantial variability in intervention effectiveness related to the type and implementation of the different types of CDSS.[23] Additionally, other reviews incorporated studies that involved the use of CDSS by multidisciplinary care members, reflecting positive clinical outcomes with shared decision making.[19, 51, 55]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Unanswered questions and future research

More research is needed to establish the effectiveness of EHR-embedded CDSS on clinical outcomes of patients with chronic diseases treated in primary care settings. Further studies involving patients with chronic diseases are needed for robust evidence. Moreover, it may not be reasonable to evaluate the intervention's efficacy based on older studies, given the exponential growth and rapid advancements in the field of CDSS.[56] Future research could include practitioner-related, cost-effectiveness, and patient-related outcomes to present an ampler perspective on the effect of this type of CDSS to inform policymaking and clinicians. Other primary care sensitive conditions (PCSCs) can be studied for a broader scope, including cancer, mental health illnesses and preventative interventions such as vaccination.[57] Research around prolonged evaluations of CDSS is needed. Longer-term studies may account for potential implementation barriers that could take place. Additionally, it would be beneficial to synthesise findings on the effect of this particular type of CDSS on patient outcomes, when operated within a multidisciplinary team of non-clinician providers. It is worth noting that patient outcomes are not always addressed efficiently in trials, among the reasons is the possibility of having subjective, composite, or surrogate outcomes, which do not necessarily translate into real-life patient outcome improvements.[58] Hence, including observational studies in updated reviews may be useful. Additionally, no studies of registries embedded with CDSS fit the eligibility criteria, suggesting a potentially greater inclination to embed CDSS with EHRs currently. There also seems to be a dearth of studies evaluating non-cardiovascular risk-related chronic diseases post-implementation of CDSS linked with EHRs, as no studies evaluating patients with COPD, osteoporosis, and arthritis were eligible.

Strengths and limitations

This systematic review focusses on interventions involving a specific type of CDSS –those that are knowledge-based, coupled with EHR systems, that provide electronic alerts, and are used by physicians in primary care to support chronic disease management. To the authors' knowledge, no such review exists considering this combined focus and given the attention these systems are increasingly gaining in primary care settings.[29] As an example of the novelty of this review, compared to previous reviews with the most similar focus to our own by Bryan and Boren[23] and Bright et al[19] , this review included studies not featured in these previous reviews, such as the studies by Schnipper et al, Gill et al, Heselmans et al and Hicks et al.[38-40, 44]Application of the eligibility criteria identified eight studies for inclusion, which may be a small number given the recent growth of interest in CDSS in healthcare settings.[59] Nevertheless, piloting the electronic search strategy and supplementing it with hand-searching of reference lists of previous reviews in the field permitted a sufficient level of confidence that all relevant articles, against the eligibility criteria, were included for synthesis. The review also contributes to the knowledge provided by a recent review that focuses on process outcomes.[24]

Some limitations arose from methodological constraints. Studies were of an average of 1-year duration, which makes it difficult to deduce long-term effects on patient outcomes. Nine common chronic diseases were included in the syntax.[28] However, there was an absence of eligible studies in five of them, which limited the number of studies for inclusion. Although a selection of nine chronic diseases was included in the syntax, these covered the most common chronic diseases, as done by Reynold et al.[28] The review could have covered other PCSCs for a broader scope, including cancer and mental health illnesses, per se.The included studies were also of low-to-moderate quality of evidence. Additionally, our review does not report on process outcomes. The

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

decision to exclude process outcomes such as practitioner-related outcomes, namely clinician adherence to CPGs was made to channel the research question to focus on where the most significant knowledge gap lies, as the effect of CDSS on patient outcomes remains largely uncertain.[19-21, 23, 24] Nevertheless, capturing those outcomes would have provided a more comprehensive assessment on the effect of CDSS for the management of patients with chronic disease in primary care. The heterogeneity in study intervention components, participant covariates, setting attributes, and CDSS characteristics rendered outcome pooling in a metanalysis not suitable.

Conclusion

This systematic review aimed to assess the impact of CDSS integrated with EHRs on patient-reported and clinical outcomes of chronic disease patients in primary care settings. A definite conclusion on the intervention effect on outcomes could not be drawn due to between-study heterogeneity and methodological risks, reflecting an immature research area. Intervention characteristics and implementation approaches may have an impact on outcomes from CDSS use and we make reference to the literature in this area that may guide and help evaluate successful implementations of CDSS in practice. Further studies are needed to formulate more robust inferences and validate the intervention's effect on patient outcomes. With the continuous rise in the prevalence of chronic diseases, it is essential to understand how enhancing primary care management using IT support could support improved outcomes. Practical applications of CDSS evaluations in the healthcare workflow and setting should be sustained to suitably determine the actual effectiveness of CDSS and support improved implementation efforts into practice. Strengthening the clinical benefit of CDSS and determining their cost-effectiveness could inform policymaking of enhanced primary healthcare of chronic disease management.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Acknowledgements

The authors would like to thank the library services team at Imperial College London. Special thanks to Ms Rebecca Jones for her support in reviewing the search strategy.

Ms KI Dharmayat acknowledges support from a PhD Studentship from the National Institute for Health Research (NIHR) under the Applied Health Research (ARC) programme for North West London, United Kingdom (the views expressed in this publication are those of the authors and not necessarily those of the NHS, the NIHR or the Department of Health). Dr AJ Vallejo-Vaz acknowledges support from a fellowship from the "Programa Beatriz Galindo" from the Ministry of Universities of Spain, and University of Seville, Spain.

Authors' statement

RI conceived the original study concept with input from NM. NM and AJV-V led the overall study design and supervised the subsequent steps of the study. MA and KD designed the search strategy with supervision from the other authors. MA and KDconducted the literature search, data extraction and quality assessment of studies. MA wrote the first draft of the manuscript. All authors participated in the interpretation of data, have critically reviewed the manuscript and provided edits and comments, and approved its final submission.

Conflict of interest

Ryan Irwin and Nikolas Mastellos are employees of Cerner UK Ltd –a health technology organisation. Dr AJ Vallejo-Vaz reports past or current participation in investigator-initiated research grants to Imperial College London from Pfizer, Amgen, Sanofi, MSD, Daiichi Sankyo and Regeneron, honoraria for lectures from Amgen, Mylan and Akcea, and Steering Committee/consulting fees from Radcliffe Cardiology and Bayer, all outside the submitted work. MS KI Dharmayat reports past or current participation in investigator-initiated research grants to Imperial College London from Amgen, Sanofi, MSD, Daiichi Sankyo and Regeneron and consulting fees from Bayer, all outside the submitted work.

Funding

No funds were received for conducting the present study.

Data Availability Statement

No additional data are available

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1: Characteristics of included studies

								Participants		Comparability			
Study	Disease	Study Design	Country	Duration	Sponsorship	Setting	Condition identification from EHR	Intervention	Control	Age	Sex	Disease Severity	Outcomes
Schnipper et al. (2010) [38]	Type II Diabetes Mellitus	RCT	United States	6 months	The Agency for Healthcare Research and Quality	Partners HealthCare System, 10 practices at Brigham and Women's Hospital and Massachusetts General Hospital (Academic)	NS[1]	3431 patients	3578 patients	Comparable	IG [4]: more females (statistically significant)	Comparable	Difference in proportion of patients meeting management goals compared to baseline.
Gill et al. (2017) 39	Type II Diabetes Mellitus	Cluster RCT	United States	1 year	Sanofi, U.S.	Primary care practicesin Delaware and Maryland (Community)	International classification of diseases (ICD-9CM)EHR code	5offices 29PCPs 4484 patients Analysed (by subgroups): HbA1c :2041 LDL-C:2793	5offices 23PCPs 1902 patients Analysed (by subgroups): HbA1c: 723patients LDL-C: 931patients	Overall: 18-75 IG: older (statistically significant)	Comparable	IG: Better glycaemic control, lower LDL-C	Difference in HbA1c[2]from baseline Difference in LDL-C [3] from baseline Odds of patients achievingHbA1c and LDL-C goals
Heselmans et al. (2020)[40]	Type II Diabetes Mellitus	Cluster RCT	Belgium	1 year	None	Belgian primary care practices using OneHealth EHR (Community)	International classification of primary care (ICPC-2) EHR code or confirming lab tests	25 offices 60 PCPs Analysed: 14offices 21 PCPs 1351 patients	26 offices 60 PCPs Analysed: 15offices 31 PCPs 2464 patients	Overall: ≥ 18 years IG: older (significant statistically)	CG[5]: more females (statistically significant)	Comparable for: mean HbA1c, mean BP, % patients meeting target for HbA1c , BP and LDL-C	Mean value differences of HbA1c, LDL-C, and blood pressure of the previous year at6 and 12 months compared to baseline
Eccles et al. (2002)[41]	Asthma and Angina	Cluster RCT	United Kingdom	1 year	UK NHS research & development	General practices in north east England. (Community)	Asthma or angina EHR code in prior 12 months	Asthma: 31 offices Angina: 29 offices	Angina: 29offices, Asthma: 31offices	≥ 18 years no details presented	NS	NS	Quality of life and condition specificoutcomes using questionnaires
Gill et al. (2009)[42]	Hyperlipidaemia	Cluster RCT	United States	1 year	None	Medical quality improvement consortium (MQIC) of primary care practices across the country (Academic and Community)	Identified as having LDL levels recorded in EMR	12 offices 49 PCPs 26,696 patients	13 offices 56 PCPs 37,454 patients	Overall: 20-79 years CG younger	CG: more males (statistically significant)	CG lower disease risk	The percentage of patients with most recent LDL-C at goal
Cobos et al. (2006)[43]	Hyperlipidaemia	Cluster RCT	Spain	1 year	Novartis	General practices in Catalonia	Identified as having total cholesterol (TC) between 200mg/dL and 400 mg/dL	20 offices 1046 patients	22 offices 1145 patients	Overall: 18-94 years	CG: more females (statistically significant)	comparable	Between-group differences in means of lipid values
Hicks et al. (2008)[44]	Hypertension	Cluster RCT	United States	1.5 years	The Agency of Healthcare Research and Quality	Primary care practices affiliated with the Brigham and Women's Hospital	Identified as having 2 or more hypertension-related outpatient visits usingEHR codes	7 offices 786 patients	7 offices 1048 patients	Comparable	Comparable>20 years	IG: significantly higher % of patients with controlled BP at first visit	Percentage of patients with controlled blood pressure Odds of blood pressure control at outcome visit
Lopez et al. (2019)-2016 [45]	Hypertension	Non- RCT	United States	1 year (2016)	Centre for Disease Control	Primary care practices in New York	Identified as having at least 1 ICD code for hypertension in EHR	7 offices	7 offices	18-85	NS	NS	Odds of blood pressure control post-intervention

[1] NS abbreviates not stated
[2] HbA1c refers to Glycated haemoglobin
[3] LDL-C refers to Low-density lipoprotein
[4] IG refers to intervention group
[3] CG refers to control group

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 2: CDSS characteristics of interventions in included studies

Study	Disease	Origin	Delivery Mode	User Response	Events	Guidelines	Integrated EHR (name)	User Training	User Adherence	Usability	Auxiliary features
Schnipper et al. (2010)[38]	Diabetes mellitus	Commercial	User- initiated	Optional	• Requestsmissing data. • Alerts outputs ofassessments of clinical care. • Recommends orders for medicationchanges, laboratory studies, appointments, and referrals	NS	Longitudinal Medical Record	Brief instructions atstart Monthly tips	Tracked use by clinician and sent customised emails	Used for 5.6% of eligible patients	Local user involvement in development process
Gill et al. (2017)[39]	Diabetes mellitus	Commercial	System- initiated	Optional	• Generates reportsfor patient-specific recommendations at thePOC	American diabetes Association European Association for the study of Diabetes & ICD 9 coding	Multiple EHRs (previously employed)	Not provided	Qualitatively assessed through informal interviews	4/5 practices reported use	Recommendations also received by non-clinicians in care team
Heselmans et al. (2020)[40]	Diabetes mellitus	Commercial	User- initiated	Unclear	• Alerts for reminders, therapeutic suggestions, and diagnosis-specific guideline links	Belgian and International Guidelines	HealthOne	Not provided	Not assessed	Not assessed	None
Eccles et al. (2002)[41]	Asthma and Angina	Local	System- initiated (4 months) then altered to user-initiated	Optional	• Alerts management and prescription options and patient referrals	NS	AAH Meditel or EMIS systems	1-day training	Measured by a usage log	Median number of system interactions was zero	Education on system guidelines
Gill et al. (2009)[42]	Hyperlipidaemia	Commercial	System- initiated	Optional	• Alerts for overdue lipidtests • Alerts if patienthas not met personalised lipidgoals • Generates diseaserisk categories	Adult Treatment Panel-III (ATP-III) guidelines	EMR (Centricity)	1-year familiarity with system	Not measured	Not measured	Generates a list of patients who have not met lipid goals, outside office visits. Listed patients were sent a letter at the start and after 6 months of the study.
Cobos et al. (2006)[43]	Hyper lipidaemia	Local	System- initiated	Optional	• Recommends therapeuticoptions • Recommendsfollow-ups • Recommends lab testing	The European Society of Cardiology and other societies for Hypercholesterolemia management (ESCHM)	Not specified	None	Monitored by CDSS	71%	None
Hicks et al. (2008)[44]	Hypertension	Local	System- initiated	Optional	Alerts for therapeutic options	American Heart Association/American College of Cardiology (AHA/ACC) 2001 guidelines	Not specified	None	Assessed prescription of a recommended drug class within 1 week of clinic visit	NS	None
Lopez et al (2019)[45]	Hypertension	Local	System- initiated	Optional	• Alerts for missing BP readings • Alerts for therapeutic options	Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure (JNC- 7) guidelines	eClinicalWorks or MD Land	2-day training at start, 7 training sessions during second year, ad-hoc sessions as needed	Semi-structured interviews	Reported all practices have used all CDSS components	Generates reports of patients by status to manage visits

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 3: Description of Outcomes

				Absolute Effects		Relative Effects		
Study	**Disease**	**Outcome**	**Outcome definition**	**Intervention**	**Usual Care**		**Conclusions**	**Quality**
Schnipper et al. (2010)[38]	Diabetes	Odds of patients meeting management goals at the end compared to baseline	Targets undefined	Not reported		**ITT**: OR = 1.02; 95% CI, 0.99 to 1.04, P not reported **PP:** OR = 1.16; 95% CI, 1.08 to 1.25; P 0<.001	**ITT**: non- significant odds of intervention patients meeting management goals compared to control patients at 6 months. **PP**: significant odds of intervention patients meeting management goals compared to control patients after follow-up period.	⊕⊕◯◯ LOW
Gill et al. (2017) [39]	Diabetes	Difference in A1C from baseline Difference in LDL-C from baseline Odds ofpatients achieving A1Cand LDL-C goals	**HbA1Cgoals**: <7%and <7.5%in patients with microvascular complications. **LDL-C goals:** < 100mg/dL	**Difference in HbA1c from baseline (%)**: −0.08% ±1.15% **Difference in LDL-C from baseline (mg/dL)**: −4.35± 24.54 mg/dL	**Difference in HbA1cfrom baseline (%)**: −0.14% ± 1.51% **Difference in LDL-C from baseline (mg/dL):** −1.70 ± 26.16 mg/dL	**Adjusted HbA1c between-group difference (%)**: 0.12%, 95% CI 0.02 to 0.22. P = 0.02 **Adjusted LDL-Cbetween-group difference (mg/dL):** 3.57 mg/dL 95% CI 1.80 to5.34. P < 0.0001 **Odds of achieving personalized HbA1cgoals:** adjusted odds ratio [aOR] 1.52, 95% CI 1.24 to 1.86; P < 0.0001 **Odds of achieving HbA1c <7%:** 1.56, 95% CI 1.27 to 1.91. P < 0.0001 **Odds of achieving HbA1c ≤9%:** 1.07, 95% CI 0.75 to 1.52; P = 0.71 **Odds of achieving LDL-C < 100mg/dL:** 1.34; 95% CI 1.11 to1.61. P = 0.002	Clinically small, yet statistically significant between-group reductions in HbA1c and LDL-C	⊕⊕⊕◯ MODERATE
Heselmans et al. (2020)[40]	Diabetes	Mean change of HbA1c Mean change in LDL-C levels Mean change inBP	NA	Not reported		**Between-group difference in HbA1c means** -0.09%, 95% CI − 0.18 to 0.01, p-value = 0.06) **Between-group change difference in systolic BP:** 0.13% (95% CI − 0.91 to 1.16) p=0.81 **Between-group change difference in diastolic BP**: 0.12% (95%CI − 1.25 to 1.49) p=0.86 **Between-group change difference in** LDL-C: 1.76% (95%CI − 0.46 to3.98) p=0.12	No significant changes in between-group means for HbA1c, BP, or LDL-C	⊕⊕⊕◯ MODERATE
Eccles et al (2002)[41]	Asthma and angina	**Patient reported outcomes using**: **generic measures**: SF36, EQ-5D **Condition specific measures**: The Seattle Angina Questionnaire, The Newcastle Asthma Symptoms Questionnaire		Not reported			No effect on any patient outcome	⊕⊕⊕◯ MODERATE

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Gill et al. (2009)[42]	Hyperlipidaemia	The percentageof patientswhose most recent low- density lipoprotein. Cholesterol (LDL-C) was at goal (%)	**Defined by the ATP III guidelines** LDL goals: < 100 forhigh-risk < 130 formoderate risk <160 for low risk patients	**High risk:** Baseline: 50% End: 53.3% P-value: <0.001 **Moderate risk**: Baseline: 64.1% End: 64.7% P value: >0.1 **Low risk:** Baseline: 88.5% End: 87.9% P value <0.05	**High risk:** Baseline: 52.3% End:56.1% P value: <0.001 **Moderaterisk**: Baseline:67.1% End: 68.5% P <0.001 **Low risk:** Baseline: 90.2% End: 90.9% P value <0.001	**High risk:** OR = 1.17 (95% CI not reported), p-value <0.05 **Moderate risk:** OR = 0.29 (95% CInot reported), p-value<0.05 **Low risk:** OR =1.74 (95% CI not reported), p-value <0.05	A significant increase in proportion of patients attaining goals for all categories except the moderate-risk intervention group. No significant difference in improvement in anycategory compared to the controlgroup	⊕⊕◯◯ LOW
Cobos et al. (2006)[43]	Hyperlipidaemia	Between- group differences in means of lipid values (mg/dL)	NA	**Total cholesterol:** 233.8 **LDL-C:**149.2 **HDL-C:**58.0 **Triglycerides:** 136.6	**Total cholesterol:** 231.0 **LDL-C:**146.5 **HDL-C:**56.3 **Triglycerides:** 135.2	**Difference, 95% CI, P- value** **Total cholesterol:** -2.8 (–1.7 to –7.3**),** 0.218 **LDL-C:** -2.7 (–1.7 to –7.1), 0.227 **HDL-C:** -1.6 (–0.6 to –3.6), 0.142 **Triglycerides:** -1.4 (–8.3 to –11.2), 0.766	No impact detected on final lipid values	⊕⊕⊕◯ MODERATE
Hicks et al. (2008)[44]	Hypertension	**Absolute effects**: Percentage ofpatients with controlled BP(%) Mean baseline andend SBP and DBP(mmHg) **Relative effects**: Odds of blood pressure control at outcomevisit Baseline and End difference in SBPand DBP (mmHg)	**Controlled BP:** <130/80mmHg for patients with diabetes or renal failure <140/90 mmHg for other patients	48% Baseline SBP: 136 mmHg End SBP: 138 mmHg Baseline DBP: 78 mmHg End DBP: 77 mmHg	45% BaselineSBP:138mmHg End SBP: 137mmHg Baseline DBP: 80 mmHg End DBP: 78 mmHg	**OR**: 0.96 Baseline between-group difference in SBP: 2mmHg (p-value) 0.59 Baseline between-group difference in DBP: 2mmg Hg (p-value): 0.07 End between-group difference in SBP: 0 mm Hg (p-value):0.67 End between-group difference DBP: 1 mmHg (p-value):0.05	No differences between intervention groups in mean systolic or diastolic blood pressure readings at (end)outcome visits	⊕⊕◯◯ LOW
Lopez et al. (2019)[45]	Hypertension	**Relative effects:** odds of blood pressure controlpost-intervention	**Controlled BP:** <140/90 mmHg for all patients	Not reported		**OR**: 1.36 95% CI (1.08 to 1.71) (p-value: 0.009)	Significant odds of improvement of BP in the intervention group compared to usualcare	⊕⊕◯◯ LOW

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Figure 1: Analytic framework
Figure 2: PRISMA flowchart detailing the systematic search results

References

1) Viswanathan M, Golin CE, Jones CD, Ashok M, Blalock SJ, Wines RCM, et al. Interventions to improve adherence to self-administered medications for chronic diseases in the United States: a systematic review. *Annals of Internal Medicine.* 2012; 157 (11): 785-795. doi: 10.7326/0003-4819-157-11-201212040-00538

2) *Health and Economic Costs of Chronic Diseases | CDC.* Available from: https://www.cdc.gov/chronicdisease/about/costs/index.htm

3) Van Oostrom SH, Picavet HSJ, de Bruin SR, et al. Multimorbidity of chronic diseases and health care utilization in general practice. *BMC family practice.* 2014; 15 61. doi: 10.1186/1471-2296-15-61

4) Lehnert T, Heider D, Leicht H et al. Review: health care utilization and costs of elderly persons with multiple chronic conditions. *Medical care research and review: MCRR.* 2011; 68 (4): 387-420. doi: 10.1177/1077558711399580

5) *WHO | Innovative Care for Chronic Conditions: Building Blocks for Action.* Available from: http://www.who.int/chp/knowledge/publications/icccreport/en/

6) Francesconi P, Ballo P, Profili F, et al. Chronic Care Model for the Management of Patients with Heart Failure in Primary Care. *Health Services Insights.* 2019; 12 1178632919866200. Doi: 10.1177/1178632919866200

7) Tellefson M, Dipnarine K, Stopka C. The chronic care model and diabetes management in US primary care settings: a systematic review. *Preventing chronic disease.* 2013; 10 E26. Doi:10.5888/pcd10.120180

8) Oprea L, Cojocaru D, Sandu A, et al. The Chronic Care Model (CCM) and the Social Gradient in Health. *Revista de Cercetare si Interventie Sociala, ISSN-e 1584-5397, Vol 41, 2013, págs 176-189*. 2013

9) Murad MH. Clinical Practice Guidelines: A Primer on Development and Dissemination. *Mayo Clinic Proceedings.* 2017; 92 (3): 423-433. doi: 10.1016/j.mayocp.2017.01.001

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

10) Fürthauer J, Flamm M, Sönnichsen A. Patient and physician related factors of adherence to evidence based guidelines in diabetes mellitus type 2, cardiovascular disease and prevention: a cross sectional study. *BMC family practice.* 2013; 14 47. doi: 10.1186/1471-2296-14-47

11) Nichol MB, Knight TK, Priest JL, et al. Nonadherence to clinical practice guidelines and medications for multiple chronic conditions in a California Medicaid population. *Journal of the American Pharmacists Association: JAPhA.* 2010; 50 (4): 496-507. doi: 10.1331/JAPhA.2010.09123

12) Wasylewicz ATM, Scheepers-Hoeks, Clinical Decision Support Systems. *Fundamentals of Clinical Data Science,* 2018. doi:10.1007/978-3-319-99713-1_11

13) Jao C. Efficient Decision Support Systems: Practice and Challenges in Biomedical Related Domain. BoD–Books on Demand; 2011 Sep 6; doi:10.5772/937

14) Haynes RB, Wilczynski NL. Effects of computerized clinical decision support systems on practitioner performance and patient outcomes: methods of a decision-maker-researcher partnership systematic review. *Implementation science: IS.* 2010; 5 12. doi: 10.1186/1748-5908-5-12

15) Reed TS, David P, Daniel CB, et al. An overview of clinical decision support systems: benefits, risks, and strategies for success. *NPJ Digital Medicine.* 2020; 3 17-17. doi: 10.1038/s41746-020-0221-y

16) Shortliffe EH, Shortliffe EH, Cimino JJ, et al. Biomedical informatics: computer applications in health care and biomedicine. *Springer;* 2014.

17) Kaushal JSA, Alison E, Sarah N, Diane H, et al. Effects of workload, work complexity, and repeated alerts on alert fatigue in a clinical decision support system. *BMC Medical Informatics and Decision Making*. 2017-04-10 2017;17(1):1-9. doi:10.1186/s12911-017-0430-8

18) Shahsavarani AM, Azad Marz Abadi E, Hakimi KalkhoranM,et al. Clinical Decision Support Systems (CDSS): State of the art Review of Literature. *International Journal of Medical Reviews.* 2015; 2 (4): 299-308.

19) Bright TJ, Wong A, Dhurjati R, et al. Effect of clinical decision-support systems: a systematic review. *Annals of Internal Medicine.* 2012; 157 (1): 29-43. doi: 10.7326/0003-4819-157-1-201207030-00450

20) Souza NM, Sebaldt RJ, Mackay JA, et al. Computerized clinical decision support systems for primary preventive care: a decision-maker-researcher partnership systematic review of

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

effects on process of care and patient outcomes.:*Centre for Reviews and Dissemination (UK)*; 2011. doi:10.1186/1748-5908-6-87

21) Garg AX, Adhikari NKJ, McDonald H, et al. Effects of computerized clinical decision support systems on practitioner performance and patient outcomes: a systematic review. *JAMA.* 2005; 293 (10): 1223-1238. doi: 10.1001/jama.293.10.1223

22) Lobach D, Sanders GD, Bright TJ, et al. *Methods.:* Agency for Healthcare Research and Quality (US); 2012.

23) Bryan C, Boren S. The use and effectiveness of electronic clinical decision support tools in the ambulatory/primary care setting: a systematic review of the literature. *Journal of Innovation in Health Informatics.* 2008; 16 (2): 79-91. doi: 10.14236/jhi.v16i2.679

24) Kwan JL, Lo L, Ferguson J, et al. Computerised clinical decision support systems and absolute improvements in care: meta-analysis of controlled clinical trials. bmj. 2020 Sep 17;370.doi:10.1136/bmj.m3216

25) Moher D, Liberati A, Tetzlaff J, et al. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. *PLOS Medicine.* 2009; 6 (7): e1000097. doi: 10.1371/journal.pmed.1000097

26) Moher AL, Douglas GA, Jennifer T, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. 2009-07-21 2009;doi:10.1136/bmj.b2700

27) Howard C. *Subject and Course Guides: Evidence Based Medicine: PICO.* Available from: https://researchguides.uic.edu/c.php?g=252338&p=3954402

28) Zwar RR, Sarah D, Iqbal H, et al. A systematic review of chronic disease management interventions in primary care. OriginalPaper. *BMC Family Practice*. 2018-01-09 2018;19(1):1-13. doi:doi:10.1186/s12875-017-0692-3

29) Jansma EP, Vries RD. Patient reported outcome measures (PROMs*)*.bmi-online search blocks.Available from: https://blocks.bmi-online.nl/catalog/248.

30) Data extraction forms. https://dplp.cochrane.org/data-extraction-forms

31) Lobach D, Sanders GD, Bright TJ, et al. *Data Abstraction Guidance.* : Agency for Healthcare Research and Quality (US); 2012.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

32) Zhao KF, Kelsey Y, Matthew CT,et al. Risk of bias tools in systematic reviews of health interventions: an analysis of PROSPERO-registered protocols. *Systematic Reviews*. 2019-11-15 2019;8(1):1-9. doi:doi:10.1186/s13643-019-1172-8

33) Higgins JP, Altman DG, GøtzschePC,et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *Bmj.* 2011; d5928 (343): doi:10.1136/bmj.d5928

34) Higgins JP, Savović J, Page MJ, Elbers RG, Sterne JA. Assessing risk of bias in a randomized trial. Cochrane Handbook for Systematic Reviews of Interventions. 2019 Sep 23:205-28.doi:10.1136/bmj.d5928

35) Ryan R, Hill S, PrictorM,et al. *Cochrane Consumers & Communication Review Group Study Quality Guide, Guide for Review Authors on Assessing Study Quality.* 2013.

36) *What is GRADE? | BMJ Best Practice.* Available from: https://bestpractice.bmj.com/info/toolkit/learn-ebm/what-is-grade/

37) Meader N, Kristel K, Llewellyn A, et al. A checklist designed to aid consistency and reproducibility of GRADE assessments: development and pilot validation. *Systematic reviews*. 07/24/2014 2014;3doi:10.1186/2046-4053-3-82

38) Schnipper JL, Linder JA, Palchuk MB, et al. Effects of documentation-based decision support on chronic disease management. *The American Journal of Managed Care.* 2010; 16 (12 Suppl HIT) (pp SP72-81): ate of Pubaton: e 2010.

39) Gill J, Kucharski K, Turk B, Pan C, Wei W. Using Electronic Clinical Decision Support in Patient-Centered Medical Homes to Improve Management of Diabetes in Primary Care: The DECIDE Study. *Journal of Ambulatory Care Management.* 2019; 42 (2): 105-115.doi:10.1097/JAC.0000000000000267

40) Heselmans A, Delvaux N, Laenen A, Van de Velde S, Ramaekers D, Kunnamo I, et al. Computerized clinical decision support system for diabetes in primary care does not improve quality of care: a cluster-randomized controlled trial. *Implementation science : IS.* 2020; 15 (1): 5.doi:10.1186/s13012-019-0955-6

41) Eccles M, McColl E, Steen N, Liaw ST. Computerised evidence-based guidelines may not improve asthma or angina management in primary care. *Evidence-Based Healthcare.* 2003; 7 (2): 81-82.

42) Gill JM, Chen YX, Glutting JJ, et al. Impact of decision support in electronic medical records on lipid management in primary care. *Population health management.* 2009; 12 (5): 221-226.doi:10.1089/pop.2009.0003. doi:10.1089/pop.2009.0003

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

43) Cobos A, Vilaseca J, Asenjo C, et al. Cost effectiveness of a clinical decision support system based on the recommendations of the European Society of Cardiology and other societies for the management of hypercholesterolemia: Report of a cluster-randomized trial. *Disease Management and Health Outcomes.* 2005; 13 (6): 421-432.doi:10.2165/00115677-200513060-00007

44) Hicks LS, Sequist TD, Ayanian JZ, et al. Impact of computerized decision support on blood pressure management and control: A randomized controlled trial. *Journal of General Internal Medicine.* 2008; 23 (4): 429-441.doi:10.1007/s11606-007-0403-1

45) Lopez PM, Divney A, Goldfeld K, et al. Feasibility and Outcomes of an Electronic Health Record Intervention to Improve Hypertension Management in Immigrant-serving Primary Care Practices. *Medical care.* 2019; 57 (Supplement 62) (pp S164-S171): ate of Pubaton: 01 Jun 2019.doi:10.1097/MLR.0000000000000994

46) Goud R, de Keizer NF, ter Riet G, et al. Effect of guideline based computerised decision support on decision making of multidisciplinary teams: cluster randomised trial in cardiac rehabilitation. *The BMJ.* 2009; 338.doi: 10.1136/bmj. b1440

47) Manns BJ, Tonelli M, Zhang J, et al. Enrolment in primary care networks: impact on outcomes and processes of care for patients with diabetes. *CMAJ: Canadian Medical Association journal = journal de l'Associationmedicalecanadienne.* 2012; 184 (2): 144.: doi: 10.1503/cmaj.110755

48) McClellan SR, Casalino LP, Shortell SM, et al. When does adoption of health information technology by physician practices lead to use by physicians within the practice? *Journal of the American Medical Informatics Association: JAMIA.* 2013; 20 (e1): e26-e32. doi: 10.1136/amiajnl-2012-001271

49) Latoszek-Berendsen A, Tange H, van den Herik, H. J.,et al. From clinical practice guidelines to computer-interpretable guidelines. A literature overview. *Methods of Information in Medicine.* 2010; 49 (6): 550-570. doi: 10.3414/ME10-01-0056

50) Hajjaj FM, Salek MS, Basra MK,et al. Non-clinical influences on clinical decision-making: a major challenge to evidence-based practice. Journal of the Royal Society of Medicine. 2010 May 1;103(5):178-87.doi:10.1258/jrsm.2010.100104

51) Varghese J, Kleine M, Gessner SI,et al. Effects of computerized decision support system implementations on patient outcomes in inpatient care: a systematic review. *Journal of the American Medical Informatics Association: JAMIA.* 2018; 25 (5): 593-602. doi: 10.1093/jamia/ocx100

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

52) Moja L, Kwag KH, Lytras T, Bertizzolo L, Brandt L, Pecoraro V, et al. Effectiveness of Computerized Decision Support Systems Linked to Electronic Health Records: A Systematic Review and Meta-Analysis. *American Journal of Public Health.* 2014; 104 (12): e12-e22. Available from: doi: 10.2105/AJPH.2014.302164

53) Groenhof TKJ, Asselbergs FW, GroenwoldRHH,et al. The effect of computerized decision support systems on cardiovascular risk factors: a systematic review and meta-analysis. *BMC medical informatics and decision making.* 2019; 19 (1): 108. doi: 10.1186/s12911-019-0824-x

54) Roshanov PS, Misra S, Gerstein HC, et al. Computerized clinical decision support systems for chronic disease management: A decision-maker-researcher partnership systematic review. *Implementation science : IS.* 2011; 6 (1): 92. doi: 10.1186/1748-5908-6-92

55) Kooij L, Groen WG, van Harten WH. The Effectiveness of Information Technology-Supported Shared Care for Patients With Chronic Disease: A Systematic Review. *Journal of Medical Internet Research.* 2017; 19 (6).doi: 10.2196/jmir.7405

56) Castaneda C, Nalley K, Mannion C, et al. Clinical decision support systems for improving diagnostic accuracy and achieving precision medicine. *Journal of Clinical Bioinformatics.* 2015; 5 .doi: 10.1186/s13336-015-0019-3

57) Gibbons RJ, Chatterjee K, Daley J,et al. ACC/AHA/ACP-ASIM guidelines for the management of patients with chronic stable angina: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines (Committee on Management of Patients With Chronic Stable Angina). *Journal of the American College of Cardiology*. 1999 Jun;33(7):2092-197. doi:10.1016/s0735-1097(99)00150-3

58) Heneghan C, Goldacre B, Mahtani KR. Why clinical trial outcomes fail to translate into benefits for patients. *Trials.* 2017 Dec;18(1):1-7. doi:10.1186/s13063-017-1870-2

59) Afzal M, Hussain M, Ali T, et al Knowledge-based query construction using the CDSS knowledge base for efficient evidence retrieval. *Sensors*. 2015 Sep;15(9):21294-314.doi:10.3390/s150921294

1. Kn, M. V, Golin CE, et al. Interventions to improve adherence to self-administered medications for chronic diseases in the United States: a systematic review. *Annals of internal medicine*. 12/04/2012 2012;157(11)doi:10.7326/0003-4819-157-11-201212040-00538
2. Health and Economic Costs of Chronic Diseases | CDC. 2021-01-12T11:38:26Z 2021;

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

3. Ca, H. vOS, Picavet HS, et al. Multimorbidity of chronic diseases and health care utilization in general practice. *BMC family practice*. 04/07/2014 2014;15doi:10.1186/1471-2296-15-61
4. Hh, T. L, Heider D, et al. Review: health care utilization and costs of elderly persons with multiple chronic conditions. *Medical care research and review : MCRR*. 2011 Aug 2011;68(4)doi:10.1177/1077558711399580
5. WHO. *Innovative Care for Chronic Conditions: Building Blocks for Action: Global Report*. 2002. 92 4 159 017 3. https://www.who.int/chp/knowledge/publications/icccglobalreport.pdf?ua=1
6. P. F, Ballo P, Profili F, Policardo L, Roti L, Zuppiroli. Chronic Care Model for the Management of Patients with Heart Failure in Primary Care. *Health services insights*. 08/19/2019 2019;12doi:10.1177/1178632919866200
7. M. S, Dipnarine K, Stopka. The chronic care model and diabetes management in US primary care settings: a systematic review. *Preventing chronic disease*. 2013 2013;10doi:10.5888/pcd10.120180
8. Oprea L, Cojocaru D, Sandu A, Bulgaru-Iliescu D. The Chronic Care Model (CCM) and the Social Gradient in Health. http://purl.org/dc/dcmitype/Text. *Revista de Cercetare si Interventie Sociala, ISSN-e 1584-5397, Vol 41, 2013, págs 176-189*. 2013 2013;doi:https://dialnet.unirioja.es/servlet/dcart?info=link&codigo=5518798&orden=0
9. Mh M. Clinical Practice Guidelines: A Primer on Development and Dissemination. 2017;doi:10.1016/j.mayocp.2017.01.001
10. J. F, Flamm M, Sönnichsen. Patient and physician related factors of adherence to evidence based guidelines in diabetes mellitus type 2, cardiovascular disease and prevention: a cross sectional study. *BMC family practice*. 04/04/2013 2013;14doi:10.1186/1471-2296-14-47
11. Nichol MB, mnichol@usc.edu, Department of Clinical P, et al. Nonadherence to clinical practice guidelines and medications for multiple chronic conditions in a California Medicaid population. *Journal of the American Pharmacists Association : Japha*. 2021;50(4):496-507. doi:10.1331/japha.2010.09123
12. Wasylewicz Atm S-H. Clinical Decision Support Systems. *Fundamentals of Clinical Data Science*. 2018;doi:10.1007/978-3-319-99713-1_11
13. Efficient Decision Support Systems - Practice and Challenges in Biomedical Related Domain. 2011-09-06 2011;doi:10.5772/937
14. B. HR, Wilczynski NL. Effects of computerized clinical decision support systems on practitioner performance and patient outcomes: methods of a decision-maker-researcher partnership systematic review. *Implementation science : IS*. 02/05/2010 2010;5doi:10.1186/1748-5908-5-12
15. Ki, T. SR, Pincock D, et al. An overview of clinical decision support systems: benefits, risks, and strategies for success. *NPJ digital medicine*. 02/06/2020 2020;3doi:10.1038/s41746-020-0221-y
16. *Computer Applications in Health Care and Biomedicine.* . Springer-Verlag; 2014.
17. Kaushal JSA, Alison E, Sarah N, Diane H, Elizabeth M, Rainu. Effects of workload, work complexity, and repeated alerts on alert fatigue in a clinical decision support system. OriginalPaper. *BMC Medical Informatics and Decision Making*. 2017-04-10 2017;17(1):1-9. doi:doi:10.1186/s12911-017-0430-8

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

18. Shahsavarani Am AMAEHKMJSQS. Clinical Decision Support Systems (CDSS): State of the art Review of Literature. *International Journal of Medical Reviews*. 2015;(2 (4)):299-308.
19. J. BT, Wong A, Dhurjati R, et al. Effect of clinical decision-support systems: a systematic review. *Annals of internal medicine*. 07/03/2012 2012;157(1)doi:10.7326/0003-4819-157-1-201207030-00450
20. M. SN, Sebaldt RJ, Mackay JA, et al. Computerized clinical decision support systems for primary preventive care: a decision-maker-researcher partnership systematic review of effects on process of care and patient outcomes. *Implementation science : IS*. 08/03/2011 2011;6doi:10.1186/1748-5908-6-87
21. Rb, X. GA, Adhikari NK, et al. Effects of computerized clinical decision support systems on practitioner performance and patient outcomes: a systematic review. *JAMA*. 03/09/2005 2005;293(10)doi:10.1001/jama.293.10.1223
22. Kendrick DL, Gillian DS, Tiffani JB, et al. *Methods*. Agency for Healthcare Research and Quality (US); 2012.
23. Sa, C. B, Boren. The use and effectiveness of electronic clinical decision support tools in the ambulatory/primary care setting: a systematic review of the literature. *Informatics in primary care*. 2008 2008;16(2)doi:10.14236/jhi.v16i2.679
24. Shojania JLK, Lisha L, Jacob F, et al. Computerised clinical decision support systems and absolute improvements in care: meta-analysis of controlled clinical trials. 2020-09-17 2020;doi:10.1136/bmj.m3216
25. D. M, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS medicine*. 07/21/2009 2009;6(7)doi:10.1371/journal.pmed.1000097
26. Moher AL, Douglas GA, Jennifer T, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. 2009-07-21 2009;doi:10.1136/bmj.b2700
27. Howard C. Subject and Course Guides: Evidence Based Medicine: PICO. University of Illinois at Chicago. Updated Mar 15, 2021. https://researchguides.uic.edu/c.php?g=252338&p=3954402
28. Zwar RR, Sarah D, Iqbal H, et al. A systematic review of chronic disease management interventions in primary care. OriginalPaper. *BMC Family Practice*. 2018-01-09 2018;19(1):1-13. doi:doi:10.1186/s12875-017-0692-3
29. Patient reported outcome measures (PROMs) - bmi-online search blocks. https://blocks.bmi-online.nl/catalog/248
30. Data extraction forms. https://dplp.cochrane.org/data-extraction-forms
31. Kendrick DL, Gillian DS, Tiffani JB, et al. Data Abstraction Guidance. Agency for Healthcare Research and Quality (US); 2012.
32. Zhao KF, Kelsey Y, Matthew CT, Linlu. Risk of bias tools in systematic reviews of health interventions: an analysis of PROSPERO-registered protocols. OriginalPaper. *Systematic Reviews*. 2019-11-15 2019;8(1):1-9. doi:doi:10.1186/s13643-019-1172-8
33. P. HJ, Altman DG, Gøtzsche PC, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ (Clinical research ed)*. 10/18/2011 2011;343doi:10.1136/bmj.d5928
34. Chapter 8: Assessing risk of bias in a randomized trial. 2021.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

35. Ryan R HSPMMJ. *Cochrane Consumers & Communication Review Group Study Quality Guide, Guide for Review Authors on Assessing Study Quality*. 2013. https://cccrg.cochrane.org/sites/cccrg.cochrane.org/files/public/uploads/StudyQualityGuide_May%202013.pdf
36. What is GRADE? | BMJ Best Practice. https://bestpractice.bmj.com/info/toolkit/learn-ebm/what-is-grade/
37. N. M, King K, Llewellyn A, et al. A checklist designed to aid consistency and reproducibility of GRADE assessments: development and pilot validation. *Systematic reviews*. 07/24/2014 2014;3doi:10.1186/2046-4053-3-82
38. L. SJ, Linder JA, Palchuk MB, et al. Effects of documentation-based decision support on chronic disease management. *The American journal of managed care*. 2010 Dec 2010;16(12 Suppl HIT)
39. J. G, Kucharski K, Turk B, Pan C, Wei. Using Electronic Clinical Decision Support in Patient-Centered Medical Homes to Improve Management of Diabetes in Primary Care: The DECIDE Study. *The Journal of ambulatory care management*. Apr/Jun 2019 2019;42(2)doi:10.1097/JAC.0000000000000267
40. Aertgeerts AH, Nicolas D, Annouschka L, et al. Computerized clinical decision support system for diabetes in primary care does not improve quality of care: a cluster-randomized controlled trial. OriginalPaper. *Implementation Science*. 2020-01-07 2020;15(1):1-14. doi:doi:10.1186/s13012-019-0955-6
41. M E, E M, N S, ST L. Computerised evidence-based guidelines may not improve asthma or angina management in primary care. 2003;
42. Mi, M. GJ, Chen YX, Glutting JJ, Diamond JJ, Lieberman. Impact of decision support in electronic medical records on lipid management in primary care. *Population health management*. 2009 Oct 2009;12(5)doi:10.1089/pop.2009.0003
43. Bergoñón AC, Jordi V, César A, et al. Cost Effectiveness of a Clinical Decision Support System Based on the Recommendations of the European Society of Cardiology and Other Societies for the Management of Hypercholesterolemia. OriginalPaper. *Disease Management and Health Outcomes*. 2012-10-07 2005;13(6):421-432. doi:doi:10.2165/00115677-200513060-00007
44. Dw, S. HL, Sequist TD, et al. Impact of computerized decision support on blood pressure management and control: a randomized controlled trial. *Journal of general internal medicine*. 2008 Apr 2008;23(4)doi:10.1007/s11606-007-0403-1
45. M. LP, Divney A, Goldfeld K, et al. Feasibility and Outcomes of an Electronic Health Record Intervention to Improve Hypertension Management in Immigrant-serving Primary Care Practices. *Medical care*. 2019 Jun 2019;57 Suppl 6 Suppl 2(Suppl 6 2)doi:10.1097/MLR.0000000000000994
46. R. G, de Keizer NF, ter Riet G, et al. Effect of guideline based computerised decision support on decision making of multidisciplinary teams: cluster randomised trial in cardiac rehabilitation. *BMJ (Clinical research ed)*. 04/27/2009 2009;338doi:10.1136/bmj.b1440
47. Br, J. MB, Tonelli M, et al. Enrolment in primary care networks: impact on outcomes and processes of care for patients with diabetes. *CMAJ : Canadian Medical Association journal = journal de l'Association medicale canadienne*. 02/07/2012 2012;184(2)doi:10.1503/cmaj.110755

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

48. Dr, R. MS, Casalino LP, Shortell SM, Rittenhouse. When does adoption of health information technology by physician practices lead to use by physicians within the practice? *Journal of the American Medical Informatics Association : JAMIA*. 2013 Jun 2013;20(e1)doi:10.1136/amiajnl-2012-001271

49. A. L-B, Tange H, van den Herik HJ, Hasman. From clinical practice guidelines to computer-interpretable guidelines. A literature overview. *Methods of information in medicine*. 2010 2010;49(6)doi:10.3414/ME10-01-0056

50. Ay, M. HF, Salek MS, Basra MK, Finlay. Non-clinical influences on clinical decision-making: a major challenge to evidence-based practice. *Journal of the Royal Society of Medicine*. 2010 May 2010;103(5)doi:10.1258/jrsm.2010.100104

51. J. V, Kleine M, Gessner SI, Sandmann S, Dugas. Effects of computerized decision support system implementations on patient outcomes in inpatient care: a systematic review. *Journal of the American Medical Informatics Association : JAMIA*. 05/01/2018 2018;25(5)doi:10.1093/jamia/ocx100

52. L. M, Kwag KH, Lytras T, et al. Effectiveness of computerized decision support systems linked to electronic health records: a systematic review and meta-analysis. *American journal of public health*. 2014 Dec 2014;104(12)doi:10.2105/AJPH.2014.302164

53. J. GTK, Asselbergs FW, Groenwold RHH, Grobbee DE, Visseren FLJ, Bots ML. The effect of computerized decision support systems on cardiovascular risk factors: a systematic review and meta-analysis. *BMC medical informatics and decision making*. 06/10/2019 2019;19(1)doi:10.1186/s12911-019-0824-x

54. S. RP, Misra S, Gerstein HC, et al. Computerized clinical decision support systems for chronic disease management: a decision-maker-researcher partnership systematic review. *Implementation science : IS*. 08/03/2011 2011;6doi:10.1186/1748-5908-6-92

55. Wh, L. K, Groen WG, van H. The Effectiveness of Information Technology-Supported Shared Care for Patients With Chronic Disease: A Systematic Review. *Journal of medical Internet research*. 06/22/2017 2017;19(6)doi:10.2196/jmir.7405

56. Suh CC, Kip N, Ciaran M, et al. Clinical decision support systems for improving diagnostic accuracy and achieving precision medicine. OriginalPaper. *Journal of Clinical Bioinformatics*. 2015-03-26 2015;5(1):1-16. doi:doi:10.1186/s13336-015-0019-3

57. Ka, Gardner TJ, Garson A, et al. ACC/AHA/ACP-ASIM guidelines for the management of patients with chronic stable angina: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines (Committee on Management of Patients With Chronic Stable Angina). *Journal of the American College of Cardiology*. 1999 Jun 1999;33(7)doi:10.1016/s0735-1097(99)00150-3

58. Mahtani CH, Ben G, Kamal R. Why clinical trial outcomes fail to translate into benefits for patients. Letter. *Trials*. 2017-03-14 2017;18(1):1-7. doi:doi:10.1186/s13063-017-1870-2

59. Kang MA, Maqbool H, Taqdir A, et al. Knowledge-Based Query Construction Using the CDSS Knowledge Base for Efficient Evidence Retrieval. Article. *Sensors*. 2015-08-28 2015;15(9):21294-21314. doi:10.3390/s150921294

For peer review only

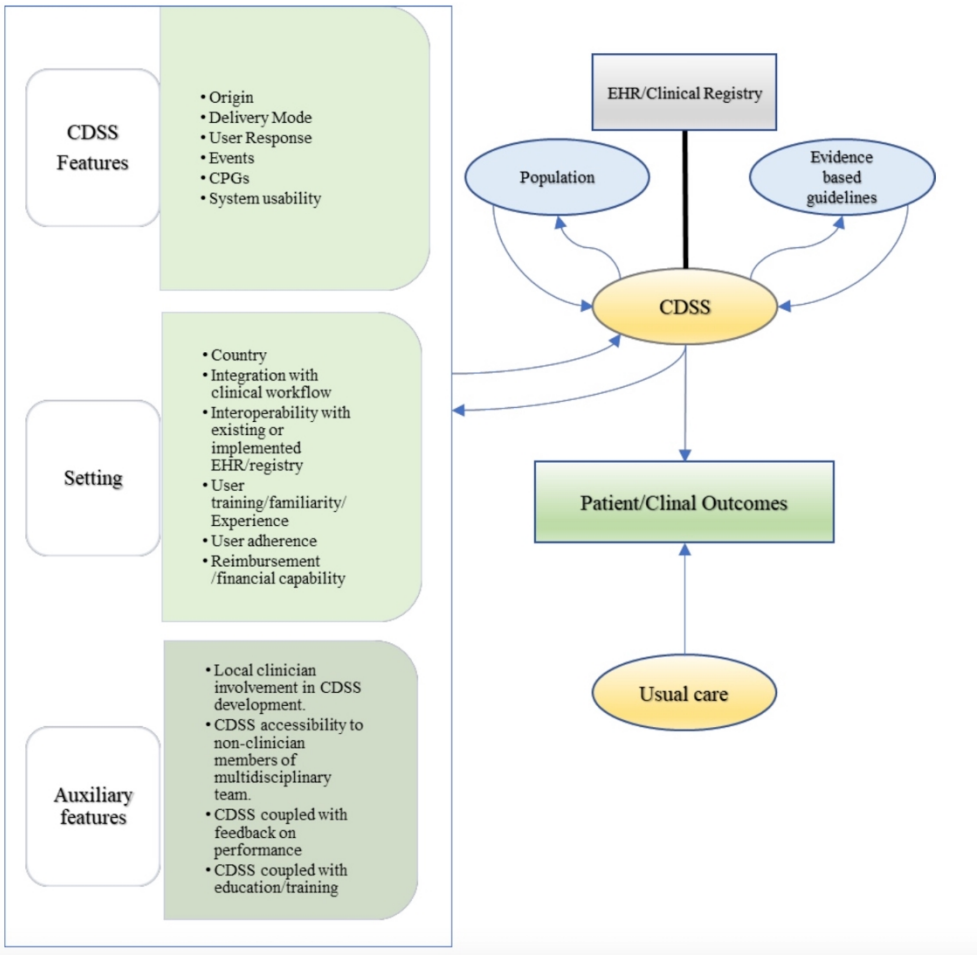


Figure 1: Analytic framework

488x471mm (300 x 300 DPI)

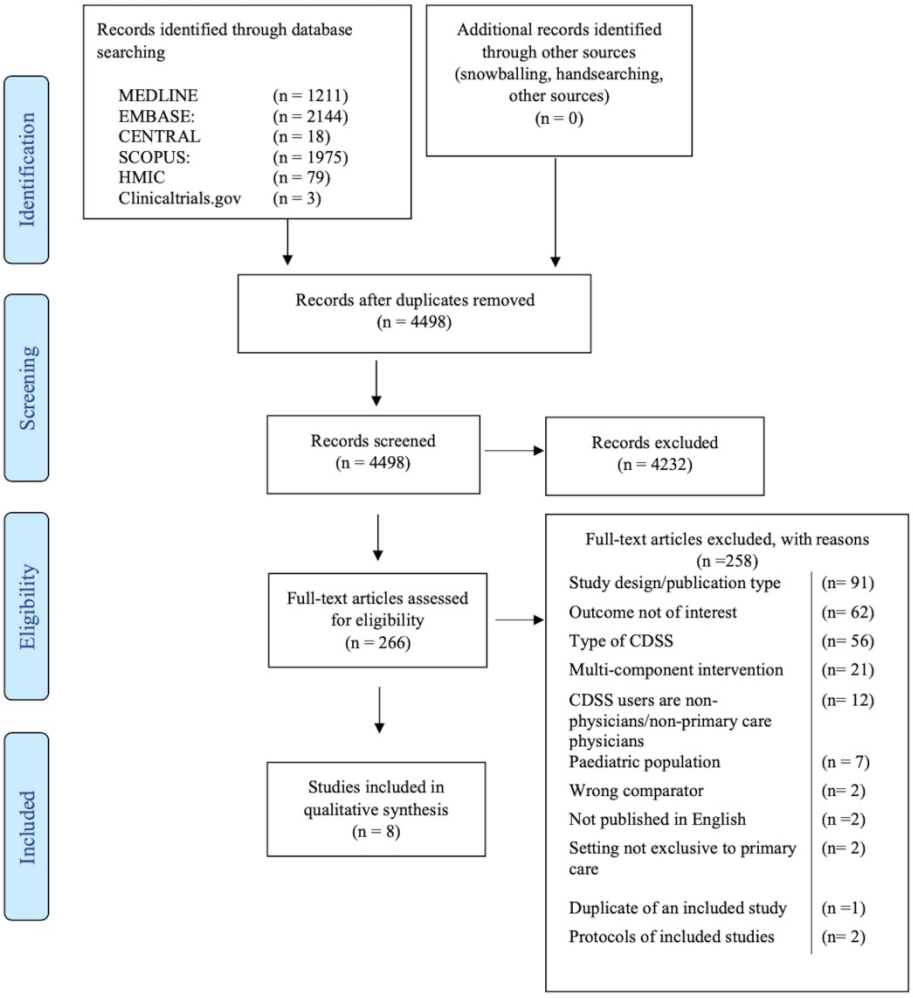


Figure 2: PRISMA flowchart detailing the systematic search results

405x436mm (300 x 300 DPI)

Supplementary Material

Supplementary Table 1: Eligibility criteria defined by PICOS

P	**Participants:** - Patients above 18 years with one or more concurrent chronic condition, regardless of sex, education, marital status, employment status, income, or other socioeconomic factors who were managed in a primary care setting using a EHR or clinical registry) supported by a computerised, knowledge-based CDSS. - Chronic conditions were defined and limited to the following conditions: o Asthma, chronic obstructive pulmonary disease (COPD), heart failure, myocardial ischemia, hypertension, type II diabetes mellitus, hyperlipidaemia, arthritis, and osteoporosis.
I	**Interventions:** Computerised knowledge based CDSS embedded EHRs or clinical registries in primary care settings as an intervention with the following inclusion and exclusion criteria: **Studies will be included if:** - They focus on evaluating a CDSS for medical management of the selected chronic diseases - The CDSS is of knowledge-based type (informed by CPGs). - Included CDSS is synchronous, meaning alerts are electronically presented to the clinician in real-time during the physician-patient encounter. - The CDSS users are primary care physicians (PCPs). - They report at least: one quantifiable patient clinical outcome/patient-related measure in adult populations in the specified chronic disease entities **Studies will be excluded if**: - Not published in English. - Not conducted in primary care settings. - Include care provided by subspecialized physicians (not PCPs). - Evaluates a CDSS for other outcome measures such as system usability, or other quality outcomes, not including patient or clinical outcomes. - Interventions that are used for other direct care purposes other than chronic disease management or non-direct care purposes, such as research purposes. - Evaluate the effect of a CDSS in other diseases not specified in the inclusion criteria. - Include a CDSS that does not produce real-time recommendations based on CPGs (non-knowledge based CDSS). - Web-based CDSS, paper-based CDSS, standalone CDSS (not integrated with EHRs), CDSS integrated with EHR but not producing electronic alerts, and other CDSS that use e-mail, phone, or pagers to notify the user. - Studies without a comparator group - Comparator group is another type of CDSS.
C	**Comparators:** Usual care not utilising CDSS
O	**Outcomes:** - Patient-related health and clinical outcomes in chronic disease management (such as HBA1C, $FEV_{1,}$ or other biomarkers of chronic diseases), morbidity, mortality, HRQoL, preventable co-morbidities, or disease activity) - Practitioner-related, patient-related, and cost-effectiveness outcomes were not included.
S	**Study types**: Randomised controlled trials (RCTs), cluster RCTs, quasi-RCTs, interrupted time series (ITS) and controlled before-and-after (CBA) studies

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Table 2: List of search terms

CDSS terms	Dashboard*, registry, clinical decision support system*, medical order entry system*, computeri?ed physician order entry system*, CPOE, order set*, computer assisted decision making, (computer$ adj3 decision$), knowledge management system*, case-finding algorithm*, advisory decision support, (computer* adj2 therap*)

AND

General Practice/Primary care	general practi*, family practice, family clinic*, family practi*, family physician*, family doctor*, primary care physician*, primary care clinic*, primary care practi*,

OR each of the below individual entities combined with AND

Chronic Disease Entities	
Asthma	asthma*, antiasthma*, wheez*, (bronch* adj3 spasm*), bronchoconstrict*, (bronch$ adj3 constrict$), ((bronchial$ or respiratory or airway$ or lung$) adj3 (hypersensitiv$ or hyperreactiv$ or allerg$ or insufficiency)),
Heart Failure/ Myocardial Ischemia	(heart* adj2 decomp*), ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*)), ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*))
Hypertension	((elevat$ or increas$ or lower or high or rais$ or rising) adj2 (arterial pressure or blood pressure or diastolic pressure or systolic pressure)), hypertens$ or prehypertens$),
Diabetes Mellitus	(IDDM or T2DM or T2D), (insulin* depend* or insulin?depend*)., (non insulin* depend* or noninsulin* depend* or non insulin?depend* or noninsulin?depend*)., diabet*.
Hyperlipidemia	LDL, (dyslipid?emia* or lipoprotein* or dyslipidprotein?emia* or hyperlipoprotein?emia), (hyperlipid?emia* or cholesterol* or lip?emia* or hypercholesterol?emia* or hypercholester?emia* or hyperlip?emia* or triglycerid* or hypertriglycerid?emia*).,
COPD	(COPD or COAD or COBD or AECB), ((obstruct$ adj3 lung$) or (obstruct$ adj 3 airway$)), (obstruct$ adj3 pulmonary), (chronic* adj3 bronchiti*), emphysema*.,
Arthritis	Osteoarthritis, undifferentiated oligoarthritis, ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$))., (joint pain or chronic joint symptoms or gonarthrosis or osteoarthrosis or astoarthrosis or degeneretive arthritis or joint diseases or arthralgia),
Osteoporosis	Densitometry, osteopenia, bone mass, bone densit*, osteoporos?s, (osteoporosis or osteopenia or (bone and (density or mass or loss)) or hepatic osteodystrophy),

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Table 3 - Full Search Strategy

Full Medline Search Strategy

1 Dashboard*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1228)
2 registry.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (109312)
3 registries.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (104552)
4 exp Registries/ (95452)
5 1 or 2 or 3 or 4 (173943)
6 exp Decision Support Systems, Clinical/ (7925)
7 clinical decision support system*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (2132)
8 medical order entry system*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (2297)
9 computeri?ed physician order entry system*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (224)
10 CPOE.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1113)
11 order set*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1009)
12 exp Decision Making, Computer-Assisted/ (144386)
13 computer assisted decision making.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (76)
14 (computer$ adj3 decision$).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (4723)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

15 knowledge management system*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (109)

16 case-finding algorithm*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (84)

17 advisory decision support.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1)

18 exp Therapy, Computer-Assisted/ (63908)

19 (computer* adj2 therap*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (8758)

20 exp Reminder Systems/ (3443)

21 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 (159108)

22 5 or 21 (331941)

23 exp Asthma/ (127501)

24 asthma*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (179286)

25 antiasthma*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1458)

26 exp Respiratory Sounds/ (12994)

27 wheez*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (13724)

28 exp Bronchial Spasm/ (4303)

29 (bronch* adj3 spasm*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (4523)

30 exp Bronchoconstriction/ (4131)

31 bronchoconstrict*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (11491)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

32 (bronch$ adj3 constrict$).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (715)

33 exp Bronchial Hyperreactivity/ (7348)

34 exp Respiratory Hypersensitivity/ (154474)

35 ((bronchial$ or respiratory or airway$ or lung$) adj3 (hypersensitiv$ or hyperreactiv$ or allerg$ or insufficiency)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (66254)

36 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 (261794)

37 exp Heart Failure/ (120835)

38 (heart* adj2 decomp*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (4261)

39 ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (664015)

40 ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (664015)

41 exp Myocardial Ischemia/ (428216)

42 exp Coronary Disease/ (216812)

43 37 or 38 or 39 or 40 or 41 or 42 (842183)

44 exp Hypertension/ (253393)

45 exp Essential Hypertension/ (2255)

46 (hypertens$ or prehypertens$).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (510526)

47 ((elevat$ or increas$ or lower or high or rais$ or rising) adj2 (arterial pressure or blood pressure or diastolic pressure or systolic pressure)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (63244)

48 44 or 45 or 46 or 47 (537839)

49 exp Diabetes Mellitus/ (423542)

For peer review only

50 diabet*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (693674)

51 (non insulin* depend* or noninsulin* depend* or non insulin?depend* or noninsulin?depend*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (12372)

52 (insulin* depend* or insulin?depend*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (29531)

53 (IDDM or T2DM or T2D).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (37329)

54 exp Diabetes Mellitus, Type 2/ (132230)

55 49 or 50 or 51 or 52 or 53 or 54 (697363)

56 exp Hyperlipidemias/ (65918)

57 exp Dyslipidemias/ (79414)

58 exp Cholesterol/ (159670)

59 (hyperlipid?emia* or cholesterol* or lip?emia* or hypercholesterol?emia* or hypercholester?emia* or hyperlip?emia* or triglycerid* or hypertriglycerid?emia*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (389543)

60 (dyslipid?emia* or lipoprotein* or dyslipidprotein?emia* or hyperlipoprotein?emia*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (209412)

61 LDL.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (101128)

62 56 or 57 or 58 or 59 or 60 or 61 (479855)

63 exp Pulmonary Disease, Chronic Obstructive/ (55377)

64 exp Lung Diseases, Obstructive/ (210481)

65 emphysema*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (35893)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

66 (chronic* adj3 bronchiti*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (11493)

67 (obstruct$ adj3 pulmonary).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (67392)

68 ((obstruct$ adj3 lung$) or (obstruct$ adj 3 airway$)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (24368)

69 (COPD or COAD or COBD or AECB).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (46979)

70 63 or 64 or 65 or 66 or 67 or 68 or 69 (256014)

71 exp Arthritis/ (258204)

72 exp Osteoarthritis/ (62805)

73 osteoarthritis.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (88630)

74 exp Joint Diseases/ (378029)

75 exp Arthralgia/ (13069)

76 (joint pain or chronic joint symptoms or gonarthrosis or osteoarthrosis or astoarthrosis or degeneretive arthritis or joint diseases or arthralgia).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (49768)

77 ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (176792)

78 undifferentiated oligoarthritis.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (36)

79 71 or 72 or 73 or 74 or 75 or 76 or 77 or 78 (466351)

80 exp Osteoporosis/ (55499)

81 (osteoporosis or osteopenia or (bone and (density or mass or loss)) or hepatic osteodystrophy).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (229156)

82 osteoporos?s.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (86861)

83 exp Bone Density/ (52984)

84 bone densit*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (70284)

85 exp Bone Demineralization, Pathologic/ (652)

86 exp Bone Diseases, Metabolic/ (76180)

87 bone mass.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (18851)

88 osteopenia.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (9555)

89 exp Densitometry/ (34775)

90 densitometry.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (20649)

91 80 or 81 or 82 or 83 or 84 or 85 or 86 or 87 or 88 or 89 or 90 (266936)

92 exp General Practitioners/ (7841)

93 exp General Practice/ (74873)

94 general practi*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (91362)

95 exp Physicians, Family/ (16348)

96 exp Family Practice/ (65161)

97 family practice.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (68324)

98 family clinic*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (470)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

99 family physician*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (14677)

100 family practi*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (69798)

101 family doctor*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (4649)

102 primary care physician*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (19694)

103 primary care clinic*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (7532)

104 exp Physicians, Primary Care/ (3421)

105 primary care practi*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (7576)

106 92 or 93 or 94 or 95 or 96 or 97 or 98 or 99 or 100 or 101 or 102 or 103 or 104 or 105 (180127)

107 22 and 106 (4326)

108 91 and 107 (50)

109 79 and 107 (76)

110 70 and 107 (184)

111 62 and 107 (206)

112 55 and 107 (568)

113 48 and 107 (239)

114 43 and 107 (343)

115 36 and 107 (148)

116 108 or 109 or 110 or 111 or 112 or 113 or 114 or 115 (1211)

117 from 116 keep 1-1211 (1211)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Full Embase Search Strategy

1 Dashboard*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (2390)

2 registry.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (208322)

3 registries.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (39715)

4 exp register/ (115531)

5 1 or 2 or 3 or 4 (266586)

6 exp clinical decision support system/ (3039)

7 clinical decision support system*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (4672)

8 medical order entry system*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (136)

9 computeri?ed physician order entry system*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (362)

10 CPOE.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (1664)

11 order set*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (2621)

12 exp decision support system/ (25185)

13 computer assisted decision making.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (116)

14 (computer$ adj3 decision$).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (2950)

15 knowledge management system*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (141)

16 case-finding algorithm*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (141)

For peer review only

17 advisory decision support.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (1)
18 exp computer assisted therapy/ (15163)
19 (computer* adj2 therap*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (5331)
20 exp reminder system/ (2611)
21 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 (48813)
22 5 or 21 (314449)
23 exp asthma/ (275547)
24 asthma*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (309894)
25 antiasthma*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (9548)
26 exp abnormal respiratory sound/ (60406)
27 wheez*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (33727)
28 exp bronchospasm/ (27974)
29 (bronch* adj3 spasm*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (861)
30 exp bronchoconstriction/ (29996)
31 bronchoconstrict*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (12976)
32 (bronch$ adj3 constrict$).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (1187)
33 exp bronchus hyperreactivity/ (12410)
34 exp respiratory tract allergy/ (316133)
35 ((bronchial$ or respiratory or airway$ or lung$) adj3 (hypersensitiv$ or hyperreactiv$ or allerg$ or insufficiency)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (55027)
36 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 (435382)
37 22 and 36 (3673)
38 exp heart failure/ (531445)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

39 (heart* adj2 decomp*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (9079)

40 ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (1152207)

41 exp heart muscle ischemia/ (94337)

42 exp coronary artery disease/ (340285)

43 38 or 39 or 40 or 41 or 42 (1486256)

44 22 and 43 (46272)

45 exp hypertension/ (772502)

46 exp essential hypertension/ (30699)

47 (hypertens$ or prehypertens$).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (988948)

48 ((elevat$ or increas$ or lower or high or rais$ or rising) adj2 (arterial pressure or blood pressure or diastolic pressure or systolic pressure)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (92975)

49 45 or 46 or 47 or 48 (1084030)

50 22 and 49 (18606)

51 exp diabetes mellitus/ (1000793)

52 diabet*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (1186708)

53 (non insulin* depend* or noninsulin* depend* or non insulin?depend* or noninsulin?depend*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (257419)

54 (insulin* depend* or insulin?depend*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (355610)

55 (IDDM or T2DM or T2D).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (63858)

56 51 or 52 or 53 or 54 or 55 (1195069)

57 22 and 56 (26835)

58 exp hyperlipidemia/ (164564)

59 exp dyslipidemia/ (70815)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

60 exp cholesterol/ (335599)

61 (hyperlipid?emia* or cholesterol* or lip?emia* or hypercholesterol?emia* or hypercholester?emia* or hyperlip?emia* or triglycerid* or hypertriglycerid?emia*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (619049)

62 (dyslipid?emia* or lipoprotein* or dyslipidprotein?emia* or hyperlipoprotein?emia*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (401478)

63 LDL.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (117119)

64 58 or 59 or 60 or 61 or 62 or 63 (764638)

65 22 and 64 (8600)

66 exp chronic obstructive lung disease/ (134392)

67 emphysema*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (58780)

68 (chronic* adj3 bronchiti*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (23183)

69 (obstruct$ adj3 pulmonary).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (82519)

70 ((obstruct$ adj 3 lung$) or (obstruct$ adj 3 airway$)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (1)

71 (COPD or COAD or COBD or AECB).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (90187)

72 66 or 67 or 68 or 69 or 70 or 71 (226781)

73 22 and 72 (3785)

74 exp arthritis/ (511276)

75 exp osteoarthritis/ (137512)

76 exp osteoarthritis/ (137512)

77 osteoarthritis.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (149492)

78 exp arthropathy/ (816046)

79 exp arthralgia/ (61579)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

80 (joint pain or chronic joint symptoms or gonarthrosis or osteoarthrosis or astoarthrosis or degeneretive arthritis or joint diseases or arthralgia).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (79510)

81 ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (304665)

82 ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (304665)

83 undifferentiated oligoarthritis.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (39)

84 74 or 75 or 76 or 77 or 78 or 79 or 80 or 81 or 82 or 83 (897748)

85 22 and 84 (13224)

86 exp osteoporosis/ (134965)

87 (osteoporosis or osteopenia or (bone and (density or mass or loss)) or hepatic osteodystrophy).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (383399)

88 osteoporos?s.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (160054)

89 exp bone density/ (93700)

90 bone densit*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (108740)

91 exp bone demineralization/ (137263)

92 bone mass.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (38017)

93 osteopenia.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (23760)

94 exp bone densitometry/ (5807)

95 densitometry.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (34380)

96 86 or 87 or 88 or 89 or 90 or 91 or 92 or 93 or 94 or 95 (411752)

97 22 and 96 (3018)

98 exp general practitioner/ (102749)

99 exp general practice/ (84760)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

100 general practi*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (208636)
101 family practice.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (9954)
102 family clinic*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (652)
103 family practi*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (12786)
104 family physician.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (6553)
105 family doctor.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (4317)
106 98 or 99 or 100 or 101 or 102 or 103 or 104 or 105 (216919)
107 97 and 106 (71)
108 85 and 106 (192)
109 73 and 106 (173)
110 65 and 106 (271)
111 57 and 106 (738)
112 50 and 106 (341)
113 44 and 106 (612)
114 37 and 106 (180)
115 107 or 108 or 109 or 110 or 111 or 112 or 113 or 114 (1655)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Full Health Management Information Consortium Search Strategy

1. Dashboard*.mp. [mp=title, other title, abstract, heading words]
2. registry.mp. [mp=title, other title, abstract, heading words]
3. registries.mp. [mp=title, other title, abstract, heading words]
4. exp Registers/ or exp Medical registers/ or exp Patient registers/ or exp Disease registers/
5. 1 or 2 or 3 or 4
6. exp Information systems/ or exp Decision support systems/
7. clinical decision support system*.mp. [mp=title, other title, abstract, heading words]
8. medical order entry system*.mp. [mp=title, other title, abstract, heading words]
9. computeri?ed physician order entry system*.mp. [mp=title, other title, abstract, heading words]
10. CPOE.mp. [mp=title, other title, abstract, heading words]
11. order set*.mp. [mp=title, other title, abstract, heading words]
12. computer assisted decision making.mp. [mp=title, other title, abstract, heading words]
13. (computer$ adj3 decision$).mp. [mp=title, other title, abstract, heading words]
14. knowledge management system*.mp. [mp=title, other title, abstract, heading words]
15. case-finding algorithm*.mp. [mp=title, other title, abstract, heading words]
16. advisory decision support.mp. [mp=title, other title, abstract, heading words]
17. exp Computer systems/
18. (computer* adj2 therap*).mp. [mp=title, other title, abstract, heading words]
19. 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18
20. 5 or 19
21. exp Asthma/
22. asthma*.mp. [mp=title, other title, abstract, heading words]
23. antiasthma*.mp. [mp=title, other title, abstract, heading words]
24. exp Bronchial spasm/ or exp Lung function/
25. wheez*.mp. [mp=title, other title, abstract, heading words]
26. exp Bronchial spasm/
27. (bronch* adj3 spasm*).mp. [mp=title, other title, abstract, heading words]
28. bronchoconstrict*.mp. [mp=title, other title, abstract, heading words]
29. (bronch$ adj3 constrict$).mp. [mp=title, other title, abstract, heading words]
30. ((bronchial$ or respiratory or airway$ or lung$) adj3 (hypersensitiv$ or hyperreactiv$ or allerg$ or insufficiency)).mp. [mp=title, other title, abstract, heading words]
31. 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30
32. exp Heart diseases/ or exp Coronary diseases/ or exp Congestive heart failure/
33. (heart* adj2 decomp*).mp. [mp=title, other title, abstract, heading words]

For peer review only

34. ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*)).mp. [mp=title, other title, abstract, heading words]
35. exp Cardiovascular diseases/
36. exp Heart diseases/ or exp Heart arrest/ or exp Myocardial infarction/
37. 32 or 33 or 34 or 35 or 36
38. exp Hypertension/
39. (hypertens$ or prehypertens$).mp. [mp=title, other title, abstract, heading words]
40. ((elevat$ or increas$ or lower or high or rais$ or rising) adj2 (arterial pressure or blood pressure or diastolic pressure or systolic pressure)).mp. [mp=title, other title, abstract, heading words]
41. exp Diabetes/
42. diabet*.mp. [mp=title, other title, abstract, heading words]
43. (non insulin* depend* or noninsulin* depend* or non insulin?depend* or noninsulin?depend*).mp. [mp=title, other title, abstract, heading words]
44. (insulin* depend* or insulin?depend*).mp. [mp=title, other title, abstract, heading words]
45. (IDDM or T2DM or T2D).mp. [mp=title, other title, abstract, heading words]
46. 41 or 42 or 43 or 44 or 45
47. exp Hyperlipidaemia/
48. exp Cholesterol measurement/ or exp Cholesterol/
49. (hyperlipid?emia* or cholesterol* or lip?emia* or hypercholesterol?emia* or hypercholester?emia* or hyperlip?emia* or triglycerid* or hypertriglycerid?emia*).mp. [mp=title, other title, abstract, heading words]
50. (dyslipid?emia* or lipoprotein* or dyslipidprotein?emia* or hyperlipoprotein?emia*).mp. [mp=title, other title, abstract, heading words]
51. LDL.mp. [mp=title, other title, abstract, heading words]
52. 47 or 48 or 49 or 50 or 51
53. exp chronic obstructive pulmonary disease/
54. emphysema*.mp. [mp=title, other title, abstract, heading words]
55. (chronic* adj3 bronchiti*).mp. [mp=title, other title, abstract, heading words]
56. (obstruct$ adj3 pulmonary).mp. [mp=title, other title, abstract, heading words]
57. ((obstruct$ adj3 lung$) or (obstruct$ adj 3 airway$)).mp. [mp=title, other title, abstract, heading words]
58. (COPD or COAD or COBD or AECB).mp. [mp=title, other title, abstract, heading words]
59. 53 or 54 or 55 or 56 or 57 or 58
60. exp Arthritis/
61. exp Osteoarthritis/

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

62. Arthritis/ or exp Rheumatoid arthritis/
63. exp Joint diseases/ or exp Bone diseases/
64. (joint pain or chronic joint symptoms or gonarthrosis or osteoarthrosis or astoarthrosis or degeneretive arthritis or joint diseases or arthralgia).mp. [mp=title, other title, abstract, heading words]
65. ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$)).mp. [mp=title, other title, abstract, heading words]
66. ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$)).mp. [mp=title, other title, abstract, heading words]
67. undifferentiated oligoarthritis.mp. [mp=title, other title, abstract, heading words]
68. 60 or 61 or 62 or 63 or 64 or 65 or 66 or 67
69. exp Osteoporosis/
70. (osteoporosis or osteopenia or (bone and (density or mass or loss)) or hepatic osteodystrophy).mp. [mp=title, other title, abstract, heading words]
71. osteoporos?s.mp. [mp=title, other title, abstract, heading words]
72. bone densit*.mp. [mp=title, other title, abstract, heading words]
73. bone mass.mp. [mp=title, other title, abstract, heading words]
74. osteopenia.mp. [mp=title, other title, abstract, heading words]
75. densitometry.mp. [mp=title, other title, abstract, heading words]
76. 69 or 70 or 71 or 72 or 73 or 74 or 75
77. exp General practitioners/
78. exp General practice/
79. general practi*.mp. [mp=title, other title, abstract, heading words]
80. family clinic*.mp. [mp=title, other title, abstract, heading words]
81. family practi*.mp. [mp=title, other title, abstract, heading words]
82. family physician*.mp. [mp=title, other title, abstract, heading words]
83. family doctor*.mp. [mp=title, other title, abstract, heading words]
84. primary care physician*.mp. [mp=title, other title, abstract, heading words]
85. primary care clinic*.mp. [mp=title, other title, abstract, heading words]
86. primary care practi*.mp. [mp=title, other title, abstract, heading words]
87. 77 or 78 or 79 or 80 or 81 or 82 or 83 or 84 or 85 or 86
88. 20 and 87
89. 31 and 88

For peer review only

90. 37 and 88
91. 38 or 39 or 40
92. 88 and 91
93. 46 and 88
94. 52 and 88
95. 59 and 88
96. 68 and 88
97. 76 and 88
98. 89 or 90 or 92 or 93 or 94 or 95 or 96 or 97

For peer review only

Full Scopus Search Strategy

(("Dashboard" OR "registry" OR "registries" OR "clinical decision support system" OR "medical order entry system" OR "CPOE" OR "decision support system" OR "computer assisted decision making" OR "case-finding algorithm" OR "advisory decision support" OR "computer assisted therapy") AND ("general practi*" OR "family practi*" OR "family clinic*" OR "family physician" OR "family doctor" OR "primary care clinic*" OR " primary care physician")) AND (("asthma*" "wheez*" OR "bronchus hyperreactivity" OR "heart failure" OR "heart insufficiency" OR "cardiac failure" OR " cardiac insufficiency" OR "myocardial infarction" OR "myocardial insufficiency" OR "myocardial ischemia" OR "cardiac disease" OR "coronary artery disease") OR ("diabet*" OR ("non insulin* depend*" OR "noninsulin* depend*" OR "non insulin depend*" OR "noninsulin?depend*") OR "IDDM" OR "T2DM" OR "T2D") OR ("hyperlipidemia" OR "dyslipidemia" OR "hyperlipid?emia*" OR "cholesterol*" OR "lip?emia*" OR "hypercholesterol?emia*" OR "hypercholester?emia*" OR "hyperlip?emia*" OR "triglycerid*" OR "hypertriglycerid?emia*" OR "dyslipid?emia*" OR "lipoprotein*" OR "dyslipidprotein?emia*" OR "hyperlipoprotein?emia*" OR "chronic obstructive lung disease" OR "emphysema*" OR (("obstruct* N/3lung*") OR ("obstruct* N/3 airway*")) OR copd OR coad OR cobd OR aecb) OR ("hypertension" OR "Prehypertension" OR (("elevat* OR increas* OR lower OR high OR rais* OR rising") W/2 ("arterial pressure" OR "blood pressure" OR "diastolic pressure" OR "systolic pressure"))) OR ("arthritis" OR "osteoarthritis" OR "joint pain" OR "chronic joint symptoms" OR "gonarthrosis" OR "osteoarthrosis" OR "arthrosis" OR "degeneretive arthritis" OR "joint diseases" OR "arthralgia" OR "rheumat*" OR "reumat*" OR "arthrit*" OR "artrit*") OR ("osteoporosis" OR "osteopenia" OR bone AND density AND loss OR "osteopenia" OR "bone mass loss")) AND NOT INDEX (medline) AND (LIMIT-TO (SRCTYPE , "j")) AND (LIMIT-TO (DOCTYPE , "ar")) AND (LIMIT-TO (SUBJAREA , "MEDI")) AND (LIMIT-TO (LANGUAGE , "English"))

For peer review only

Supplementary Table 4: Risk of bias in included studies as per the Cochrane tool

Trial	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data	Selective outcome reporting	Other bias	Selective recruitment of cluster members
Shnipper et al.(2010)	✓	✗	!	!	!	✓	✓	NA
Gill et al. (2017)	✗	✓	✓	✓	✗	✗	✓	✓
Hesselman et al.(2020)	✓	✓	✓	✓	✗	✓	✓	✓
Eccles et al. (2002)	✓	✓	✓	✓	✗	✗	✓	✓
Gill et al. (2009)	✗	✓	!	!	✗	✓	✓	✓
Cobos et al. (2006)	✓	✓	!	!	✓	✓	✓	✓
Hicks et al. (2008)	✗	✓	!	!	✗	✓	✓	✓
Lopez et al. (2019)	✗	✗	✗	✗	✓	✓	✓	✓

Key
✓ Low risk
✗ High risk
! Unclear

For peer review only

Supplementary Table 5: Adapted risk of bias table detailing methodological risks and characteristics. PCPs refers to primary care practitioners (clinicians)

	Randomization				Baseline Comparability				Blinding			Withdrawal		
	Truly Random	Allocation Concealment	Number Stated		Presented	Achieved	Inlcusion/Exclusion Criteria Specified		Assessors	Patients	PCPs	>80% in final analysis	Reason stated	Intention to treat
Trial			Patients	PCPs			Patients	PCPs						
Shnipper et al.(2010)	✓	✗	✓	✗	✓	✓	✓	✓	NS	NS	NA	✗	NS	✓
Gill et al. (2017)	✗	✗	✓	✓	✓	✗	✓	✓	✓	✗	NA	✗	✓	✗
Hesselman et al.(2020)	✓	✓	✓	✓	✗	✓	✓	✓	✓	✓	NA	✗	✓	✓
Eccles et al. (2002)	✓	✓	✓	✓	!	!	✓	✓	✓	✓	NA	✓	✓	✓
Gill et al. (2009)	!	✗	✓	✓	✓	✗	✓	✓	✗	✗	NA	!	✓	✗
Cobos et al. (2006)	✓	✓	✓	✓	✓	✓	✓	✓	NS	NS	NA	✗	✓	✓
Hicks et al. (2008)	!	✗	!	✓	✗	✓	✓	✓	NS	NS	NA	✗	✓	✓
Lopez et al. (2019)	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗	NA	!	NA	!

Key	
✓	Adequately addressed
!	Partially addressed
✗	Not adequately addressed
NS	Not stated
NA	Not applicable

For peer review only

Risk of Bias Assessment: Type 2 Diabetes Mellitus

Referring to **Supplementary Tables 4 and 5**, Schnipper et al was rated at 'low risk' for random sequence generation as a random number generation on Excel was used [1].Allocation concealment was rated as 'high' after considering the randomisation an open allocation method. Blinding of personnel was not cited. Thus, it was rated as unclear. There was no mention of withdrawals; hence, the risk was unclear for incomplete outcome reporting. All outcomes set be studied were reported, and no other risks related to study funding were detected.

Gill et al's study was rated at 'high risk' for incomplete and selective outcome reporting due to analysing subgroups of patients for A1C and LDL-C for the narrated study limitations in the following section [2].Randomisation was at 'high risk' since clusters of practices were allocated based on the number of clinicians. Although the randomisation method could be considered open to predictability, eligibility criteria for patients within clusters of practices were pre-defined, which could not have been subverted by allocation knowledge. All other elements were considered at 'low risk.'

Heselmans et al's study was at 'low risk' for all elements except incomplete outcome data due to the withdrawal of a significant number of practices, with less than 80% of practices remaining after randomisation for analysis[3]. Reasons for system concerns were attributed to withdrawals.

3.2.2 Risk of Bias Assessment: Asthma and Angina

Referring to **Supplementary Tables 4 and 5**, the risk of bias in Eccles et al. 's study was 'low' for all elements due to the use of a computerised randomisation process and blinding of assessors[4]. However, less than 80% of eligible patients completed the provided questionnaires, which could have introduced attrition bias. Baseline characteristics of patients and practitioners were not sufficiently addressed, but there was a mention that participating practices were of similar size and training. Selective recruitment of clusters was adequately prevented by recruiting practices and enrolling all patients who met the inclusion criteria.

Risk of Bias Assessment: Hyperlipidaemia

Referring to Supplementary T**ables 4 and 5**, Cobos et al. 's study was rated at 'low-risk' for random sequence generation and allocation concealment due to the use of a random number table to randomise, which does not allow predictable sequences[5]. There was no mention of blinding of assessors, statisticians, or patients. Although <80% of randomised patients had follow-up visits and LDL-C follow-up tests, the study was rated as 'low risk' for attrition and reporting bias because the ITT analysis yielded insignificant outcomes as did the PP analyses. Study funding did not seem to introduce methodological bias since the outcome was patient-reported. Selective recruitment of clusters was adequately prevented by recruiting practices and enrolling all patients of randomised practices who met the inclusion criteria.

Gill et al 's trial was rated at 'high risk' for random sequence generation but 'low risk' for allocation concealment due to randomising clusters of practices based on practice similarities, which is not preferable randomisation[6]. The Eligibility criteria for patients within clusters of practices were pre-defined, which could not have been subverted by allocation knowledge. Blinding of assessors and

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

statisticians was not mentioned. The number of patients analysed was not reported. However, all outcomes set to be studied were undertaken. The first author is a consultant researcher at the company that holds the concerned EHR's licensing rights, but this conflict of interest did not seem to reflect in methodological risks. Similarly, selective recruitment of clusters was prevented.

Risk of Bias Assessment: Hypertension

Referring to **Supplementary Tables 4 and 5,** Hick's et al 's study was rated 'high risk' for random sequence generation, as clusters of practices were randomised by size and type of practice, which may not be a desirable method of randomisation [7]. Although randomisation could have been open to predictability, patient eligibility criteria practices were pre-defined, which could not have been subverted by allocation knowledge. Thus, the study was rated at 'low risk' for allocation concealment. Blinding of assessors or patients was not mentioned. Thus, the study was rated 'unclear' for blinding participants and outcome assessments. Due to the high number of randomised patients that were withdrawn due to stated reasons, it was rated at 'high risk' for attrition bias. All outcomes were explored, and there was a 'low risk' for selective recruitment of cluster members.

Lopez et al. 's study was rated at 'high risk' for random sequence generation and allocation concealment due to its non-randomised design[8]. There was no mention of blinding personnel, hence it was rated 'unclear' for blinding. There was no mention of withdrawals, and all outcomes were reported. Thus, it was rated as 'low risk' for attrition and selective outcome reporting. No other risks related to funding were significant.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Table 6: Quality of Evidence from individual studies evaluated using GRADE

Study	Outcome	Risk of bias	Level of inconsist-ency	Level of indirect-ness	Level of imprecise-on	Publication bias	Quality	Reason for downgrading
Schnipper et al. (2010)	Difference in proportion of patients meeting management goals at the end compared to the beginning of the study	severe	NA	some	low	NA	⊕⊕◯◯ LOW	**RoB:** High risk for allocation concealment and unclear whether personnel blinding occurred **Indirectness:** population identification and management goals were not pre-defined
Gill et al. (2017)	• Difference in HbA1c from baseline • Difference in LDL-C from baseline • % patients achieving HbA1c goals	some	NA	low	low	NA	⊕⊕⊕◯ MODERATE	**RoB:** High risk for random sequence generation and selective outcome reporting
Heselmans et al. (2020)	• Difference between the mean value of HbA1c, LDL-C, and blood pressure of the previous year at the moment of the study start and the mean value of HbA1c, LDL-C, and blood pressure of the previous year at 6 and 12 months.	low	NA	some	low	NA	⊕⊕⊕◯ MODERATE	**Indirectness:** patients had reasonable HbA1c mean values, which does not necessarily reflect diabetic populations with a higher severity of illness
Eccles et al. (2002)	Quality of life and condition specific outcomes using validated questionnaires	some	NA	low	low	NA	⊕⊕⊕◯ MODERATE	**RoB:** High risk for incomplete and selective outcome reporting
Gill et al. (2009)	The percentage of patients with most recent LDL-C at goal	severe	NA	low	low	NA	⊕⊕◯◯ LOW	**RoB:** High risk for random sequence generation and incomplete outcome data. Unclear for personnel blinding
Cobos et al. (2006)	• Between-group differences in means of lipid values	some	NA	some	low	NA	⊕⊕⊕◯ MODERATE	**ROB**: Unclear for blinding of personnel **Indirectness:** excludes general populations with more severe elevations of LDL
Hicks et al. (2008)	• Percentage of patients with controlled blood pressure • Odds of blood pressure control at outcome visit	severe	NA	low	low	NA	⊕⊕◯◯ LOW	**RoB:** High risk for random sequence allocation, incomplete outcome reporting and unclear for personnel blinding
Lopez et al. (2019)	• Odds of blood pressure control post-intervention	severe	NA	low	low	NA	⊕⊕◯◯ LOW	**RoB:** NRCT, non-randomised and high risk for blinding of personnel

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

References

1) Schnipper JL, Linder JA, Palchuk MB, et al. Effects of documentation-based decision support on chronic disease management. *The American Journal of Managed Care.* 2010; 16 (12 Suppl HIT) (pp SP72-81): ate of Pubaton: e 2010.

2) Gill J, Kucharski K, Turk B, Pan C, Wei W. Using Electronic Clinical Decision Support in Patient-Centered Medical Homes to Improve Management of Diabetes in Primary Care: The DECIDE Study. *Journal of Ambulatory Care Management.* 2019; 42 (2): 105-115. doi:10.1097/JAC.0000000000000267

3) Heselmans A, Delvaux N, Laenen A, Van de Velde S, Ramaekers D, Kunnamo I, et al. Computerized clinical decision support system for diabetes in primary care does not improve quality of care: a cluster-randomized controlled trial. *Implementation science : IS.* 2020; 15 (1): 5. doi:10.1186/s13012-019-0955-6

4) Eccles M, McColl E, Steen N, Liaw ST. Computerised evidence-based guidelines may not improve asthma or angina management in primary care. *Evidence-Based Healthcare.* 2003; 7 (2): 81-82.

5) Cobos A, Vilaseca J, Asenjo C, et al. Cost effectiveness of a clinical decision support system based on the recommendations of the European Society of Cardiology and other societies for the management of hypercholesterolemia: Report of a cluster-randomized trial. *Disease Management and Health Outcomes.* 2005; 13 (6): 421-432. doi:10.2165/00115677-200513060-00007

6) Gill JM, Chen YX, Glutting JJ, et al. Impact of decision support in electronic medical records on lipid management in primary care. *Population health management.* 2009; 12 (5): 221-226. doi:10.1089/pop.2009.0003. doi:10.1089/pop.2009.0003

7) Hicks LS, Sequist TD, Ayanian JZ, et al. Impact of computerized decision support on blood pressure management and control: A randomized controlled trial. *Journal of General Internal Medicine.* 2008; 23 (4): 429-441. doi:10.1007/s11606-007-0403-1

8) Lopez PM, Divney A, Goldfeld K, et al. Feasibility and Outcomes of an Electronic Health Record Intervention to Improve Hypertension Management in Immigrant-serving Primary Care Practices. *Medical care.* 2019; 57 (Supplement 62) (pp S164-S171): ate of Pubaton: 01 Jun 2019. doi:10.1097/MLR.0000000000000994

For Peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 3/4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 3/4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 4
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 5, supplementary material table 3
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 5
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 5
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 5/6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 5/6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 6
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 6
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g., tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 6
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 6
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 6
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 6
Reporting bias	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 6

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 6
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 6/7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 6/7
Study characteristics	17	Cite each included study and present its characteristics.	Page 6/7
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 7 & supplementary section
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Page 7/8/9
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page 7 & supplementary section
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 7/8/9
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 7 & supplementary section
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page 7
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 10/11/12
	23b	Discuss any limitations of the evidence included in the review.	Page10/11/12
	23c	Discuss any limitations of the review processes used.	Page10/11/12
	23d	Discuss implications of the results for practice, policy, and future research.	Page10/11/12
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 2
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 2
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Page 2
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 14
Competing	26	Declare any competing interests of review authors.	Page 14

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
interests			
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 14

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

For peer review only

BMJ Open

Effect of computerised, knowledge-based, clinical decision support systems on patient-reported and clinical outcomes of patients with chronic disease managed in primary care settings: a systematic review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-054659.R2
Article Type:	Original research
Date Submitted by the Author:	22-Nov-2021
Complete List of Authors:	El Asmar, Marie line; Imperial College London, Self Care Academic Research Unit (SCARU) Dharmayat, Kanika ; Imperial College London, Imperial Centre for Cardiovascular Disease Prevention (ICCP), Department of Primary Care and Public Health Vallejo-Vaz, Antonio J.; Imperial College London, Imperial Centre for Cardiovascular Disease Prevention (ICCP), Department of Primary Care and Public Health, School of Public Health Irwin, Ryan; University of Birmingham, Institute of Applied Health Research Mastellos, Nikolaos; Imperial College London, Department of Primary Care and Public Health, School of Public Health
Primary Subject Heading:	General practice / Family practice
Secondary Subject Heading:	General practice / Family practice, Health informatics, Evidence based practice
Keywords:	GENERAL MEDICINE (see Internal Medicine), PRIMARY CARE, PREVENTIVE MEDICINE, Health informatics < BIOTECHNOLOGY & BIOINFORMATICS

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Effect of computerised, knowledge-based, clinical decision support systems on patient-reported and clinical outcomes of patients with chronic disease managed in primary care settings: a systematic review

Marie Line El Asmar,[1] MD, MPH, Kanika I Dharmayat,[2] MPH, BSc (hons), Antonio J. Vallejo-Vaz,[2,3,4] MD, PHD, Ryan Irwin,[5,*] PhD MPA MSc BSc (hons), Nikolaos Mastellos[6,*] PhD, MSc, BSc (hons)

1- Self Care Academic Research Unit (SCARU), Imperial College London. London, United Kingdom.
2- Imperial Centre for Cardiovascular Disease Prevention (ICCP), Department of Primary Care and Public Health, School of Public Health, Imperial College London. London, United Kingdom.
3- Department of Medicine, Faculty of Medicine, University of Seville. Seville, Spain.
4- Clinical Epidemiology and Vascular Risk, Instituto de Biomedicina de Sevilla, IBiS/Hospital Universitario Virgen del Rocío/Universidad de Sevilla/CSIC. Seville, Spain.
5- Institute of Applied Health Research, University of Birmingham, United Kingdom.
6- Department of Primary Care and Public Health, School of Public Health, Imperial College London. London, United Kingdom.

* Joint principal/last authors. These authors contributed equally to the work.

Corresponding author:
Dr Marie Line El Asmar, MD, MPH
me719@ic.ac.uk

For peer review only

Key words
Clinical decision support system
Electronic health records
Primary care
Chronic disease
Health outcomes

Word Count:

- Abstract: 300 words
- Main text: 4,646 words

Number of Figures: 2
Number of Tables: 3
Number of references: 59

Abstract

Objectives: Chronic diseases are the leading cause of disability globally. Most chronic disease management occurs in primary care with outcomes varying across primary care providers. Computerised Clinical Decision Support Systems (CDSS) have been shown to positively affect clinician behaviour by improving adherence to clinical guidelines. This study provides a summary of the available evidence on the effect of CDSS embedded in electronic health records on patient-reported and clinical outcomes of adult patients with chronic disease managed in primary care.

Design and eligibility criteria: Systematic review including randomised controlled trials (RCTs), cluster RCTs, quasi-RCTs, interrupted time series, and controlled before-and-after studies , assessing the effect of CDSS (vs. usual care) on patient-reported or clinical outcomes of adult patients with selected common chronic diseases (asthma, chronic obstructive pulmonary disease, heart failure, myocardial ischemia, hypertension, diabetes mellitus, hyperlipidaemia, arthritis, and osteoporosis) managed in primary care.

Data sources: Medline, Embase, Cochrane Central Register of Controlled Trials, Scopus, Health Management Information Consortium, and trial register clinicaltrials.gov, searched from inception to June 24, 2020.

Data extraction and synthesis: Screening, data extraction, and quality assessment were performed by two reviewers independently. The Cochrane Risk of Bias tool was used for quality appraisal.

Results: From 5,430 articles, eight studies met the inclusion criteria. Studies were heterogeneous in population characteristics, intervention components, and outcome measurements and focused on diabetes, asthma, hyperlipidaemia, and hypertension. Most outcomes were clinical with one study reporting on patient-reported outcomes. Quality of the evidence was impacted by methodological biases of studies.

Conclusions: There is inconclusive evidence in support of CDSS. A firm inference on the intervention effect was not possible due to methodological biases and study heterogeneity. Further research is needed to provide evidence on the intervention effect and the interplay between healthcare setting features, CDSS characteristics, and implementation processes.

Review registration: Prospero ID: CRD42020218184.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Strengths and limitations of this study

- In contrast to other reviews, this review focusses on interventions involving CDSSs that are knowledge-based, embedded within EHR systems, provide electronic alerts, and are used by physicians in primary care.
- The electronic search strategy was piloted, and hand-searching of reference lists of previous reviews permitted a sufficient level of confidence that all relevant articles were included.
- Some limitations arose related to studies being of 1-year duration on average, making it difficult to deduce long-term effects on patient outcomes.
- The scope of the review was restricted to nine chronic diseases; however, the selection included the most common chronic diseases, yet there were no eligible studies in five of them.
- The included studies were of low-to-moderate quality of evidence, and heterogeneity among the included studies limited the ability to conduct a meta-analysis.

Introduction

Non-communicable chronic diseases are the leading cause of disability and mortality worldwide, affecting one in three adults globally.[1] Rising healthcare expenditure because of chronic disease prevalence has led to an ever-growing demand for effective primary care services for the long-term management of chronic conditions.[2-4] Considering that health systems have mainly adapted to providing periodic care for acute problems, an even greater need to achieve efficient, comprehensive and coordinated long-term care persists.[5] Models such as the Chronic Care Model (CCM) have been proposed to improve the quality of care in chronic care systems.[6] This model recommends 'decision support' as one of six applicable changes in primary care settings that promote better patient outcomes, if successfully implemented.[6-8] Within this and other models of primary care and chronic condition management, clinicians are encouraged to provide patient care according to evidence-based clinical practice guidelines (CPGs) to support improved patient outcomes.[9,10] Given the frequent updates to CPGs resulting from rapid advancements in medical knowledge, a major challenge is the maintenance of clinicians' knowledge of evidence-based CPGs.[10] Consequently, clinicians' adherence to CPGs may be variable and frequently insufficient, which may result in sub-optimal disease management outcomes.[10,11]

Clinical decision support is an umbrella term that includes various care structures, processes and tools.[12] Clinical decision support systems (CDSS) range from non-computerised to computerised systems and basic to advanced systems.[12,13] Currently, most attention is driven toward computerised advanced CDSS, which perform complex decision-making functions from electronic health record (EHR) patient data.[12] Such systems are termed knowledge-based, as they use software algorithms to generate actionable patient-specific outputs in the form of recommendations to the practitioner using conditional (IF-THEN) rule statements derived from CPGs.[14 15] They also differ in their technical and clinical interface, whether passive (providing user-initiated prompts to generate clinical advice, e.g., by clicking buttons), or active (automatic system-initiated prompts).[12,16,17] CDSS also differ in system-user communication, which could be consulting (i.e., provide appropriate 'next-steps' clinical recommendations) or of critiquing nature (verify decisions taken by practitioners).[12,16] Given the differences in interfaces, the scope of this

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

review includes computerised, knowledge-based CDSS embedded in EHRs that are of either passive or active, consulting or critiquing, nature.

Although CDSS are ubiquitous, few have been successfully adopted.[18] Previous reviews appraised the efficacy of different types of CDSS in primary care, presenting a greater consensus on the useful role of CDSS on practitioner-related outcomes such as guideline adherence.[19-23] However, the impact of CDSS on patient outcomes remains largely uncertain.[19-23] *Bryan et al* demonstrated a positive association between the use of CDSS in primary care and patient outcomes.[23] However, a need to further evaluate the effect of CDSS on patient outcomes was identified, particularly in primary care and specifically for chronic disease management. The novel focus of this review is in the combined focus on the effect of computerised, knowledge-based, clinical decision support systems *and* their effect on patient-reported *and* clinical outcomes of patients *and* with chronic disease *and* managed in primary care settings. This particular focus was chosen to inform more specific and efficacious implementation strategies in primary care management of chronic disease and attract the industry's consideration of CDSS use in primary care settings, which a more generalised focus on multiple care settings, multiple diseases beyond just chronic disease, or lack of focus on patient-reported outcomes may not achieve. There remains a need to assess the capabilities of CDSS in distinctive clinical professional areas and disease entities, particularly on patient outcomes.[18] A recent review on EHR-embedded CDSS focused on process delivery (practitioner outcomes) and was not limited to primary care settings and chronic conditions.[24] Despite obvious heterogeneity, clinically meaningful improvements in care processes were shown.[24] However, it was unclear under what clinical circumstances these improvements could be replicated in every clinical context to affect clinical outcomes presumptively.[24]

In this systematic review, patient outcomes refer to quantifiable clinical biomarkers such as glycated haemoglobin (HbA1c) and low-density lipoprotein-cholesterol (LDL-C) levels, clinical outcomes such as mortality and morbidity, and patient-reported outcomes, such as health-related quality of life (HRQoL). To our knowledge, and in the view of rapid upgrades in the world of CDSS, no review exists solely on the effect of CDSS embedded within EHRs on patient outcomes for patients with chronic disease managed in primary care. This review aims to fill this knowledge gap by collating all the available evidence on this classification of CDSS.

Methods

Study design and eligibility criteria

A systematic review of the existing literature was conducted following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines.[25,26] The research question, inclusion, and exclusion criteria were defined using the PICOS framework.[27] Nine common chronic diseases were selected as part of the inclusion criteria, following a review by *Reynolds et al,*[28] namely asthma, chronic obstructive pulmonary disease (COPD), heart failure, myocardial ischemia, hypertension, type 2 diabetes mellitus, hyperlipidaemia, arthritis, and osteoporosis. Studies reporting on patient-related and clinical outcomes managed by primary care clinicians using CDSS-EHR compared to usual care were included. Studies involving participants <18 years of age, multidisciplinary (non-clinician) system users, and other varieties of CDSS (not

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

linked with EHR) were excluded. A detailed view of the criteria is provided in Supplementary Table 1.

Figure 1 depicts an analytic framework of the intervention adapted from a framework published by *Lobach et al*.[22] The framework provides a theoretical context for possible features that could underpin the success, or lack of success, of a complex intervention such as implementing CDSS in a primary care setting. The framework guided the extraction of relevant data from studies to inform the narrative synthesis.

Search Strategy

Electronic databases Medline, Embase, Scopus, Cochrane Central Register of Controlled Trials (CENTRAL) and Health Management Information Consortium (HMIC), and one clinical trial registry (clinicaltrials.gov), were searched from inception to June 24th 2020. Cochrane reviews were screened for relevant search terms and subject headings (MeSH) terms under the following categories 'registries supported by CDSS' AND 'General Practice/Primary care' OR each of the nine chronic diseases combined with AND 'Asthma', 'heart failure/myocardial ischemia', 'hypertension', 'diabetes mellitus', 'hyperlipidaemia', 'COPD', 'arthritis', 'osteoporosis' (Supplementary Table 2 and 3). The search was limited to articles in English.

Terms for patient-reported outcome measures (PROM) were excluded. Existing PROM filters are not comprehensive, and the use of those filters in the search strategy could increase the risk of missing relevant studies.[29] However, we manually screened papers to identify any "patient-reported outcomes" measured in studies. PROM may be described in various ways, thus attempting to list specific patient outcomes for each entity could introduce bias and lead to missing relevant studies. Previous systematic reviews around CDSS in primary care were also searched for potentially eligible studies.[19-21,23,24]

Study Screening and Selection

Search results were exported to RefWorks, then screened using Covidence. After removal of duplicates, screening was conducted by two reviewers independently (MA, KID). Disagreements were resolved by consensus within the review group. After an initial screening of titles and abstracts, the full text of potentially relevant articles were assessed.

Data Extraction

A data extraction form was adapted using the 'Cochrane data collection form for intervention reviews: randomised controlled trials (RCTs) and non-RCTs' and aided by *Lobach et al*'s data abstraction guidance.[30,31] The form was piloted and two reviewers (MA, KID) completed data extraction independently.

Quality Assessment of Individual Studies

Risk of bias was assessed using the Cochrane risk of bias (RoB) tool,[32,33] by two reviewers independently (MA, KD). The methodological risk of bias was assessed per the following elements

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

in RCTs: selection bias, performance bias, detection bias, attrition bias, reporting bias, and other biases.[33,34]

Quasi-RCTs were reported as high risk for random sequence generation. Cluster-RCTs were assessed for an additional element 'selective recruitment of cluster participants.' All judgments were made with guidance from the Cochrane Handbook and the guidelines of the Cochrane Consumers and Communication Review Group.[33-35]

Studies were considered as high risk if they scored 'high' or 'unclear' for either random sequence generation or allocation concealment, based on the rising evidence that these two elements are particularly significant sources of bias.[34] A summary 'risk of bias' figure was generated for all risk assessments of elements with a narrative commentary for each study. Additionally, a detailed adapted table was generated to visualise sub-elements that fit under each element in the Cochrane RoB tool.[34]

Quality Assessment of Outcomes

The GRADE (Grading of Recommendations, Assessment, Development and Evaluations) tool was used to assess the quality of outcomes from individual studies.[36] The tool involves five elements: risk of bias, inconsistency, indirectness, imprecision, and publication bias. The tool is primarily used for meta-analyses; however, it can be adapted for use in narrative reviews. Study outcomes were appraised, ranked as high, moderate, low or very low and justifications were provided, in line with Cochrane guidance.[37]

Data Synthesis

Treatment effect was reported as odds ratios (OR) with 95% confidence intervals (CIs) and p-values for dichotomous outcomes, where available. Continuous data were presented as mean differences with 95% CIs and p-values.

Heterogeneity in the results was anticipated due to the wide spectrum of possible patient-related outcomes reported by the studies. Methodological heterogeneity was assessed using Cochrane's RoB tool. Heterogeneity in included studies varied in terms of intervention components, population characteristics, and outcome measurements; thus, conducting a meta-analysis was deemed not appropriate by the four-person author group.

Patient and Public Involvement

No patients were involved.

Ethics Approval

The present work relied on already published aggregated data and ethics committee approval was unnecessary.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Results

The search identified 5,430 articles. Following removal of duplicates, 4,498 articles remained for screening by title and abstract. Of these, 266 articles were selected for full-text review and 10 articles met the inclusion criteria. Among them, two articles were published protocols of included articles; thus, 8 different studies were finally included. The detailed search results are presented in the PRISMA flowchart (Figure 2).

Study Characteristics

Characteristics of included studies are presented in Table 1. Included studies comprised of patients in four disease areas: T2DM, asthma, hyperlipidaemia, and hypertension.[38-45] No studies on COPD, heart failure, myocardial ischemia, arthritis, and osteoporosis were identified. Six studies were cluster-RCTs, one study was parallel-group RCT, and one was quasi-RCT of stepped-wedge design. Most studies were conducted in the United States, and three studies took place in the United Kingdom, Spain, and Belgium. Studies, both among the disease conditions and within each disease entity, differed appreciably in participant covariates, setting attributes, and CDSS characteristics. These factors, compounded with the heterogeneity of the studies (as described above in the methods section) suggested that the reported outcomes were not fit for statistical pooling, hence, these factors justified not conducted a meta-analysis. Follow-up ranged from 6 months to 1.5 years. No study incorporated financial reimbursement before the implementation of interventions. Further details are shown in Table 1.

CDSS Characteristics

CDSS interventions were characterised based on their origin (commercial/local), delivery mode (user or system-initiated), user response (optional/mandatory), events (system functions), coupled EHR, and auxiliary features (local clinician involvement in system development process, system accessibility to other team members, provision of performance feedback, or coupled with user education). Additional information such as user-training, user adherence, and reported usability were recorded. Details are provided in Table 2.

Risk of Bias and Quality of the Evidence

Supplementary Tables 4 and 5 provide risk assessments details. Overall, the included studies were impacted by risk of bias, mainly due to incomplete outcome data and uncertainty on blinding of outcome assessment and personnel. In most studies, it was unclear if outcome assessors were blinded. Additionally, blinding of physicians (CDSS users) was not possible across all studies, as they were the intended system users. Most studies reported substantial withdrawal rates due to poor follow-up. These limitations resulted in analysing subgroups (rather than the initially intended full cohorts) in some trials, hence most studies were considered of high risk for incomplete outcome data.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Overall, quality of the evidence was rated as either moderate or low, with the most impact resulting from methodological risks of bias within studies. Supplementary Table 6 presents the quality of evidence evaluated using GRADE, with reasons stated for quality downgrading.

Narrative Synthesis

Diabetes Mellitus

Three studies assessed the effect of CDSS on clinical and patient-reported outcomes in T2DM patients.[38-40]

Gill *et al* retrieved slight, yet statistically significant, reduced HbA1c and LDL-C with the CDSS intervention compared to the control group, with a small between-group difference in HbA1c and LDL-C levels of 0.12% and 3.57 mg/dL, respectively (Table 3).[39]The reported reductions in HbA1c and LDL-C were clinically small; however, statistically, the intervention group had 52% higher odds of achieving personalised HbA1c levels, 56% higher odds of achieving HbA1c <7.0%, and 34% higher odds of achieving an LDL-C <100 mg/dL compared to usual care patients over 1 year of follow-up. Nevertheless, compared to the control group, the intervention group had better-controlled baseline levels of HbA1c and LDL-C. The study by *Gill et al* involved a system-initiated "push" CDSS which was integrated into EHRs already operating at participating practices. Reported limitations were of EHR-CDSS interoperability difficulties, which resulted in missing patient lab values.[39] Other missing lab values also resulted from patients not returning for lab tests. In view of these limitations, a subset of randomised patients without missing HbA1c and LDL-C values were analysed (Table 3).

Schnipper *et al* (RCT) reported the odds of patients meeting management goals (Table 3).[38] Intention-to-treat analysis showed non-significant changes in patients' odds of meeting management goals; a per-protocol analysis, however, showed slightly significant odds of attaining management goals in the intervention group compared to the control group (Table 3). Among the three studies, this intervention is the only one that involved user training (Table 2)**.** Auxiliary features also included feedback on performance and local user (physician) involvement in the CDSS development process; yet system usability was reported to be used for only 5.6% of patients (Table 2).

Heselmans *et al* (cluster RCT) reported non-significant between-group differences in HbA1c, LDL-C, diastolic (DBP) and systolic (SBP) blood pressure (BP) (Table 3).[40] Levels of HbA1c were relatively well-controlled**,** with reasonable mean values for HbA1c in both groups at baseline, especially in the control group (Table 1).[40] Subsequent analysis of a subgroup of 601 patients with HbA1c >7.0% retrieved a statistically significant between-group difference in HbA1c favouring the intervention, though not clinically relevant (-0.40% [95% CI -0.70 to -0.09]). The same analysis retrieved non-significant differences in LDL-C and BP. User adherence and usability were not measured.

In all three studies, there was no mention of the provision of reimbursement for the implementation of the CDSS. All CDSS were commercial, and user response to the system alerts was optional.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Asthma and Angina

One study assessed the impact of CDSS on self-reported quality of life and condition-specific outcomes in patients with asthma and angina.[41] The angina intervention group acted as the control group for the asthma intervention group and vice versa (justified by authors to eliminate a possible Hawthorne effect in practitioner behaviour on the primary outcome). Questionnaires were administered in three rounds (a year before the intervention, beginning of the intervention, and a year after the intervention), which showed non-significant differences in both intervention groups post-intervention (Table 3). Completion of the three rounds of questionnaires occurred in 46% and 35% of patients with angina and asthma, respectively. The implemented CDSS was locally produced and embedded in two widely used health systems (Table 2). The CDSS was modified four months into the study from system to user-initiated, as requested by participating physicians. User training was limited, and system usability was very low, measured using a log with a median of zero user-system interaction during most of the study (Table 2).

Hyperlipidaemia

Two studies involved participants with hyperlipidaemia.[42,43]

The study by *Gill et al* involved implementing a commercial CDSS in participating practices of one institution.[42] All patients enlisted under participating practices with recorded LDL-C values were enrolled. A high correlation in outcomes was found at baseline and end measurements, causing patients not meeting LDL-C levels after randomisation to be analysed only. The control group also had a lower risk of disease and was significantly younger. Patients were stratified into three guideline-based risk groups, and the percentage of patients reaching the pre-defined goals at baseline and end were recorded (Table 1). All categories in the intervention and control groups showed a significant increase in the proportion of patients attaining lipid goals, except in the moderate-risk group. Calculated differences were not presented; however, p-values of differences within the intervention and control groups suggested no significant improvement in any risk category.

In *Cobos et al*, despite a usability of 71%, the intervention had no superiority over usual care in terms of between-group differences in lipid values (Table 3).[43] The study reported >25% patient lost to follow-up for clinic visits or LDL-C testing (Table 1).

Hypertension

Two studies assessed the effect of CDSS on BP in hypertensive patients.[44,45] Both studies employed CDSS that were system-initiated with optional user responses (Table 2).

Hicks et al reported no differences between intervention groups in BP control (Table 3).[44] Differences in baseline and end SBP and DBP between groups were not significant. Nonetheless, the intervention group had a significantly higher percentage of patients with controlled BP (Table 1). No user training was provided, and system usability was not measured (Table 2). Less than 80% of recruited patients were followed-up due to withdrawals caused by the study's inclusion

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

criteria, which required patients to have a recorded ethnicity in the EHR. Subsequently, patients without BP recordings were considered uncontrolled. However, a subset analysis excluding these patients still retrieved a non-significant difference in BP values.

Lopez et al study was a quasi-RCT and reported significant odds of BP improvement in the intervention group compared to the control group (Table 3).[45] Notably, extensive training was provided for participating clinicians compared to all other studies, and semi-structured interviews revealed that all practices appeared to use the system components (Table 1).

Discussion

Main findings

As primary care providers are increasingly required to monitor and manage chronic diseases, there is a need to assess the capabilities of CDSS on patient outcomes. This systematic review included the eight available studies on the effect of computerised, knowledge-based, EHR-embedded CDSS on clinical and patient-reported outcomes. Several study designs were included in the eligibility criteria; however, studies fit for inclusion were trials, namely RCTs, cluster-RCTs, and non-RCTs (supplementary table 1, table 1). Two studies reported a statistically significant impact of CDSS on patient outcomes,[39,45] one reporting clinically small yet significant reductions in $HbA1_C$ and LDL-C and another reporting significant increases in the proportion of hypertensive patients who achieve goals, which may be clinically relevant.[45] Another two studies reported statistically significant findings but in subgroup analyses.[38,40] One study reported statistical significance in partial outcomes and three studies did not report any statistically significant impact of CDSS on outcomes at all.[42-44] Overall, evidence was not conclusive to imply a differential effect, either positive or negative, of clinician use of CDSS on patient or clinical outcomes compared to usual care, mainly due to the quality of evidence impacted by the prominent risk of bias in the included studies. The systematic review also noted a limited availability of studies evaluating the CDSS of interest directed to clinicians in primary care for the management of the common chronic diseases. This is especially true for COPD, arthritis, and osteoporosis, which were eligible conditions, yet no such studies were available.

Implications of the review

The systematic review infers the importance of considering how the interplay of factors such as the nature of a health care setting (academic, community) and CDSS interact with clinicians (or users per se). System-initiated CDSS seemed to have a better reflection on patient outcomes compared to user-initiated systems; however, this relationship was confounded by many factors such as poor process measures, namely low system usability and EHR-CDSS interoperability difficulties. A common limitation reported across most studies was poor patient follow-up, which highlights the importance of patient behaviour as a confounding factor to the performance of CDSS interventions in clinical settings. Some interventions included helpful auxiliary features such as involvement of local physicians in CDSS development and feedback on performance, yet these features did not seem to reflect positively on patient outcomes. Understanding user (clinician) experiences qualitatively in the context of using such CDSS to manage chronic diseases in primary care may be of benefit to inform on their practical use. This review assessed CDSS functionality

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

when used by clinicians; however, CDSS accessibility to other members of a multidisciplinary team may have an added positive effect which could translate into better patient outcomes. Multidisciplinary teams in primary care have been shown to aid successful implementation of CDSS by increasing adherence to CPGs and improving patient outcomes in the absence of CDSS.[46,47]

The healthcare setting and intervention implementation itself may dictate the success or failure of an intervention. Large academic care settings may be better equipped with IT and financial resources than community centres to ensure successful implementations of CDSS.[48] Included studies revealed limitations due to poor implementation of the CDSS, namely interoperability issues, lack of IT support, poor usability, and lack of user training, which could have hindered the intervention from reflecting significant outcomes compared to usual care. Models have been offered to help explicate how structural elements of healthcare settings and various levels of healthcare organisations levelling up to the industry may influence clinician behaviour and uptake of health IT to influence patient outcomes.[17,49] Finally, it is imperative to consider the effect of non-clinical factors such as patient behaviour outside clinic visits, their perceptions, and adherence to clinical advice.[50]

Comparison with previous reviews

Previous reviews of CDSS of multiple types have indicated impact on patient outcomes in some instances. A recent review showed that CDSS systems achieve small to moderate effects on processes of care, yet the extent of translation to patient outcomes and under what clinical contexts and conditions this was achieved was not determined.[24] Previous reviews were not limited to primary care settings and included various other CDSS, namely paper-based, web-based, standalone electronic systems, and patient-directed systems that do not necessarily alert a clinician at the point-of-care.[19,21,51] However, reviews assessing the impact of various types of CDSS on chronic disease patient outcomes in primary care identified inconclusive results, with a need to define more clearly the exact role of CDSS compared to usual care in subsequent studies.[20,23] Reviews evaluating patient outcomes noted conceptual between-study heterogeneity, as this review did, especially when several diseases and clinical outcomes were considered.[52] In terms of patient outcomes, CDSS seem to affect clinical outcomes for cardiovascular risk factors such as hyperlipidaemia and diabetes.[11,53] Nevertheless, a much more recent review on the effect of CDSS on cardiovascular risks revealed no definitive clinical benefits after their implementation.[53] On the other hand, the evidence is less robust for other non-cardiovascular risk-related chronic conditions, such as cancer, and other primary care sensitive cases (PCSC) such as vaccination, which were not part of the eligibility criteria in this review.[54]

A review of CDSS embedded within EHRs in all-purpose healthcare settings on overall mortality and morbidity suggested no effect of CDSS on mortality. However, it noted that potential benefits or risks would highly depend on setting and disease characteristics, which is an evident implication in this review.[52] A review by *Bryan et al* validated the use of CDSS in primary care to improve outcomes; however, they reported substantial variability in intervention effectiveness related to the type and implementation of the different types of CDSS.[23] Additionally, other reviews incorporated studies that involved the use of CDSS by multidisciplinary care members, reflecting positive clinical outcomes with shared decision making.[19,51,55]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Unanswered questions and future research

More research is needed to establish the effectiveness of EHR-embedded CDSS on clinical outcomes of patients with chronic diseases treated in primary care settings. Further studies involving patients with chronic diseases are needed for robust evidence. Moreover, it may not be reasonable to evaluate the intervention's efficacy based on older studies, given the exponential growth and rapid advancements in the field of CDSS.[56] Future research could include practitioner-related, cost-effectiveness, and patient-related outcomes to present an ampler perspective on the effect of this type of CDSS to inform policymaking and clinicians. Other primary care sensitive conditions (PCSCs) can be studied for a broader scope, including cancer, mental health illnesses and preventative interventions such as vaccination.[57] Research around prolonged evaluations of CDSS is needed. Longer-term studies may account for potential implementation barriers that could take place. Additionally, it would be beneficial to synthesise findings on the effect of this particular type of CDSS on patient outcomes, when operated within a multidisciplinary team of non-clinician providers. It is worth noting that patient outcomes are not always addressed efficiently in trials, among the reasons is the possibility of having subjective, composite, or surrogate outcomes, which do not necessarily translate into real-life patient outcome improvements.[58] Hence, including observational studies in updated reviews may be useful. Additionally, no studies of registries embedded with CDSS fit the eligibility criteria, suggesting a potentially greater inclination to embed CDSS with EHRs currently. There also seems to be a dearth of studies evaluating non-cardiovascular risk-related chronic diseases post-implementation of CDSS linked with EHRs, as no studies evaluating patients with COPD, osteoporosis, and arthritis were eligible.

Strengths and limitations

This systematic review focusses on interventions involving a specific type of CDSS –those that are knowledge-based, coupled with EHR systems, that provide electronic alerts, and are used by physicians in primary care to support chronic disease management. To the authors' knowledge, no such review exists considering this combined focus and given the attention these systems are increasingly gaining in primary care settings.[29] As an example of the novelty of this review, compared to previous reviews with the most similar focus to our own by Bryan and Boren[23] and Bright et al[19] , this review included studies not featured in these previous reviews, such as the studies by Schnipper et al, Gill et al, Heselmans et al and Hicks et al.[38-40, 44]Application of the eligibility criteria identified eight studies for inclusion, which may be a small number given the recent growth of interest in CDSS in healthcare settings.[59] Nevertheless, piloting the electronic search strategy and supplementing it with hand-searching of reference lists of previous reviews in the field permitted a sufficient level of confidence that all relevant articles, against the eligibility criteria, were included for synthesis. The review also contributes to the knowledge provided by a recent review that focuses on process outcomes.[24]

Some limitations arose from methodological constraints. Studies were of an average of 1-year duration, which makes it difficult to deduce long-term effects on patient outcomes. Nine common chronic diseases were included in the syntax.[28] However, there was an absence of eligible studies in five of them, which limited the number of studies for inclusion. Although a selection of nine chronic diseases was included in the syntax, these covered the most common chronic diseases, as done by Reynold et al.[28] The review could have covered other PCSCs for a broader scope, including cancer and mental health illnesses, per se.The included studies were also of low-to-moderate quality of evidence. Additionally, our review does not report on process outcomes. The

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

decision to exclude process outcomes such as practitioner-related outcomes, namely clinician adherence to CPGs was made to channel the research question to focus on where the most significant knowledge gap lies, as the effect of CDSS on patient outcomes remains largely uncertain.[19-21,23,24] Nevertheless, capturing those outcomes would have provided a more comprehensive assessment on the effect of CDSS for the management of patients with chronic disease in primary care. The heterogeneity in study intervention components, participant covariates, setting attributes, and CDSS characteristics rendered outcome pooling in a metanalysis not suitable.

Conclusion

This systematic review aimed to assess the impact of CDSS integrated with EHRs on patient-reported and clinical outcomes of chronic disease patients in primary care settings. A definite conclusion on the intervention effect on outcomes could not be drawn due to between-study heterogeneity and methodological risks, reflecting an immature research area. Intervention characteristics and implementation approaches may have an impact on outcomes from CDSS use and we make reference to the literature in this area that may guide and help evaluate successful implementations of CDSS in practice. Further studies are needed to formulate more robust inferences and validate the intervention's effect on patient outcomes. With the continuous rise in the prevalence of chronic diseases, it is essential to understand how enhancing primary care management using IT support could support improved outcomes. Practical applications of CDSS evaluations in the healthcare workflow and setting should be sustained to suitably determine the actual effectiveness of CDSS and support improved implementation efforts into practice. Strengthening the clinical benefit of CDSS and determining their cost-effectiveness could inform policymaking of enhanced primary healthcare of chronic disease management.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Acknowledgements

The authors would like to thank the library services team at Imperial College London. Special thanks to Ms Rebecca Jones for her support in reviewing the search strategy.

Ms KI Dharmayat acknowledges support from a PhD Studentship from the National Institute for Health Research (NIHR) under the Applied Health Research (ARC) programme for North West London, United Kingdom (the views expressed in this publication are those of the authors and not necessarily those of the NHS, the NIHR or the Department of Health). Dr AJ Vallejo-Vaz acknowledges support from a fellowship from the "Programa Beatriz Galindo" from the Ministry of Universities of Spain, and University of Seville, Spain.

Authors' statement

RI conceived the original study concept with input from NM. NM and AJV-V led the overall study design and supervised the subsequent steps of the study. MA and KD designed the search strategy with supervision from the other authors. MA and KDconducted the literature search, data extraction and quality assessment of studies. MA wrote the first draft of the manuscript. All authors participated in the interpretation of data, have critically reviewed the manuscript and provided edits and comments, and approved its final submission.

Conflict of interest

Ryan Irwin and Nikolas Mastellos are employees of Cerner UK Ltd –a health technology organisation. Dr AJ Vallejo-Vaz reports past or current participation in investigator-initiated research grants to Imperial College London from Pfizer, Amgen, Sanofi, MSD, Daiichi Sankyo and Regeneron, honoraria for lectures from Amgen, Mylan and Akcea, and Steering Committee/consulting fees from Radcliffe Cardiology and Bayer, all outside the submitted work. MS KI Dharmayat reports past or current participation in investigator-initiated research grants to Imperial College London from Amgen, Sanofi, MSD, Daiichi Sankyo and Regeneron and consulting fees from Bayer, all outside the submitted work.

Funding

No funds were received for conducting the present study.

Data Availability Statement

No additional data are available

For peer review only

Table 1: Characteristics of included studies

							Participants		Comparability			
Study	Disease	Study Design	Duration	Sponsorship	Setting	Condition identification from EHR	Intervention	Control	Age	Sex	Disease Severity	Target Outcomes
Schnipper et al. (2010) [38] United States	Type II Diabetes Mellitus	RCT	6 months	The Agency for Healthcare Research and Quality	Partners HealthCare System, 10 practices at Brigham and Women's Hospital and Massachusetts General Hospital (Academic)	NS[1]	3431 patients	3578 patients	Comparable	IG [4]: more females (statistically significant)	Comparable	Patients meeting management goals
Gill et al. (2017) [39] United States	Type II Diabetes Mellitus	Cluster RCT	1 year	Sanofi, U.S.	Primary care practices in Delaware and Maryland (Community)	International classification of diseases (ICD-9CM) EHR code	5offices 29PCPs 4484 patients Analysed (by subgroups): HbA1c :2041 LDL-C:2793	5offices 23PCPs 1902 patients Analysed (by subgroups): HbA1c: 723patients LDL-C: 931patients	Overall: 18-75 IG: older (statistically significant)	Comparable	IG: Better glycaemic control, lower LDL-C	HbA1c[2]and LDL-C [3]
Heselmans et al. (2020)[40] Belgium	Type II Diabetes Mellitus	Cluster RCT	1 year	None	Belgian primary care practices using OneHealth EHR (Community)	International classification of primary care (ICPC-2) EHR code or confirming lab tests	25 offices 60 PCPs Analysed: 14offices 21 PCPs 1351 patients	26 offices 60 PCPs Analysed: 15offices 31 PCPs 2464 patients	Overall: ≥ 18 years IG: older (significant statistically)	CG[5]: more females (statistically significant)	Comparable for: mean HbA1c, mean BP, % patients meeting target for HbA1c , BP and LDL-C	HbA1c, LDL-C, and blood pressure
Eccles et al. (2002)[41] United Kingdom	Asthma and Angina	Cluster RCT	1 year	UK NHS research & development	General practices in north east England. (Community)	Asthma or angina EHR code in prior 12 months	Asthma: 31 offices Angina: 29 offices	Angina: 29offices, Asthma: 31offices	≥ 18 years no details presented	NS	NS	Quality of life and condition specific outcomes
Gill et al. (2009) [42] United States	Hyperlipidaemia	Cluster RCT	1 year	None	Medical quality improvement consortium (MQIC) of primary care practices across the country (Academic and Community)	Identified as having LDL levels recorded in EMR	12 offices 49 PCPs 26,696 patients	13 offices 56 PCPs 37,454 patients	Overall: 20-79 years CG younger	CG: more males (statistically significant)	CG lower disease risk	LDL-C
Cobos et al. (2006)[43] Spain	Hyperlipidaemia	Cluster RCT	1 year	Novartis	General practices in Catalonia	Identified as having total cholesterol (TC) between 200mg/dL and 400 mg/dL	20 offices 1046 patients	22 offices 1145 patients	Overall: 18-94 years	CG: more females (statistically significant)	comparable	Lipid values
Hicks et al. (2008)[44] United States	Hypertension	Cluster RCT	1.5 years	The Agency of Healthcare Research and Quality	Primary care practices affiliated with the Brigham and Women's Hospital	Identified as having 2 or more hypertension- related outpatient visits using EHR codes	7 offices 786 patients	7 offices 1048 patients	Comparable	Comparable>20 years	IG: significantly higher % of patients with controlled BP at first visit	Blood pressure control
Lopez et al. (2019)-2016 [45] United States	Hypertension	Non- RCT	1 year (2016)	Centre for Disease Control	Primary care practices in New York	Identified as having at least 1 ICD code for hypertension in EHR	7 offices	7 offices	18-85	NS	NS	Blood pressure control

[1] NS=not stated
[2] HbA1c=glycated haemoglobin
[3] LDL-C=low-density lipoprotein cholesterol

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[4] IG=intervention group
[3] CG=control group

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 2: CDSS characteristics of interventions in included studies

Study	Disease	Origin	Delivery Mode	User Response	Events	Guidelines	Integrated EHR (name)	User Training	User Adherence	Usability	Auxiliary features
Schnipper et al. (2010)[38]	Diabetes mellitus	Commercial	User- initiated	Optional	• Requests missing data. • Alerts outputs of assessments of clinical care. • Recommends orders for medication changes, laboratory studies, appointments, and referrals	NS	Longitudinal Medical Record	Brief instructions at start Monthly tips	Tracked use by clinician and sent customised emails	Used for 5.6% of eligible patients	Local user involvement in development process
Gill et al. (2017)[39]	Diabetes mellitus	Commercial	System- initiated	Optional	• Generates reports for patient-specific recommendations	American diabetes Association European Association for the study of Diabetes & ICD 9 coding	Multiple EHRs (previously employed)	Not provided	Qualitatively assessed through informal interviews	4/5 practices reported use	Recommendations also received by non-clinicians in care team
Heselmans et al. (2020)[40]	Diabetes mellitus	Commercial	User- initiated	Unclear	• Alerts for reminders, therapeutic suggestions, and diagnosis-specific guideline links	Belgian and International Guidelines	HealthOne	Not provided	Not assessed	Not assessed	None
Eccles et al. (2002)[41]	Asthma and Angina	Local	System- initiated (4 months) then altered to user-initiated	Optional	• Alerts management and prescription options and patient referrals	NS	AAH Meditel or EMIS systems	1-day training	Measured by a usage log	Median number of system interactions was zero	Education on system guidelines
Gill et al. (2009)[42]	Hyperlipidaemia	Commercial	System- initiated	Optional	• Alerts for overdue lipid tests • Alerts if patient has not met personalised lipid goals • Generates disease risk categories	Adult Treatment Panel-III (ATP-III) guidelines	EMR (Centricity)	1-year familiarity with system	Not measured	Not measured	Generates a list of patients who have not met lipid goals, outside office visits. Listed patients were sent a letter at the start and after 6 months of the study.
Cobos et al. (2006)[43]	Hyper lipidaemia	Local	System- initiated	Optional	• Recommends therapeutic options • Recommends follow-ups • Recommends lab testing	The European Society of Cardiology and other societies for Hypercholesterolemia management (ESCHM)	Not specified	None	Monitored by CDSS	71%	None
Hicks et al. (2008)[44]	Hypertension	Local	System- initiated	Optional	Alerts for therapeutic options	American Heart Association/American College of Cardiology (AHA/ACC) 2001 guidelines	Not specified	None	Assessed prescription of a recommended drug class within 1 week of clinic visit	NS	None
Lopez et al (2019)[45]	Hypertension	Local	System- initiated	Optional	• Alerts for missing BP readings • Alerts for therapeutic options	Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure (JNC- 7) guidelines	eClinicalWorks or MD Land	2-day training at start, 7 training sessions during second year, ad-hoc sessions as needed	Semi-structured interviews	Reported all practices have used all CDSS components	Generates reports of patients by status to manage visits

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 3: The effects of CDSS on patient health outcomes

				Absolute Effects		Relative Effects		
Study	**Disease**	**Outcome**	**Outcome definition**	**Intervention**	**Usual Care**		**Conclusions**	**Quality**
Schnipper et al. (2010)[38]	Diabetes	Odds of patients meeting management goals at the end compared to baseline	Targets undefined	Not reported		**ITT**: OR = 1.02; 95% CI, 0.99 to 1.04, P not reported **PP:** OR = 1.16; 95% CI, 1.08 to 1.25; P 0<.001	**ITT**: non- significant odds of intervention patients meeting management goals compared to control patients at 6 months. **PP**: significant odds of intervention patients meeting management goals compared to control patients after follow-up period.	⊕⊕◯◯ LOW
Gill et al. (2017) [39]	Diabetes	Difference in A1C from baseline Difference in LDL-C from baseline Odds of patients achieving A1C and LDL-C goals	**HbA1c goals**: <7%and <7.5%in patients with microvascular complications. **LDL-C goals:** < 100mg/dL	**Difference in HbA1c from baseline (%)**: −0.08% ±1.15% **Difference in LDL-C from baseline (mg/dL)**: −4.35± 24.54 mg/dL	**Difference in HbA1c from baseline (%)**: −0.14% ± 1.51% **Difference in LDL-C from baseline (mg/dL):** −1.70 ± 26.16 mg/dL	**Adjusted HbA1c between-group difference (%)**: 0.12%, 95% CI 0.02 to 0.22. P = 0.02 **Adjusted LDL-C between-group difference (mg/dL):** 3.57 mg/dL 95% CI 1.80 to5.34. P < 0.0001 **Odds of achieving personalized HbA1c goals:** adjusted odds ratio [aOR] 1.52, 95% CI 1.24 to 1.86; P < 0.0001 **Odds of achieving HbA1c <7%:** 1.56, 95% CI 1.27 to 1.91. P < 0.0001 **Odds of achieving HbA1c ≤9%:** 1.07, 95% CI 0.75 to 1.52; P = 0.71 **Odds of achieving LDL-C < 100mg/dL:** 1.34; 95% CI 1.11 to1.61. P = 0.002	Clinically small, yet statistically significant between-group reductions in HbA1c and LDL-C	⊕⊕⊕◯ MODERATE
Heselmans et al. (2020)[40]	Diabetes	Mean change of HbA1c Mean change in blood pressure Mean change in LDL-C levels	NA	Not reported		**Between-group difference in HbA1c means** -0.09%, 95% CI − 0.18 to 0.01, p-value = 0.06) **Between-group change difference in systolic BP:** 0.13% (95% CI − 0.91 to 1.16) p=0.81 **Between-group change difference in diastolic BP**: 0.12% (95%CI − 1.25 to 1.49) p=0.86 **Between-group change difference in LDL-C:** 1.76% (95%CI − 0.46 to3.98) p=0.12	No significant changes in between-group means for HbA1c, BP, or LDL-C	⊕⊕⊕◯ MODERATE
Eccles et al (2002)[41]	Asthma and angina	**Patient reported outcomes using**: **generic measures**: SF36, EQ-5D **Condition specific measures**: The Seattle Angina Questionnaire, The Newcastle Asthma Symptoms Questionnaire		Not reported			No effect on any patient outcome	⊕⊕⊕◯ MODERATE

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Gill et al. (2009)[42]	Hyperlipidaemia	The percentage of patients whose most recent low- density lipoprotein. Cholesterol (LDL-C) was at goal (%)	**Defined by the ATP III guidelines** LDL goals: < 100 for high-risk < 130 for moderate risk <160 for low risk patients	**High risk:** Baseline: 50% End: 53.3% P-value: <0.001 **Moderate risk**: Baseline: 64.1% End: 64.7% P value: >0.1 **Low risk:** Baseline: 88.5% End: 87.9% P value <0.05	**High risk:** Baseline: 52.3% End:56.1% P value: <0.001 **Moderate risk**: Baseline:67.1% End: 68.5% P <0.001 **Low risk:** Baseline: 90.2% End: 90.9% P value <0.001	**High risk:** OR = 1.17 (95% CI not reported), p-value <0.05 **Moderate risk:** OR = 0.29 (95% CI not reported), p-value<0.05 **Low risk:** OR =1.74 (95% CI not reported), p-value <0.05	A significant increase in proportion of patients attaining goals for all categories except the moderate-risk intervention group. No significant difference in improvement in any category compared to the control group	⊕⊕◯◯ LOW
Cobos et al. (2006)[43]	Hyperlipidaemia	Between- group differences in means of lipid values (mg/dL)	NA	**Total cholesterol:** 233.8 **LDL-C:**149.2 **HDL-C:**58.0 **Triglycerides:** 136.6	**Total cholesterol:** 231.0 **LDL-C:**146.5 **HDL-C:**56.3 **Triglycerides:** 135.2	**Difference, 95% CI, P- value** **Total cholesterol:** -2.8 (–1.7 to –7.3**),** 0.218 **LDL-C:** -2.7 (–1.7 to –7.1), 0.227 **HDL-C:** -1.6 (–0.6 to –3.6), 0.142 **Triglycerides:** -1.4 (–8.3 to –11.2), 0.766	No impact detected on final lipid values	⊕⊕⊕◯ MODERATE
Hicks et al. (2008)[44]	Hypertension	**Absolute effects**: Percentage of patients with controlled BP (%) Mean baseline and end SBP and DBP (mmHg) **Relative effects**: Odds of blood pressure control at outcome visit Baseline and End difference in SBP and DBP (mmHg)	**Controlled BP:** <130/80mmHg for patients with diabetes or renal failure <140/90 mmHg for other patients	48% Baseline SBP: 136 mmHg End SBP: 138 mmHg Baseline DBP: 78 mmHg End DBP: 77 mmHg	45% BaselineSBP:138mmHg End SBP: 137mmHg Baseline DBP: 80 mmHg End DBP: 78 mmHg	**OR**: 0.96 Baseline between-group difference in SBP: 2mmHg (p-value) 0.59 Baseline between-group difference in DBP: 2mmg Hg (p-value): 0.07 End between-group difference in SBP: 0 mm Hg (p-value):0.67 End between-group difference DBP: 1 mmHg (p-value):0.05	No differences between intervention groups in mean systolic or diastolic blood pressure readings at (end)outcome visits	⊕⊕◯◯ LOW
Lopez et al. (2019)[45]	Hypertension	**Relative effects:** odds of blood pressure control post-intervention	**Controlled BP:** <140/90 mmHg for all patients	Not reported		**OR**: 1.36 95% CI (1.08 to 1.71) (p-value: 0.009)	Significant odds of improvement of BP in the intervention group compared to usual care	⊕⊕◯◯ LOW

For peer review only

Figure 1: Analytic framework

Figure 2: PRISMA flowchart detailing the systematic search results

References

1) Viswanathan M, Golin CE, Jones CD, Ashok M, Blalock SJ, Wines RCM, et al. Interventions to improve adherence to self-administered medications for chronic diseases in the United States: a systematic review. *Annals of Internal Medicine.* 2012; 157 (11): 785-795. doi: 10.7326/0003-4819-157-11-201212040-00538

2) *Health and Economic Costs of Chronic Diseases | CDC.* Available from: https://www.cdc.gov/chronicdisease/about/costs/index.htm

3) Van Oostrom SH, Picavet HSJ, de Bruin SR, et al. Multimorbidity of chronic diseases and health care utilization in general practice. *BMC family practice.* 2014; 15 61. doi: 10.1186/1471-2296-15-61

4) Lehnert T, Heider D, Leicht H et al. Review: health care utilization and costs of elderly persons with multiple chronic conditions. *Medical care research and review: MCRR.* 2011; 68 (4): 387-420. doi: 10.1177/1077558711399580

5) *WHO | Innovative Care for Chronic Conditions: Building Blocks for Action.* Available from: http://www.who.int/chp/knowledge/publications/icccreport/en/

6) Francesconi P, Ballo P, Profili F, et al. Chronic Care Model for the Management of Patients with Heart Failure in Primary Care. *Health Services Insights.* 2019; 12 1178632919866200. Doi: 10.1177/1178632919866200

7) Tellefson M, Dipnarine K, Stopka C. The chronic care model and diabetes management in US primary care settings: a systematic review. *Preventing chronic disease.* 2013; 10 E26. Doi:10.5888/pcd10.120180

8) Oprea L, Cojocaru D, Sandu A, et al. The Chronic Care Model (CCM) and the Social Gradient in Health. *Revista de Cercetare si Interventie Sociala, ISSN-e 1584-5397, Vol 41, 2013, págs 176-189*. 2013

9) Murad MH. Clinical Practice Guidelines: A Primer on Development and Dissemination. *Mayo Clinic Proceedings.* 2017; 92 (3): 423-433. doi: 10.1016/j.mayocp.2017.01.001

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

10) Fürthauer J, Flamm M, Sönnichsen A. Patient and physician related factors of adherence to evidence based guidelines in diabetes mellitus type 2, cardiovascular disease and prevention: a cross sectional study. *BMC family practice.* 2013; 14 47. doi: 10.1186/1471-2296-14-47

11) Nichol MB, Knight TK, Priest JL, et al. Nonadherence to clinical practice guidelines and medications for multiple chronic conditions in a California Medicaid population. *Journal of the American Pharmacists Association: JAPhA.* 2010; 50 (4): 496-507. doi: 10.1331/JAPhA.2010.09123

12) Wasylewicz ATM, Scheepers-Hoeks, Clinical Decision Support Systems. *Fundamentals of Clinical Data Science,* 2018. doi:10.1007/978-3-319-99713-1_11

13) Jao C. Efficient Decision Support Systems: Practice and Challenges in Biomedical Related Domain. BoD–Books on Demand; 2011 Sep 6; doi:10.5772/937

14) Haynes RB, Wilczynski NL. Effects of computerized clinical decision support systems on practitioner performance and patient outcomes: methods of a decision-maker-researcher partnership systematic review. *Implementation science: IS.* 2010; 5 12. doi: 10.1186/1748-5908-5-12

15) Reed TS, David P, Daniel CB, et al. An overview of clinical decision support systems: benefits, risks, and strategies for success. *NPJ Digital Medicine.* 2020; 3 17-17. doi: 10.1038/s41746-020-0221-y

16) Shortliffe EH, Shortliffe EH, Cimino JJ, et al. Biomedical informatics: computer applications in health care and biomedicine. *Springer;* 2014.

17) Kaushal JSA, Alison E, Sarah N, Diane H, et al. Effects of workload, work complexity, and repeated alerts on alert fatigue in a clinical decision support system. *BMC Medical Informatics and Decision Making*. 2017-04-10 2017;17(1):1-9. doi:10.1186/s12911-017-0430-8

18) Shahsavarani AM, Azad Marz Abadi E, Hakimi KalkhoranM,et al. Clinical Decision Support Systems (CDSS): State of the art Review of Literature. *International Journal of Medical Reviews.* 2015; 2 (4): 299-308.

19) Bright TJ, Wong A, Dhurjati R, et al. Effect of clinical decision-support systems: a systematic review. *Annals of Internal Medicine.* 2012; 157 (1): 29-43. doi: 10.7326/0003-4819-157-1-201207030-00450

20) Souza NM, Sebaldt RJ, Mackay JA, et al. Computerized clinical decision support systems for primary preventive care: a decision-maker-researcher partnership systematic review of

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

effects on process of care and patient outcomes.:*Centre for Reviews and Dissemination (UK)*; 2011. doi:10.1186/1748-5908-6-87

21) Garg AX, Adhikari NKJ, McDonald H, et al. Effects of computerized clinical decision support systems on practitioner performance and patient outcomes: a systematic review. *JAMA.* 2005; 293 (10): 1223-1238. doi: 10.1001/jama.293.10.1223

22) Lobach D, Sanders GD, Bright TJ, et al. *Methods.:* Agency for Healthcare Research and Quality (US); 2012.

23) Bryan C, Boren S. The use and effectiveness of electronic clinical decision support tools in the ambulatory/primary care setting: a systematic review of the literature. *Journal of Innovation in Health Informatics.* 2008; 16 (2): 79-91. doi: 10.14236/jhi.v16i2.679

24) Kwan JL, Lo L, Ferguson J, et al. Computerised clinical decision support systems and absolute improvements in care: meta-analysis of controlled clinical trials. bmj. 2020 Sep 17;370.doi:10.1136/bmj.m3216

25) Moher D, Liberati A, Tetzlaff J, et al. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. *PLOS Medicine.* 2009; 6 (7): e1000097. doi: 10.1371/journal.pmed.1000097

26) Moher AL, Douglas GA, Jennifer T, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. 2009-07-21 2009;doi:10.1136/bmj.b2700

27) Howard C. *Subject and Course Guides: Evidence Based Medicine: PICO.* Available from: https://researchguides.uic.edu/c.php?g=252338&p=3954402

28) Zwar RR, Sarah D, Iqbal H, et al. A systematic review of chronic disease management interventions in primary care. OriginalPaper. *BMC Family Practice*. 2018-01-09 2018;19(1):1-13. doi:doi:10.1186/s12875-017-0692-3

29) Jansma EP, Vries RD. Patient reported outcome measures (PROMs*)*.bmi-online search blocks.Available from: https://blocks.bmi-online.nl/catalog/248.

30) Data extraction forms. https://dplp.cochrane.org/data-extraction-forms

31) Lobach D, Sanders GD, Bright TJ, et al. *Data Abstraction Guidance.* : Agency for Healthcare Research and Quality (US); 2012.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

32) Zhao KF, Kelsey Y, Matthew CT,et al. Risk of bias tools in systematic reviews of health interventions: an analysis of PROSPERO-registered protocols. *Systematic Reviews*. 2019-11-15 2019;8(1):1-9. doi:doi:10.1186/s13643-019-1172-8

33) Higgins JP, Altman DG, GøtzschePC,et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *Bmj.* 2011; d5928 (343): doi:10.1136/bmj.d5928

34) Higgins JP, Savović J, Page MJ, Elbers RG, Sterne JA. Assessing risk of bias in a randomized trial. Cochrane Handbook for Systematic Reviews of Interventions. 2019 Sep 23:205-28.doi:10.1136/bmj.d5928

35) Ryan R, Hill S, PrictorM,et al. *Cochrane Consumers & Communication Review Group Study Quality Guide, Guide for Review Authors on Assessing Study Quality.* 2013.

36) *What is GRADE? | BMJ Best Practice.* Available from: https://bestpractice.bmj.com/info/toolkit/learn-ebm/what-is-grade/

37) Meader N, Kristel K, Llewellyn A, et al. A checklist designed to aid consistency and reproducibility of GRADE assessments: development and pilot validation. *Systematic reviews*. 07/24/2014 2014;3doi:10.1186/2046-4053-3-82

38) Schnipper JL, Linder JA, Palchuk MB, et al. Effects of documentation-based decision support on chronic disease management. *The American Journal of Managed Care.* 2010; 16 (12 Suppl HIT) (pp SP72-81): ate of Pubaton: e 2010.

39) Gill J, Kucharski K, Turk B, Pan C, Wei W. Using Electronic Clinical Decision Support in Patient-Centered Medical Homes to Improve Management of Diabetes in Primary Care: The DECIDE Study. *Journal of Ambulatory Care Management.* 2019; 42 (2): 105-115.doi:10.1097/JAC.0000000000000267

40) Heselmans A, Delvaux N, Laenen A, Van de Velde S, Ramaekers D, Kunnamo I, et al. Computerized clinical decision support system for diabetes in primary care does not improve quality of care: a cluster-randomized controlled trial. *Implementation science : IS.* 2020; 15 (1): 5.doi:10.1186/s13012-019-0955-6

41) Eccles M, McColl E, Steen N, Liaw ST. Computerised evidence-based guidelines may not improve asthma or angina management in primary care. *Evidence-Based Healthcare.* 2003; 7 (2): 81-82.

42) Gill JM, Chen YX, Glutting JJ, et al. Impact of decision support in electronic medical records on lipid management in primary care. *Population health management.* 2009; 12 (5): 221-226.doi:10.1089/pop.2009.0003. doi:10.1089/pop.2009.0003

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

43) Cobos A, Vilaseca J, Asenjo C, et al. Cost effectiveness of a clinical decision support system based on the recommendations of the European Society of Cardiology and other societies for the management of hypercholesterolemia: Report of a cluster-randomized trial. *Disease Management and Health Outcomes.* 2005; 13 (6): 421-432.doi:10.2165/00115677-200513060-00007

44) Hicks LS, Sequist TD, Ayanian JZ, et al. Impact of computerized decision support on blood pressure management and control: A randomized controlled trial. *Journal of General Internal Medicine.* 2008; 23 (4): 429-441.doi:10.1007/s11606-007-0403-1

45) Lopez PM, Divney A, Goldfeld K, et al. Feasibility and Outcomes of an Electronic Health Record Intervention to Improve Hypertension Management in Immigrant-serving Primary Care Practices. *Medical care.* 2019; 57 (Supplement 62) (pp S164-S171): ate of Pubaton: 01 Jun 2019.doi:10.1097/MLR.0000000000000994

46) Goud R, de Keizer NF, ter Riet G, et al. Effect of guideline based computerised decision support on decision making of multidisciplinary teams: cluster randomised trial in cardiac rehabilitation. *The BMJ.* 2009; 338.doi: 10.1136/bmj. b1440

47) Manns BJ, Tonelli M, Zhang J, et al. Enrolment in primary care networks: impact on outcomes and processes of care for patients with diabetes. *CMAJ: Canadian Medical Association journal = journal de l'Associationmedicalecanadienne.* 2012; 184 (2): 144.: doi: 10.1503/cmaj.110755

48) McClellan SR, Casalino LP, Shortell SM, et al. When does adoption of health information technology by physician practices lead to use by physicians within the practice? *Journal of the American Medical Informatics Association: JAMIA.* 2013; 20 (e1): e26-e32. doi: 10.1136/amiajnl-2012-001271

49) Latoszek-Berendsen A, Tange H, van den Herik, H. J.,et al. From clinical practice guidelines to computer-interpretable guidelines. A literature overview. *Methods of Information in Medicine.* 2010; 49 (6): 550-570. doi: 10.3414/ME10-01-0056

50) Hajjaj FM, Salek MS, Basra MK,et al. Non-clinical influences on clinical decision-making: a major challenge to evidence-based practice. Journal of the Royal Society of Medicine. 2010 May 1;103(5):178-87.doi:10.1258/jrsm.2010.100104

51) Varghese J, Kleine M, Gessner SI,et al. Effects of computerized decision support system implementations on patient outcomes in inpatient care: a systematic review. *Journal of the American Medical Informatics Association: JAMIA.* 2018; 25 (5): 593-602. doi: 10.1093/jamia/ocx100

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

52) Moja L, Kwag KH, Lytras T, Bertizzolo L, Brandt L, Pecoraro V, et al. Effectiveness of Computerized Decision Support Systems Linked to Electronic Health Records: A Systematic Review and Meta-Analysis. *American Journal of Public Health.* 2014; 104 (12): e12-e22. Available from: doi: 10.2105/AJPH.2014.302164

53) Groenhof TKJ, Asselbergs FW, GroenwoldRHH,et al. The effect of computerized decision support systems on cardiovascular risk factors: a systematic review and meta-analysis. *BMC medical informatics and decision making.* 2019; 19 (1): 108. doi: 10.1186/s12911-019-0824-x

54) Roshanov PS, Misra S, Gerstein HC, et al. Computerized clinical decision support systems for chronic disease management: A decision-maker-researcher partnership systematic review. *Implementation science : IS.* 2011; 6 (1): 92. doi: 10.1186/1748-5908-6-92

55) Kooij L, Groen WG, van Harten WH. The Effectiveness of Information Technology-Supported Shared Care for Patients With Chronic Disease: A Systematic Review. *Journal of Medical Internet Research.* 2017; 19 (6).doi: 10.2196/jmir.7405

56) Castaneda C, Nalley K, Mannion C, et al. Clinical decision support systems for improving diagnostic accuracy and achieving precision medicine. *Journal of Clinical Bioinformatics.* 2015; 5 .doi: 10.1186/s13336-015-0019-3

57) Gibbons RJ, Chatterjee K, Daley J,et al. ACC/AHA/ACP-ASIM guidelines for the management of patients with chronic stable angina: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines (Committee on Management of Patients With Chronic Stable Angina). *Journal of the American College of Cardiology*. 1999 Jun;33(7):2092-197. doi:10.1016/s0735-1097(99)00150-3

58) Heneghan C, Goldacre B, Mahtani KR. Why clinical trial outcomes fail to translate into benefits for patients. *Trials.* 2017 Dec;18(1):1-7. doi:10.1186/s13063-017-1870-2

59) Afzal M, Hussain M, Ali T, et al Knowledge-based query construction using the CDSS knowledge base for efficient evidence retrieval. *Sensors*. 2015 Sep;15(9):21294-314.doi:10.3390/s150921294

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

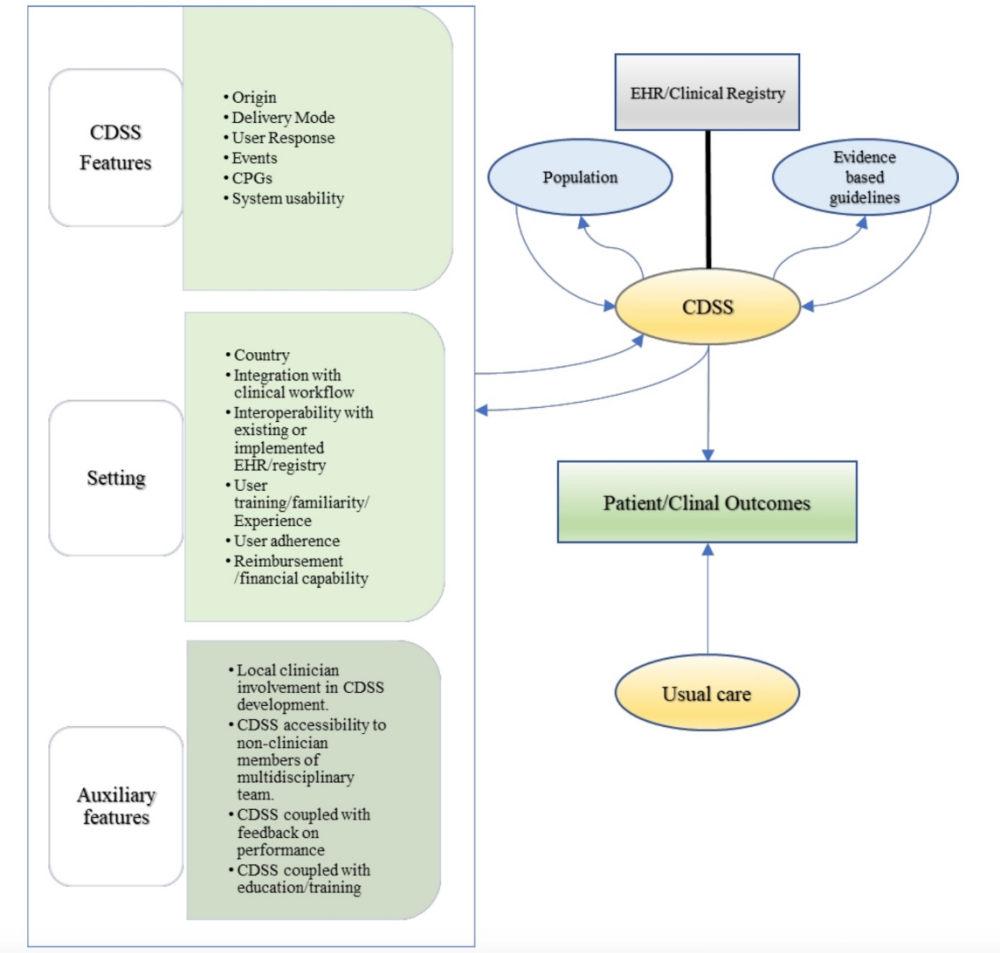


Figure 1: Analytic framework

488x471mm (300 x 300 DPI)

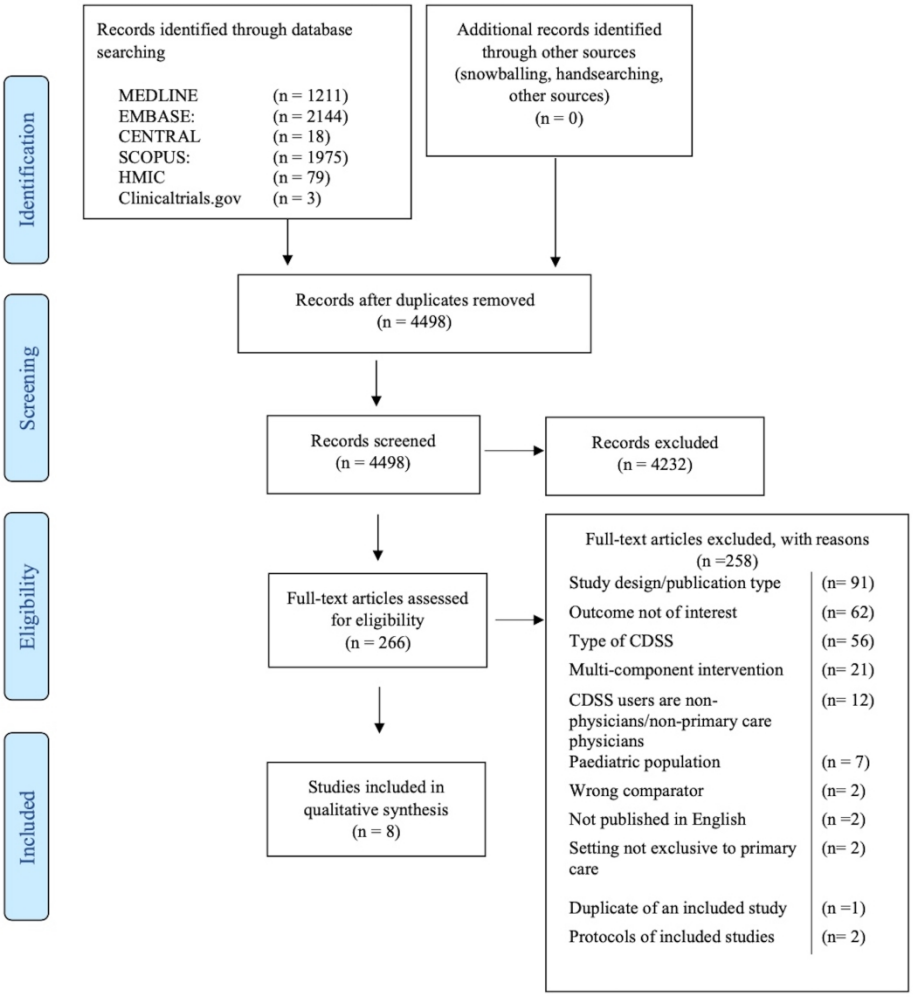


Figure 2: PRISMA flowchart detailing the systematic search results

405x436mm (300 x 300 DPI)

Supplementary Material

Supplementary Table 1: Eligibility criteria defined by PICOS

P	**Participants:** - Patients above 18 years with one or more concurrent chronic condition, regardless of sex, education, marital status, employment status, income, or other socioeconomic factors who were managed in a primary care setting using a EHR or clinical registry) supported by a computerised, knowledge-based CDSS. - Chronic conditions were defined and limited to the following conditions: o Asthma, chronic obstructive pulmonary disease (COPD), heart failure, myocardial ischemia, hypertension, type II diabetes mellitus, hyperlipidaemia, arthritis, and osteoporosis.
I	**Interventions:** Computerised knowledge based CDSS embedded EHRs or clinical registries in primary care settings as an intervention with the following inclusion and exclusion criteria: **Studies will be included if:** - They focus on evaluating a CDSS for medical management of the selected chronic diseases - The CDSS is of knowledge-based type (informed by CPGs). - Included CDSS is synchronous, meaning alerts are electronically presented to the clinician in real-time during the physician-patient encounter. - The CDSS users are primary care physicians (PCPs). - They report at least: one quantifiable patient clinical outcome/patient-related measure in adult populations in the specified chronic disease entities **Studies will be excluded if**: - Not published in English. - Not conducted in primary care settings. - Include care provided by subspecialized physicians (not PCPs). - Evaluates a CDSS for other outcome measures such as system usability, or other quality outcomes, not including patient or clinical outcomes. - Interventions that are used for other direct care purposes other than chronic disease management or non-direct care purposes, such as research purposes. - Evaluate the effect of a CDSS in other diseases not specified in the inclusion criteria. - Include a CDSS that does not produce real-time recommendations based on CPGs (non-knowledge based CDSS). - Web-based CDSS, paper-based CDSS, standalone CDSS (not integrated with EHRs), CDSS integrated with EHR but not producing electronic alerts, and other CDSS that use e-mail, phone, or pagers to notify the user. - Studies without a comparator group - Comparator group is another type of CDSS.
C	**Comparators:** Usual care not utilising CDSS
O	**Outcomes:** - Patient-related health and clinical outcomes in chronic disease management (such as HBA1C, $FEV_{1,}$ or other biomarkers of chronic diseases), morbidity, mortality, HRQoL, preventable co-morbidities, or disease activity) - Practitioner-related, patient-related, and cost-effectiveness outcomes were not included.
S	**Study types**: Randomised controlled trials (RCTs), cluster RCTs, quasi-RCTs, interrupted time series (ITS) and controlled before-and-after (CBA) studies

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Table 2: List of search terms

CDSS terms	Dashboard*, registry, clinical decision support system*, medical order entry system*, computeri?ed physician order entry system*, CPOE, order set*, computer assisted decision making, (computer$ adj3 decision$), knowledge management system*, case-finding algorithm*, advisory decision support, (computer* adj2 therap*)

AND

General Practice/Primary care	general practi*, family practice, family clinic*, family practi*, family physician*, family doctor*, primary care physician*, primary care clinic*, primary care practi*,

OR each of the below individual entities combined with AND

Chronic Disease Entities	
Asthma	asthma*, antiasthma*, wheez*, (bronch* adj3 spasm*), bronchoconstrict*, (bronch$ adj3 constrict$), ((bronchial$ or respiratory or airway$ or lung$) adj3 (hypersensitiv$ or hyperreactiv$ or allerg$ or insufficiency)),
Heart Failure/ Myocardial Ischemia	(heart* adj2 decomp*), ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*)), ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*))
Hypertension	((elevat$ or increas$ or lower or high or rais$ or rising) adj2 (arterial pressure or blood pressure or diastolic pressure or systolic pressure)), hypertens$ or prehypertens$),
Diabetes Mellitus	(IDDM or T2DM or T2D), (insulin* depend* or insulin?depend*)., (non insulin* depend* or noninsulin* depend* or non insulin?depend* or noninsulin?depend*)., diabet*.
Hyperlipidemia	LDL, (dyslipid?emia* or lipoprotein* or dyslipidprotein?emia* or hyperlipoprotein?emia), (hyperlipid?emia* or cholesterol* or lip?emia* or hypercholesterol?emia* or hypercholester?emia* or hyperlip?emia* or triglycerid* or hypertriglycerid?emia*).,
COPD	(COPD or COAD or COBD or AECB), ((obstruct$ adj3 lung$) or (obstruct$ adj 3 airway$)), (obstruct$ adj3 pulmonary), (chronic* adj3 bronchiti*), emphysema*.,
Arthritis	Osteoarthritis, undifferentiated oligoarthritis, ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$))., (joint pain or chronic joint symptoms or gonarthrosis or osteoarthrosis or astoarthrosis or degeneretive arthritis or joint diseases or arthralgia),
Osteoporosis	Densitometry, osteopenia, bone mass, bone densit*, osteoporos?s, (osteoporosis or osteopenia or (bone and (density or mass or loss)) or hepatic osteodystrophy),

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Table 3 - Full Search Strategy

Full Medline Search Strategy

1 Dashboard*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1228)
2 registry.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (109312)
3 registries.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (104552)
4 exp Registries/ (95452)
5 1 or 2 or 3 or 4 (173943)
6 exp Decision Support Systems, Clinical/ (7925)
7 clinical decision support system*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (2132)
8 medical order entry system*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (2297)
9 computeri?ed physician order entry system*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (224)
10 CPOE.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1113)
11 order set*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1009)
12 exp Decision Making, Computer-Assisted/ (144386)
13 computer assisted decision making.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (76)
14 (computer$ adj3 decision$).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (4723)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

15 knowledge management system*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (109)

16 case-finding algorithm*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (84)

17 advisory decision support.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1)

18 exp Therapy, Computer-Assisted/ (63908)

19 (computer* adj2 therap*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (8758)

20 exp Reminder Systems/ (3443)

21 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 (159108)

22 5 or 21 (331941)

23 exp Asthma/ (127501)

24 asthma*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (179286)

25 antiasthma*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (1458)

26 exp Respiratory Sounds/ (12994)

27 wheez*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (13724)

28 exp Bronchial Spasm/ (4303)

29 (bronch* adj3 spasm*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (4523)

30 exp Bronchoconstriction/ (4131)

31 bronchoconstrict*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (11491)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

32 (bronch$ adj3 constrict$).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (715)

33 exp Bronchial Hyperreactivity/ (7348)

34 exp Respiratory Hypersensitivity/ (154474)

35 ((bronchial$ or respiratory or airway$ or lung$) adj3 (hypersensitiv$ or hyperreactiv$ or allerg$ or insufficiency)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (66254)

36 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 (261794)

37 exp Heart Failure/ (120835)

38 (heart* adj2 decomp*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (4261)

39 ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (664015)

40 ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (664015)

41 exp Myocardial Ischemia/ (428216)

42 exp Coronary Disease/ (216812)

43 37 or 38 or 39 or 40 or 41 or 42 (842183)

44 exp Hypertension/ (253393)

45 exp Essential Hypertension/ (2255)

46 (hypertens$ or prehypertens$).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (510526)

47 ((elevat$ or increas$ or lower or high or rais$ or rising) adj2 (arterial pressure or blood pressure or diastolic pressure or systolic pressure)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (63244)

48 44 or 45 or 46 or 47 (537839)

49 exp Diabetes Mellitus/ (423542)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

50 diabet*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (693674)

51 (non insulin* depend* or noninsulin* depend* or non insulin?depend* or noninsulin?depend*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (12372)

52 (insulin* depend* or insulin?depend*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (29531)

53 (IDDM or T2DM or T2D).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (37329)

54 exp Diabetes Mellitus, Type 2/ (132230)

55 49 or 50 or 51 or 52 or 53 or 54 (697363)

56 exp Hyperlipidemias/ (65918)

57 exp Dyslipidemias/ (79414)

58 exp Cholesterol/ (159670)

59 (hyperlipid?emia* or cholesterol* or lip?emia* or hypercholesterol?emia* or hypercholester?emia* or hyperlip?emia* or triglycerid* or hypertriglycerid?emia*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (389543)

60 (dyslipid?emia* or lipoprotein* or dyslipidprotein?emia* or hyperlipoprotein?emia*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (209412)

61 LDL.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (101128)

62 56 or 57 or 58 or 59 or 60 or 61 (479855)

63 exp Pulmonary Disease, Chronic Obstructive/ (55377)

64 exp Lung Diseases, Obstructive/ (210481)

65 emphysema*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (35893)

For peer review only

66 (chronic* adj3 bronchiti*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (11493)

67 (obstruct$ adj3 pulmonary).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (67392)

68 ((obstruct$ adj3 lung$) or (obstruct$ adj 3 airway$)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (24368)

69 (COPD or COAD or COBD or AECB).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (46979)

70 63 or 64 or 65 or 66 or 67 or 68 or 69 (256014)

71 exp Arthritis/ (258204)

72 exp Osteoarthritis/ (62805)

73 osteoarthritis.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (88630)

74 exp Joint Diseases/ (378029)

75 exp Arthralgia/ (13069)

76 (joint pain or chronic joint symptoms or gonarthrosis or osteoarthrosis or astoarthrosis or degeneretive arthritis or joint diseases or arthralgia).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (49768)

77 ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (176792)

78 undifferentiated oligoarthritis.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (36)

79 71 or 72 or 73 or 74 or 75 or 76 or 77 or 78 (466351)

80 exp Osteoporosis/ (55499)

81 (osteoporosis or osteopenia or (bone and (density or mass or loss)) or hepatic osteodystrophy).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (229156)

82 osteoporos?s.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (86861)

83 exp Bone Density/ (52984)

84 bone densit*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (70284)

85 exp Bone Demineralization, Pathologic/ (652)

86 exp Bone Diseases, Metabolic/ (76180)

87 bone mass.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (18851)

88 osteopenia.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (9555)

89 exp Densitometry/ (34775)

90 densitometry.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (20649)

91 80 or 81 or 82 or 83 or 84 or 85 or 86 or 87 or 88 or 89 or 90 (266936)

92 exp General Practitioners/ (7841)

93 exp General Practice/ (74873)

94 general practi*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (91362)

95 exp Physicians, Family/ (16348)

96 exp Family Practice/ (65161)

97 family practice.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (68324)

98 family clinic*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (470)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

99 family physician*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (14677)

100 family practi*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (69798)

101 family doctor*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (4649)

102 primary care physician*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (19694)

103 primary care clinic*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (7532)

104 exp Physicians, Primary Care/ (3421)

105 primary care practi*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (7576)

106 92 or 93 or 94 or 95 or 96 or 97 or 98 or 99 or 100 or 101 or 102 or 103 or 104 or 105 (180127)

107 22 and 106 (4326)

108 91 and 107 (50)

109 79 and 107 (76)

110 70 and 107 (184)

111 62 and 107 (206)

112 55 and 107 (568)

113 48 and 107 (239)

114 43 and 107 (343)

115 36 and 107 (148)

116 108 or 109 or 110 or 111 or 112 or 113 or 114 or 115 (1211)

117 from 116 keep 1-1211 (1211)

For peer review only

Full Embase Search Strategy

1 Dashboard*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (2390)

2 registry.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (208322)

3 registries.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (39715)

4 exp register/ (115531)

5 1 or 2 or 3 or 4 (266586)

6 exp clinical decision support system/ (3039)

7 clinical decision support system*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (4672)

8 medical order entry system*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (136)

9 computeri?ed physician order entry system*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (362)

10 CPOE.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (1664)

11 order set*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (2621)

12 exp decision support system/ (25185)

13 computer assisted decision making.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (116)

14 (computer$ adj3 decision$).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (2950)

15 knowledge management system*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (141)

16 case-finding algorithm*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (141)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

17 advisory decision support.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (1)

18 exp computer assisted therapy/ (15163)

19 (computer* adj2 therap*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (5331)

20 exp reminder system/ (2611)

21 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 (48813)

22 5 or 21 (314449)

23 exp asthma/ (275547)

24 asthma*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (309894)

25 antiasthma*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (9548)

26 exp abnormal respiratory sound/ (60406)

27 wheez*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (33727)

28 exp bronchospasm/ (27974)

29 (bronch* adj3 spasm*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (861)

30 exp bronchoconstriction/ (29996)

31 bronchoconstrict*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (12976)

32 (bronch$ adj3 constrict$).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (1187)

33 exp bronchus hyperreactivity/ (12410)

34 exp respiratory tract allergy/ (316133)

35 ((bronchial$ or respiratory or airway$ or lung$) adj3 (hypersensitiv$ or hyperreactiv$ or allerg$ or insufficiency)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (55027)

36 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 (435382)

37 22 and 36 (3673)

38 exp heart failure/ (531445)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

39 (heart* adj2 decomp*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (9079)

40 ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (1152207)

41 exp heart muscle ischemia/ (94337)

42 exp coronary artery disease/ (340285)

43 38 or 39 or 40 or 41 or 42 (1486256)

44 22 and 43 (46272)

45 exp hypertension/ (772502)

46 exp essential hypertension/ (30699)

47 (hypertens$ or prehypertens$).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (988948)

48 ((elevat$ or increas$ or lower or high or rais$ or rising) adj2 (arterial pressure or blood pressure or diastolic pressure or systolic pressure)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (92975)

49 45 or 46 or 47 or 48 (1084030)

50 22 and 49 (18606)

51 exp diabetes mellitus/ (1000793)

52 diabet*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (1186708)

53 (non insulin* depend* or noninsulin* depend* or non insulin?depend* or noninsulin?depend*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (257419)

54 (insulin* depend* or insulin?depend*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (355610)

55 (IDDM or T2DM or T2D).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (63858)

56 51 or 52 or 53 or 54 or 55 (1195069)

57 22 and 56 (26835)

58 exp hyperlipidemia/ (164564)

59 exp dyslipidemia/ (70815)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

60 exp cholesterol/ (335599)

61 (hyperlipid?emia* or cholesterol* or lip?emia* or hypercholesterol?emia* or hypercholester?emia* or hyperlip?emia* or triglycerid* or hypertriglycerid?emia*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (619049)

62 (dyslipid?emia* or lipoprotein* or dyslipidprotein?emia* or hyperlipoprotein?emia*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (401478)

63 LDL.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (117119)

64 58 or 59 or 60 or 61 or 62 or 63 (764638)

65 22 and 64 (8600)

66 exp chronic obstructive lung disease/ (134392)

67 emphysema*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (58780)

68 (chronic* adj3 bronchiti*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (23183)

69 (obstruct$ adj3 pulmonary).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (82519)

70 ((obstruct$ adj 3 lung$) or (obstruct$ adj 3 airway$)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (1)

71 (COPD or COAD or COBD or AECB).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (90187)

72 66 or 67 or 68 or 69 or 70 or 71 (226781)

73 22 and 72 (3785)

74 exp arthritis/ (511276)

75 exp osteoarthritis/ (137512)

76 exp osteoarthritis/ (137512)

77 osteoarthritis.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (149492)

78 exp arthropathy/ (816046)

79 exp arthralgia/ (61579)

For peer review only

80 (joint pain or chronic joint symptoms or gonarthrosis or osteoarthrosis or astoarthrosis or degeneretive arthritis or joint diseases or arthralgia).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (79510)

81 ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (304665)

82 ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (304665)

83 undifferentiated oligoarthritis.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (39)

84 74 or 75 or 76 or 77 or 78 or 79 or 80 or 81 or 82 or 83 (897748)

85 22 and 84 (13224)

86 exp osteoporosis/ (134965)

87 (osteoporosis or osteopenia or (bone and (density or mass or loss)) or hepatic osteodystrophy).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (383399)

88 osteoporos?s.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (160054)

89 exp bone density/ (93700)

90 bone densit*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (108740)

91 exp bone demineralization/ (137263)

92 bone mass.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (38017)

93 osteopenia.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (23760)

94 exp bone densitometry/ (5807)

95 densitometry.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (34380)

96 86 or 87 or 88 or 89 or 90 or 91 or 92 or 93 or 94 or 95 (411752)

97 22 and 96 (3018)

98 exp general practitioner/ (102749)

99 exp general practice/ (84760)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

100 general practi*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (208636)

101 family practice.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (9954)

102 family clinic*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (652)

103 family practi*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (12786)

104 family physician.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (6553)

105 family doctor.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] (4317)

106 98 or 99 or 100 or 101 or 102 or 103 or 104 or 105 (216919)

107 97 and 106 (71)

108 85 and 106 (192)

109 73 and 106 (173)

110 65 and 106 (271)

111 57 and 106 (738)

112 50 and 106 (341)

113 44 and 106 (612)

114 37 and 106 (180)

115 107 or 108 or 109 or 110 or 111 or 112 or 113 or 114 (1655)

For peer review only

Full Health Management Information Consortium Search Strategy

1. Dashboard*.mp. [mp=title, other title, abstract, heading words]
2. registry.mp. [mp=title, other title, abstract, heading words]
3. registries.mp. [mp=title, other title, abstract, heading words]
4. exp Registers/ or exp Medical registers/ or exp Patient registers/ or exp Disease registers/
5. 1 or 2 or 3 or 4
6. exp Information systems/ or exp Decision support systems/
7. clinical decision support system*.mp. [mp=title, other title, abstract, heading words]
8. medical order entry system*.mp. [mp=title, other title, abstract, heading words]
9. computeri?ed physician order entry system*.mp. [mp=title, other title, abstract, heading words]
10. CPOE.mp. [mp=title, other title, abstract, heading words]
11. order set*.mp. [mp=title, other title, abstract, heading words]
12. computer assisted decision making.mp. [mp=title, other title, abstract, heading words]
13. (computer$ adj3 decision$).mp. [mp=title, other title, abstract, heading words]
14. knowledge management system*.mp. [mp=title, other title, abstract, heading words]
15. case-finding algorithm*.mp. [mp=title, other title, abstract, heading words]
16. advisory decision support.mp. [mp=title, other title, abstract, heading words]
17. exp Computer systems/
18. (computer* adj2 therap*).mp. [mp=title, other title, abstract, heading words]
19. 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18
20. 5 or 19
21. exp Asthma/
22. asthma*.mp. [mp=title, other title, abstract, heading words]
23. antiasthma*.mp. [mp=title, other title, abstract, heading words]
24. exp Bronchial spasm/ or exp Lung function/
25. wheez*.mp. [mp=title, other title, abstract, heading words]
26. exp Bronchial spasm/
27. (bronch* adj3 spasm*).mp. [mp=title, other title, abstract, heading words]
28. bronchoconstrict*.mp. [mp=title, other title, abstract, heading words]
29. (bronch$ adj3 constrict$).mp. [mp=title, other title, abstract, heading words]
30. ((bronchial$ or respiratory or airway$ or lung$) adj3 (hypersensitiv$ or hyperreactiv$ or allerg$ or insufficiency)).mp. [mp=title, other title, abstract, heading words]
31. 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30
32. exp Heart diseases/ or exp Coronary diseases/ or exp Congestive heart failure/
33. (heart* adj2 decomp*).mp. [mp=title, other title, abstract, heading words]

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

34. ((heart* or cardiac* or myocard*) adj2 (fail* or insuff* or infarct* or ishcem* or disease*)).mp. [mp=title, other title, abstract, heading words]
35. exp Cardiovascular diseases/
36. exp Heart diseases/ or exp Heart arrest/ or exp Myocardial infarction/
37. 32 or 33 or 34 or 35 or 36
38. exp Hypertension/
39. (hypertens$ or prehypertens$).mp. [mp=title, other title, abstract, heading words]
40. ((elevat$ or increas$ or lower or high or rais$ or rising) adj2 (arterial pressure or blood pressure or diastolic pressure or systolic pressure)).mp. [mp=title, other title, abstract, heading words]
41. exp Diabetes/
42. diabet*.mp. [mp=title, other title, abstract, heading words]
43. (non insulin* depend* or noninsulin* depend* or non insulin?depend* or noninsulin?depend*).mp. [mp=title, other title, abstract, heading words]
44. (insulin* depend* or insulin?depend*).mp. [mp=title, other title, abstract, heading words]
45. (IDDM or T2DM or T2D).mp. [mp=title, other title, abstract, heading words]
46. 41 or 42 or 43 or 44 or 45
47. exp Hyperlipidaemia/
48. exp Cholesterol measurement/ or exp Cholesterol/
49. (hyperlipid?emia* or cholesterol* or lip?emia* or hypercholesterol?emia* or hypercholester?emia* or hyperlip?emia* or triglycerid* or hypertriglycerid?emia*).mp. [mp=title, other title, abstract, heading words]
50. (dyslipid?emia* or lipoprotein* or dyslipidprotein?emia* or hyperlipoprotein?emia*).mp. [mp=title, other title, abstract, heading words]
51. LDL.mp. [mp=title, other title, abstract, heading words]
52. 47 or 48 or 49 or 50 or 51
53. exp chronic obstructive pulmonary disease/
54. emphysema*.mp. [mp=title, other title, abstract, heading words]
55. (chronic* adj3 bronchiti*).mp. [mp=title, other title, abstract, heading words]
56. (obstruct$ adj3 pulmonary).mp. [mp=title, other title, abstract, heading words]
57. ((obstruct$ adj3 lung$) or (obstruct$ adj 3 airway$)).mp. [mp=title, other title, abstract, heading words]
58. (COPD or COAD or COBD or AECB).mp. [mp=title, other title, abstract, heading words]
59. 53 or 54 or 55 or 56 or 57 or 58
60. exp Arthritis/
61. exp Osteoarthritis/

For peer review only

62. Arthritis/ or exp Rheumatoid arthritis/
63. exp Joint diseases/ or exp Bone diseases/
64. (joint pain or chronic joint symptoms or gonarthrosis or osteoarthrosis or astoarthrosis or degeneretive arthritis or joint diseases or arthralgia).mp. [mp=title, other title, abstract, heading words]
65. ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$)).mp. [mp=title, other title, abstract, heading words]
66. ((rheumatoid or reumatoid or revmatoid or rheumatic or reumatic or revmatic or rheumat$ or reumat$ or revmarthrit$) adj3 (arthrit$ or artrit$ or diseas$ or condition$ or nodule$)).mp. [mp=title, other title, abstract, heading words]
67. undifferentiated oligoarthritis.mp. [mp=title, other title, abstract, heading words]
68. 60 or 61 or 62 or 63 or 64 or 65 or 66 or 67
69. exp Osteoporosis/
70. (osteoporosis or osteopenia or (bone and (density or mass or loss)) or hepatic osteodystrophy).mp. [mp=title, other title, abstract, heading words]
71. osteoporos?s.mp. [mp=title, other title, abstract, heading words]
72. bone densit*.mp. [mp=title, other title, abstract, heading words]
73. bone mass.mp. [mp=title, other title, abstract, heading words]
74. osteopenia.mp. [mp=title, other title, abstract, heading words]
75. densitometry.mp. [mp=title, other title, abstract, heading words]
76. 69 or 70 or 71 or 72 or 73 or 74 or 75
77. exp General practitioners/
78. exp General practice/
79. general practi*.mp. [mp=title, other title, abstract, heading words]
80. family clinic*.mp. [mp=title, other title, abstract, heading words]
81. family practi*.mp. [mp=title, other title, abstract, heading words]
82. family physician*.mp. [mp=title, other title, abstract, heading words]
83. family doctor*.mp. [mp=title, other title, abstract, heading words]
84. primary care physician*.mp. [mp=title, other title, abstract, heading words]
85. primary care clinic*.mp. [mp=title, other title, abstract, heading words]
86. primary care practi*.mp. [mp=title, other title, abstract, heading words]
87. 77 or 78 or 79 or 80 or 81 or 82 or 83 or 84 or 85 or 86
88. 20 and 87
89. 31 and 88

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

90. 37 and 88
91. 38 or 39 or 40
92. 88 and 91
93. 46 and 88
94. 52 and 88
95. 59 and 88
96. 68 and 88
97. 76 and 88
98. 89 or 90 or 92 or 93 or 94 or 95 or 96 or 97

For peer review only

Full Scopus Search Strategy

((*"Dashboard"* OR *"registry"* OR *"registries"* OR *"clinical decision support system"* OR *"medical order entry system"* OR *"CPOE"* OR *"decision support system"* OR *"computer assisted decision making"* OR *"case-finding algorithm"* OR *"advisory decision support"* OR *"computer assisted therapy"*) AND (*"general practi*"* OR *"family practi*"* OR *"family clinic*"* OR *"family physician"* OR *"family doctor"* OR *"primary care clinic*"* OR *" primary care physician"*)) AND ((*"asthma*"* *"wheez*"* OR *"bronchus hyperreactivity"* OR *"heart failure"* OR *"heart insufficiency"* OR *"cardiac failure"* OR *" cardiac insufficiency"* OR *"myocardial infarction"* OR *"myocardial insufficiency"* OR *"myocardial ischemia"* OR *"cardiac disease"* OR *"coronary artery disease"*) OR (*"diabet*"* OR (*"non insulin* depend*"* OR *"noninsulin* depend*"* OR *"non insulin depend*"* OR *"noninsulin?depend*"*) OR *"IDDM"* OR *"T2DM"* OR *"T2D"*) OR (*"hyperlipidemia"* OR *"dyslipidemia"* OR *"hyperlipid?emia*"* OR *"cholesterol*"* OR *"lip?emia*"* OR *"hypercholesterol?emia*"* OR *"hypercholester?emia*"* OR *"hyperlip?emia*"* OR *"triglycerid*"* OR *"hypertriglycerid?emia*"* OR *"dyslipid?emia*"* OR *"lipoprotein*"* OR *"dyslipidprotein?emia*"* OR *"hyperlipoprotein?emia*"* OR *"chronic obstructive lung disease"* OR *"emphysema*"* OR ((*"obstruct* N/3lung*"*) OR (*"obstruct* N/3 airway*"*)) OR *copd* OR *coad* OR *cobd* OR *aecb*) OR (*"hypertension"* OR *"Prehypertension"* OR ((*"elevat* OR increas* OR lower OR high OR rais* OR rising"*) W/2 (*"arterial pressure"* OR *"blood pressure"* OR *"diastolic pressure"* OR *"systolic pressure"*))) OR (*"arthritis"* OR *"osteoarthritis"* OR *"joint pain"* OR *"chronic joint symptoms"* OR *"gonarthrosis"* OR *"osteoarthrosis"* OR *"arthrosis"* OR *"degeneretive arthritis"* OR *"joint diseases"* OR *"arthralgia"* OR *"rheumat*"* OR *"reumat*"* OR *"arthrit*"* OR *"artrit*"*) OR (*"osteoporosis"* OR *"osteopenia"* OR *bone* AND *density* AND *loss* OR *"osteopenia"* OR *"bone mass loss"*)) AND NOT INDEX (*medline*) AND (LIMIT-TO (SRCTYPE , *"j"*)) AND (LIMIT-TO (DOCTYPE , *"ar"*)) AND (LIMIT-TO (SUBJAREA , *"MEDI"*)) AND (LIMIT-TO (LANGUAGE , *"English"*))

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Table 4: Risk of bias in included studies as per the Cochrane tool

Trial	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data	Selective outcome reporting	Other bias	Selective recruitment of cluster members
Shnipper et al.(2010)	✓	✗	!	!	!	✓	✓	NA
Gill et al. (2017)	✗	✓	✓	✓	✗	✗	✓	✓
Hesselman et al.(2020)	✓	✓	✓	✓	✗	✓	✓	✓
Eccles et al. (2002)	✓	✓	✓	✓	✗	✗	✓	✓
Gill et al. (2009)	✗	✓	!	!	✗	✓	✓	✓
Cobos et al. (2006)	✓	✓	!	!	✓	✓	✓	✓
Hicks et al. (2008)	✗	✓	!	!	✗	✓	✓	✓
Lopez et al. (2019)	✗	✗	✗	✗	✓	✓	✓	✓

Key
✓ Low risk
✗ High risk
! Unclear

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Table 5: Adapted risk of bias table detailing methodological risks and characteristics. PCPs refers to primary care practitioners (clinicians)

	Randomization				Baseline Comparability				Blinding			Withdrawal		
Trial	Truly Random	Allocation Concealment	Number Stated		Presented	Achieved	Inlcusion/Exclusion Criteria Specified		Assessors	Patients	PCPs	>80% in final analysis	Reason stated	Intention to treat
			Patients	PCPs			Patients	PCPs						
Shnipper et al.(2010)	✓	✗	✓	✗	✓	✓	✓	✓	NS	NS	NA	✗	NS	✓
Gill et al. (2017)	✗	✗	✓	✓	✓	✗	✓	✓	✓	✗	NA	✗	✓	✗
Hesselman et al.(2020)	✓	✓	✓	✓	✗	✓	✓	✓	✓	✓	NA	✗	✓	✓
Eccles et al. (2002)	✓	✓	✓	✓	!	!	✓	✓	✓	✓	NA	✓	✓	✓
Gill et al. (2009)	!	✗	✓	✓	✓	✗	✓	✓	✗	✗	NA	!	✓	✗
Cobos et al. (2006)	✓	✓	✓	✓	✓	✓	✓	✓	NS	NS	NA	✗	✓	✓
Hicks et al. (2008)	!	✗	!	✓	✗	✓	✓	✓	NS	NS	NA	✗	✓	✓
Lopez et al. (2019)	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗	NA	!	NA	!

Key	
✓	Adequately addressed
!	Partially addressed
✗	Not adequately addressed
NS	Not stated
NA	Not applicable

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Risk of Bias Assessment: Type 2 Diabetes Mellitus

Referring to **Supplementary Tables 4 and 5**, Schnipper et al was rated at 'low risk' for random sequence generation as a random number generation on Excel was used [1].Allocation concealment was rated as 'high' after considering the randomisation an open allocation method. Blinding of personnel was not cited. Thus, it was rated as unclear. There was no mention of withdrawals; hence, the risk was unclear for incomplete outcome reporting. All outcomes set be studied were reported, and no other risks related to study funding were detected.

Gill et al's study was rated at 'high risk' for incomplete and selective outcome reporting due to analysing subgroups of patients for A1C and LDL-C for the narrated study limitations in the following section [2].Randomisation was at 'high risk' since clusters of practices were allocated based on the number of clinicians. Although the randomisation method could be considered open to predictability, eligibility criteria for patients within clusters of practices were pre-defined, which could not have been subverted by allocation knowledge. All other elements were considered at 'low risk.'

Heselmans et al's study was at 'low risk' for all elements except incomplete outcome data due to the withdrawal of a significant number of practices, with less than 80% of practices remaining after randomisation for analysis[3]. Reasons for system concerns were attributed to withdrawals.

3.2.2 Risk of Bias Assessment: Asthma and Angina

Referring to **Supplementary Tables 4 and 5**, the risk of bias in Eccles et al. 's study was 'low' for all elements due to the use of a computerised randomisation process and blinding of assessors[4]. However, less than 80% of eligible patients completed the provided questionnaires, which could have introduced attrition bias. Baseline characteristics of patients and practitioners were not sufficiently addressed, but there was a mention that participating practices were of similar size and training. Selective recruitment of clusters was adequately prevented by recruiting practices and enrolling all patients who met the inclusion criteria.

Risk of Bias Assessment: Hyperlipidaemia

Referring to Supplementary T**ables 4 and 5**, Cobos et al. 's study was rated at 'low-risk' for random sequence generation and allocation concealment due to the use of a random number table to randomise, which does not allow predictable sequences[5]. There was no mention of blinding of assessors, statisticians, or patients. Although <80% of randomised patients had follow-up visits and LDL-C follow-up tests, the study was rated as 'low risk' for attrition and reporting bias because the ITT analysis yielded insignificant outcomes as did the PP analyses. Study funding did not seem to introduce methodological bias since the outcome was patient-reported. Selective recruitment of clusters was adequately prevented by recruiting practices and enrolling all patients of randomised practices who met the inclusion criteria.

Gill et al 's trial was rated at 'high risk' for random sequence generation but 'low risk' for allocation concealment due to randomising clusters of practices based on practice similarities, which is not preferable randomisation[6]. The Eligibility criteria for patients within clusters of practices were pre-defined, which could not have been subverted by allocation knowledge. Blinding of assessors and

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

statisticians was not mentioned. The number of patients analysed was not reported. However, all outcomes set to be studied were undertaken. The first author is a consultant researcher at the company that holds the concerned EHR's licensing rights, but this conflict of interest did not seem to reflect in methodological risks. Similarly, selective recruitment of clusters was prevented.

Risk of Bias Assessment: Hypertension

Referring to **Supplementary Tables 4 and 5,** Hick's et al 's study was rated 'high risk' for random sequence generation, as clusters of practices were randomised by size and type of practice, which may not be a desirable method of randomisation [7]. Although randomisation could have been open to predictability, patient eligibility criteria practices were pre-defined, which could not have been subverted by allocation knowledge. Thus, the study was rated at 'low risk' for allocation concealment. Blinding of assessors or patients was not mentioned. Thus, the study was rated 'unclear' for blinding participants and outcome assessments. Due to the high number of randomised patients that were withdrawn due to stated reasons, it was rated at 'high risk' for attrition bias. All outcomes were explored, and there was a 'low risk' for selective recruitment of cluster members.

Lopez et al. 's study was rated at 'high risk' for random sequence generation and allocation concealment due to its non-randomised design[8]. There was no mention of blinding personnel, hence it was rated 'unclear' for blinding. There was no mention of withdrawals, and all outcomes were reported. Thus, it was rated as 'low risk' for attrition and selective outcome reporting. No other risks related to funding were significant.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Table 6: Quality of Evidence from individual studies evaluated using GRADE

Study	Outcome	Risk of bias	Level of inconsist-ency	Level of indirect-ness	Level of imprecise-on	Publication bias	Quality	Reason for downgrading
Schnipper et al. (2010)	Difference in proportion of patients meeting management goals at the end compared to the beginning of the study	severe	NA	some	low	NA	⊕⊕◯◯ LOW	**RoB:** High risk for allocation concealment and unclear whether personnel blinding occurred **Indirectness:** population identification and management goals were not pre-defined
Gill et al. (2017)	• Difference in HbA1c from baseline • Difference in LDL-C from baseline • % patients achieving HbA1c goals	some	NA	low	low	NA	⊕⊕⊕◯ MODERATE	**RoB:** High risk for random sequence generation and selective outcome reporting
Heselmans et al. (2020)	• Difference between the mean value of HbA1c, LDL-C, and blood pressure of the previous year at the moment of the study start and the mean value of HbA1c, LDL-C, and blood pressure of the previous year at 6 and 12 months.	low	NA	some	low	NA	⊕⊕⊕◯ MODERATE	**Indirectness:** patients had reasonable HbA1c mean values, which does not necessarily reflect diabetic populations with a higher severity of illness
Eccles et al. (2002)	Quality of life and condition specific outcomes using validated questionnaires	some	NA	low	low	NA	⊕⊕⊕◯ MODERATE	**RoB:** High risk for incomplete and selective outcome reporting
Gill et al. (2009)	The percentage of patients with most recent LDL-C at goal	severe	NA	low	low	NA	⊕⊕◯◯ LOW	**RoB:** High risk for random sequence generation and incomplete outcome data. Unclear for personnel blinding
Cobos et al. (2006)	• Between-group differences in means of lipid values	some	NA	some	low	NA	⊕⊕⊕◯ MODERATE	**ROB**: Unclear for blinding of personnel **Indirectness:** excludes general populations with more severe elevations of LDL
Hicks et al. (2008)	• Percentage of patients with controlled blood pressure • Odds of blood pressure control at outcome visit	severe	NA	low	low	NA	⊕⊕◯◯ LOW	**RoB:** High risk for random sequence allocation, incomplete outcome reporting and unclear for personnel blinding
Lopez et al. (2019)	• Odds of blood pressure control post-intervention	severe	NA	low	low	NA	⊕⊕◯◯ LOW	**RoB:** NRCT, non-randomised and high risk for blinding of personnel

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

References

1) Schnipper JL, Linder JA, Palchuk MB, et al. Effects of documentation-based decision support on chronic disease management. *The American Journal of Managed Care.* 2010; 16 (12 Suppl HIT) (pp SP72-81): ate of Pubaton: e 2010.

2) Gill J, Kucharski K, Turk B, Pan C, Wei W. Using Electronic Clinical Decision Support in Patient-Centered Medical Homes to Improve Management of Diabetes in Primary Care: The DECIDE Study. *Journal of Ambulatory Care Management.* 2019; 42 (2): 105-115. doi:10.1097/JAC.0000000000000267

3) Heselmans A, Delvaux N, Laenen A, Van de Velde S, Ramaekers D, Kunnamo I, et al. Computerized clinical decision support system for diabetes in primary care does not improve quality of care: a cluster-randomized controlled trial. *Implementation science : IS.* 2020; 15 (1): 5. doi:10.1186/s13012-019-0955-6

4) Eccles M, McColl E, Steen N, Liaw ST. Computerised evidence-based guidelines may not improve asthma or angina management in primary care. *Evidence-Based Healthcare.* 2003; 7 (2): 81-82.

5) Cobos A, Vilaseca J, Asenjo C, et al. Cost effectiveness of a clinical decision support system based on the recommendations of the European Society of Cardiology and other societies for the management of hypercholesterolemia: Report of a cluster-randomized trial. *Disease Management and Health Outcomes.* 2005; 13 (6): 421-432. doi:10.2165/00115677-200513060-00007

6) Gill JM, Chen YX, Glutting JJ, et al. Impact of decision support in electronic medical records on lipid management in primary care. *Population health management.* 2009; 12 (5): 221-226. doi:10.1089/pop.2009.0003. doi:10.1089/pop.2009.0003

7) Hicks LS, Sequist TD, Ayanian JZ, et al. Impact of computerized decision support on blood pressure management and control: A randomized controlled trial. *Journal of General Internal Medicine.* 2008; 23 (4): 429-441. doi:10.1007/s11606-007-0403-1

8) Lopez PM, Divney A, Goldfeld K, et al. Feasibility and Outcomes of an Electronic Health Record Intervention to Improve Hypertension Management in Immigrant-serving Primary Care Practices. *Medical care.* 2019; 57 (Supplement 62) (pp S164-S171): ate of Pubaton: 01 Jun 2019. doi:10.1097/MLR.0000000000000994

For Peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 3/4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 3/4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 4
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 5, supplementary material table 3
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 5
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 5
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 5/6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 5/6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 6
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 6
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g., tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 6
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 6
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 6
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 6
Reporting bias	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 6

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 6
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 6/7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 6/7
Study characteristics	17	Cite each included study and present its characteristics.	Page 6/7
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 7 & supplementary section
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Page 7/8/9
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page 7 & supplementary section
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 7/8/9
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 7 & supplementary section
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page 7
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 10/11/12
	23b	Discuss any limitations of the evidence included in the review.	Page10/11/12
	23c	Discuss any limitations of the review processes used.	Page10/11/12
	23d	Discuss implications of the results for practice, policy, and future research.	Page10/11/12
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 2
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 2
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Page 2
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 14
Competing	26	Declare any competing interests of review authors.	Page 14

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
interests			
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 14

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

For peer review only